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(54) **SEMICONDUCTOR DEVICE AND METHOD  
FOR FABRICATING THE SAME**

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**H10D 64/27** (2025.01)

(52) **U.S. Cl.**

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(2025.01); **H10D 62/875** (2025.01); **H10D**  
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**H10B 12/00** (2023.01)

**H10D 1/68** (2025.01)

**H10D 62/10** (2025.01)

(57)

**ABSTRACT**

Disclosed are highly integrated memory cells and a semiconductor device including the highly integrated memory cells. The semiconductor device includes a first pillar and a second pillar spaced apart from each other by a shield trench, each pillar including an inner side that defines inner walls of the shield trench and an outer side that faces the inner side; a shield gate formed in the shield trench; a first main gate formed on the outer side of the first pillar; a second main gate formed on the outer side of the second pillar; a bit line formed on lower portions of the first and second pillars; and a capacitor formed on an upper portion of each of the first and second pillars.

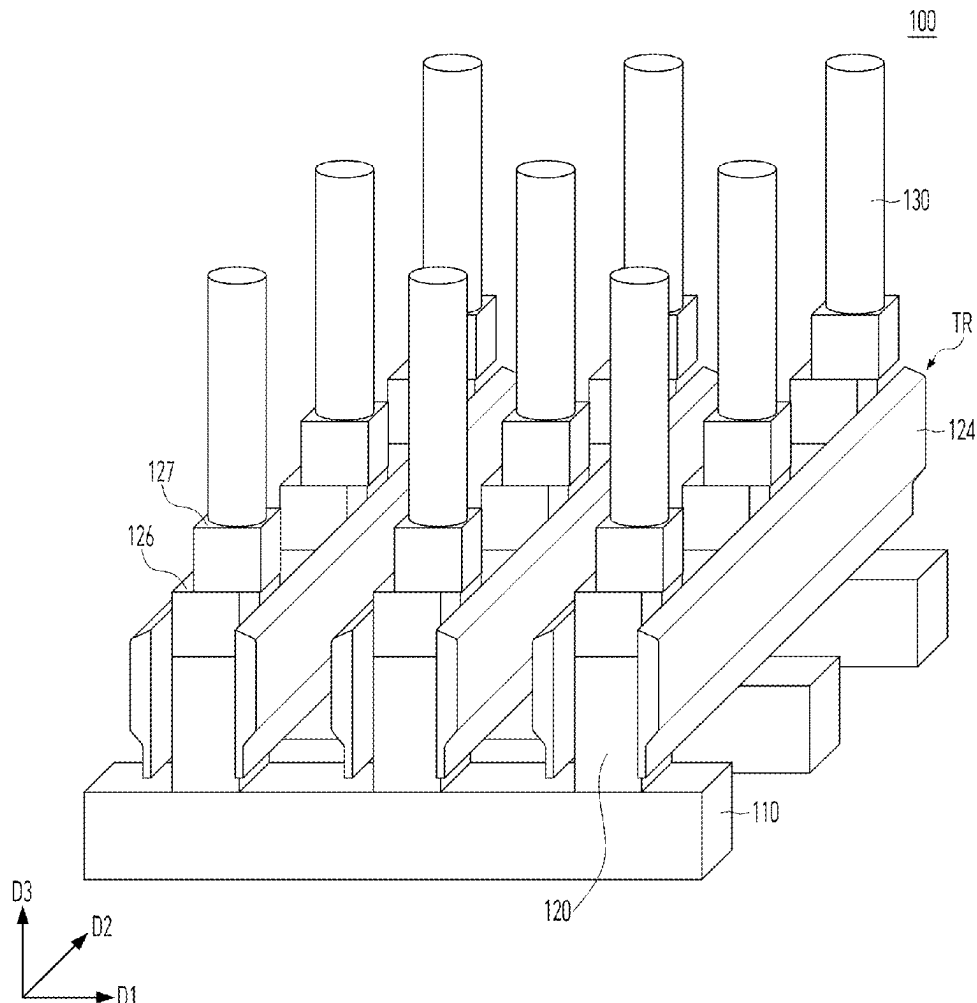


FIG. 1A

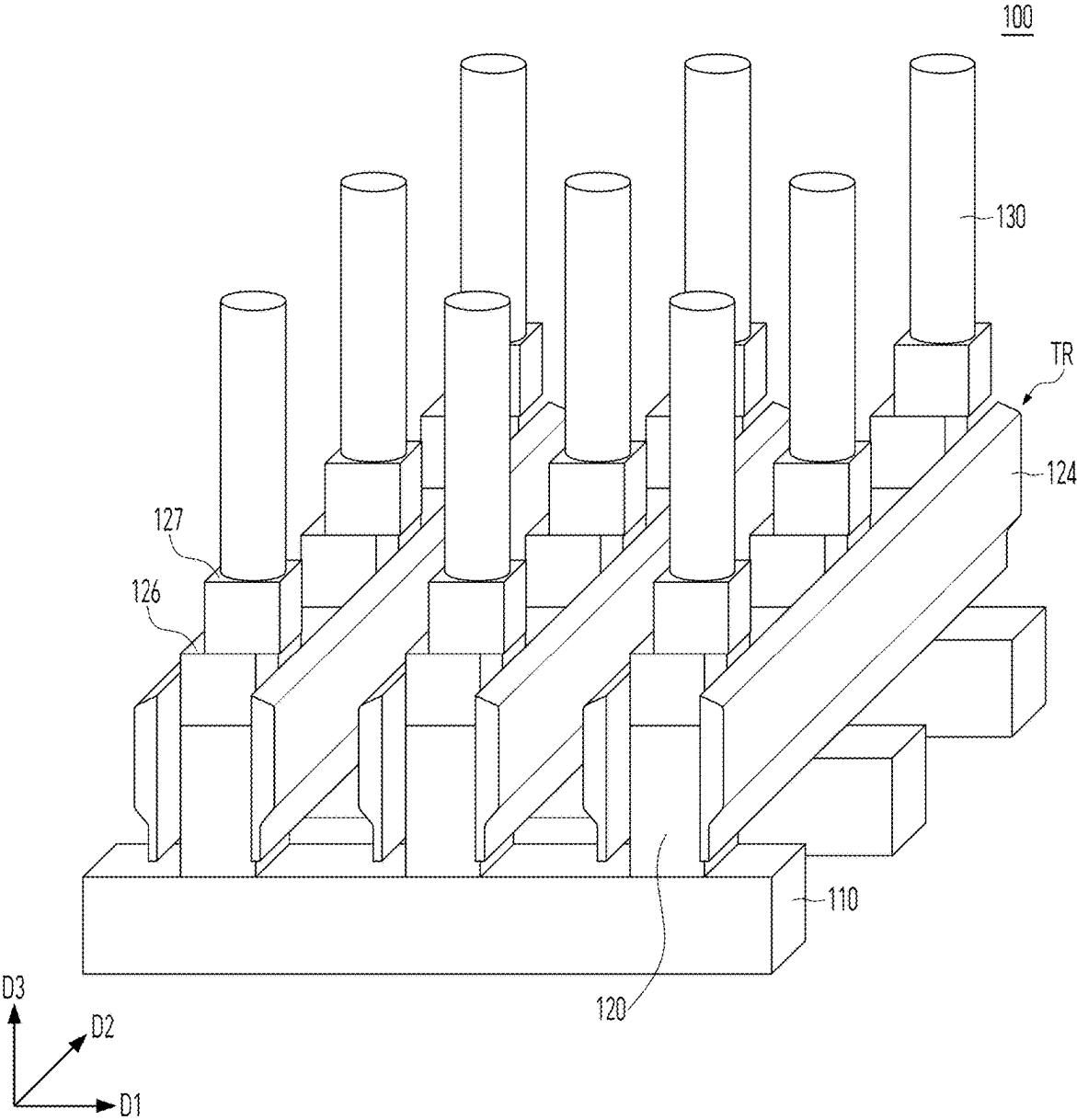


FIG. 1B

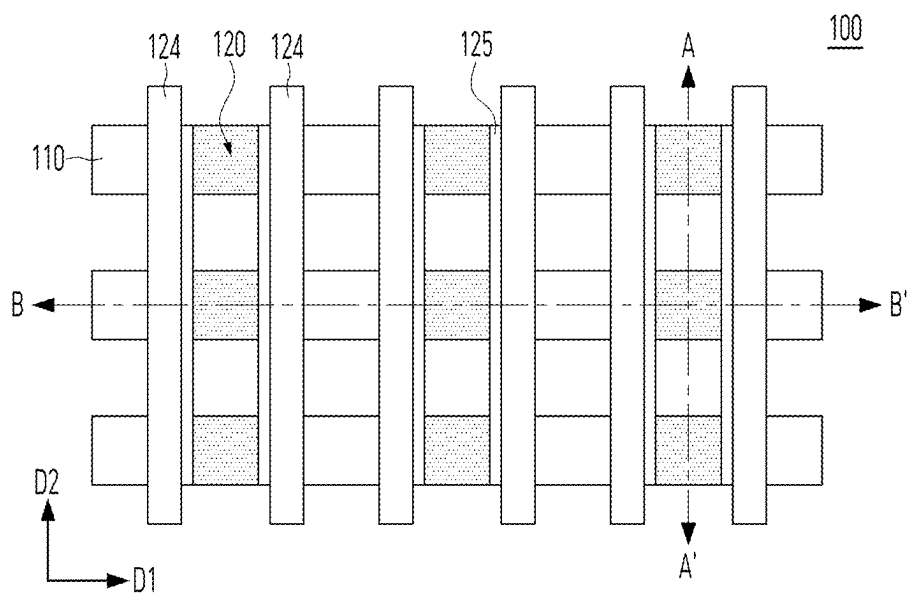


FIG. 1C

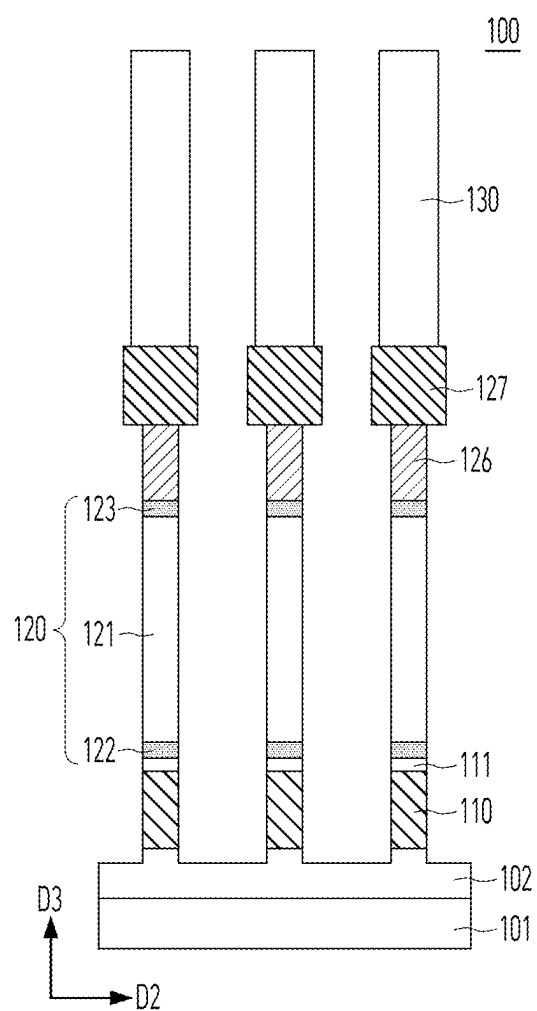




FIG. 1E

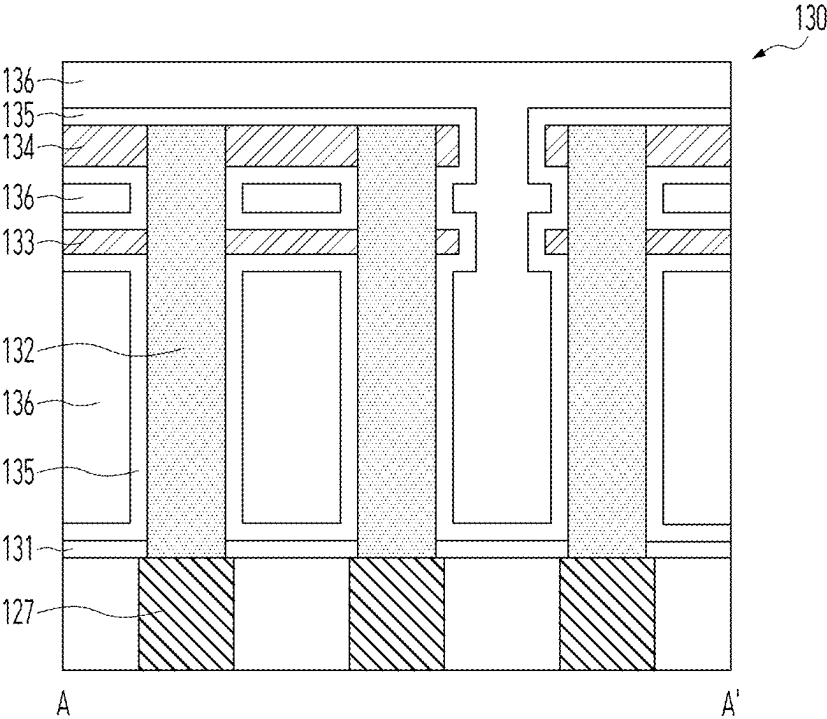


FIG. 2A

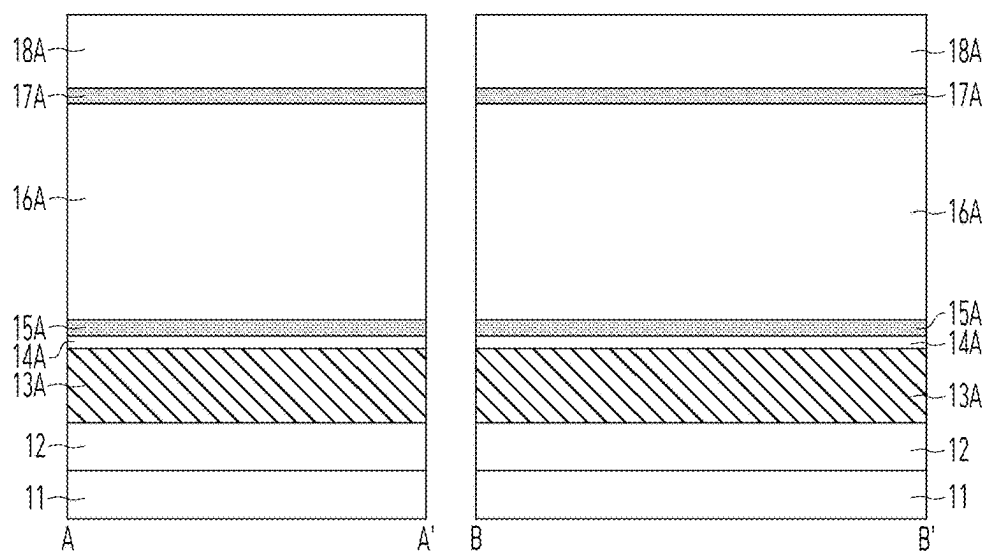


FIG. 2B

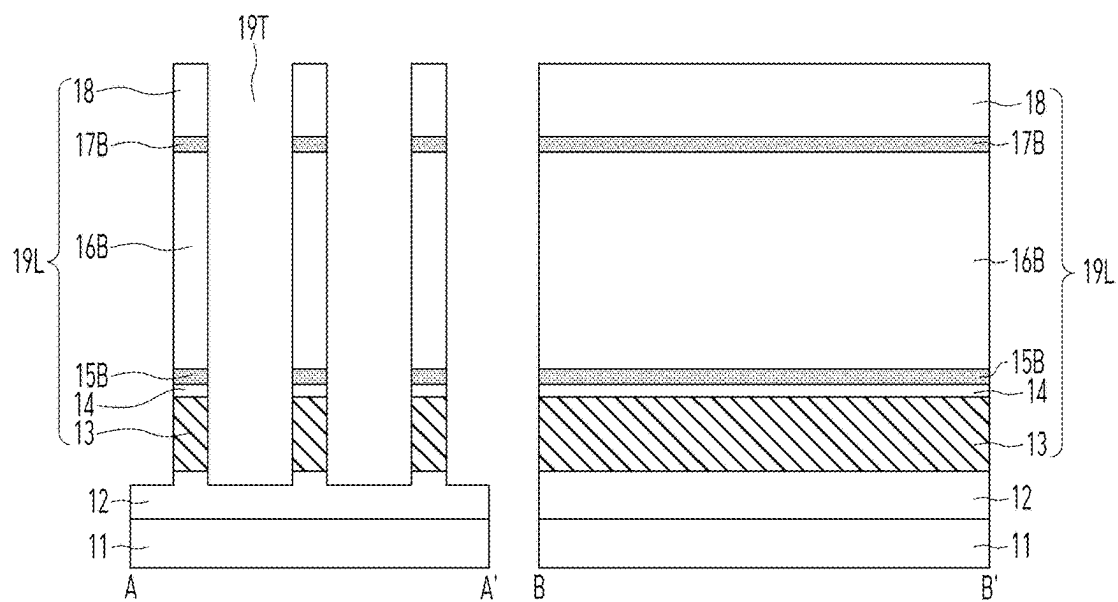


FIG. 2C

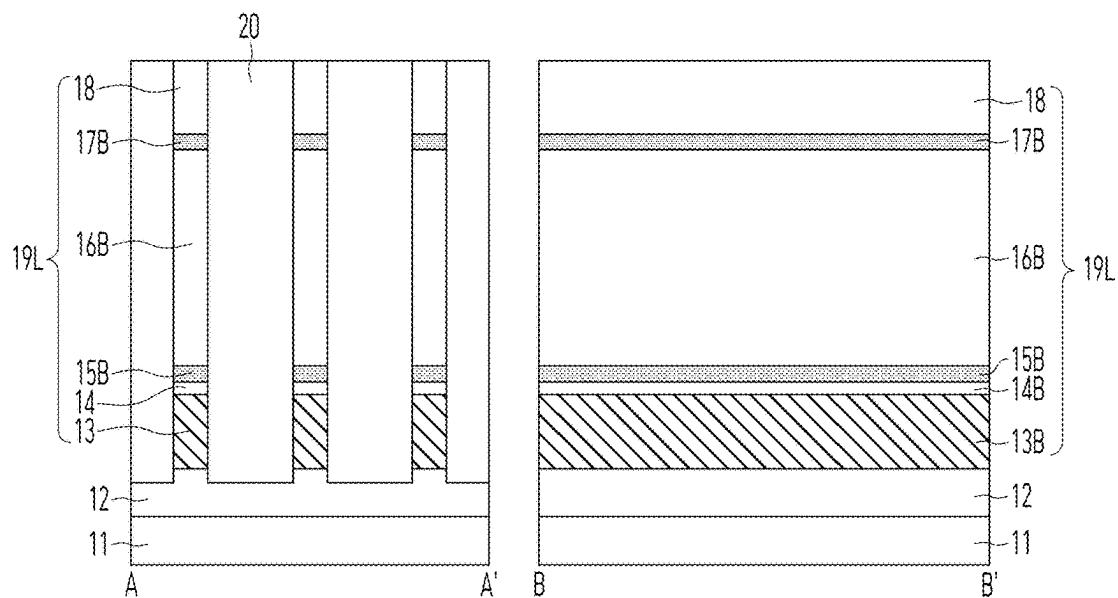


FIG. 2D

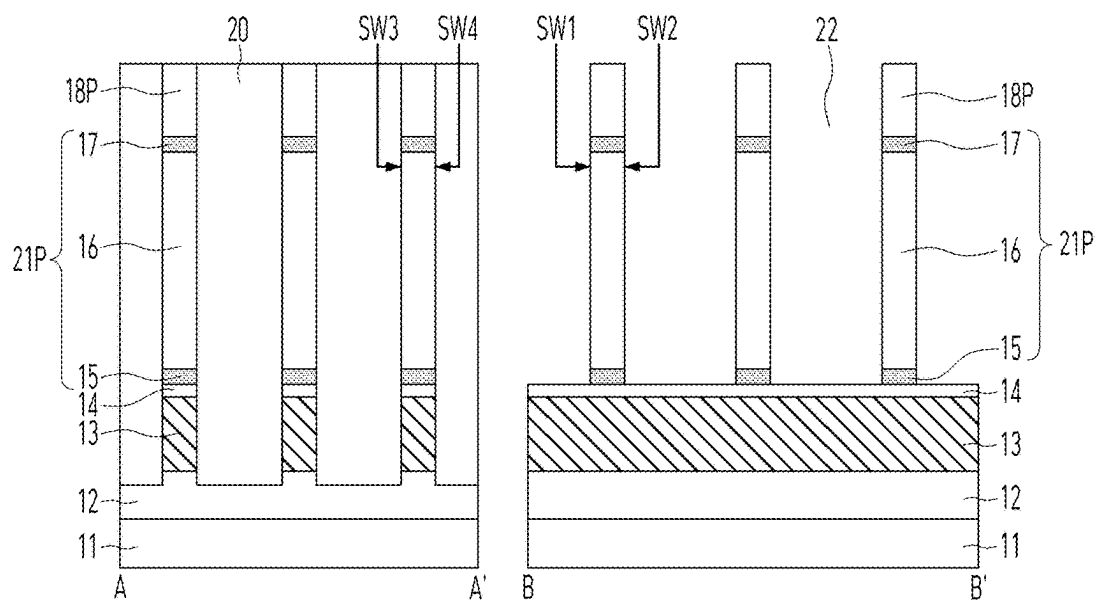




FIG. 2E

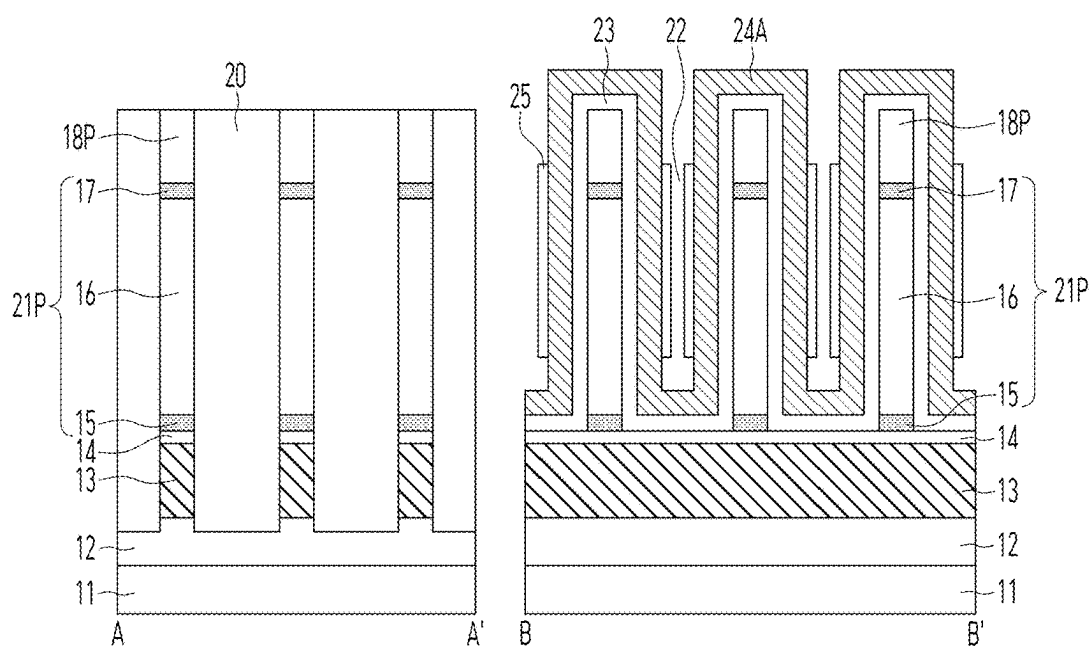


FIG. 2F

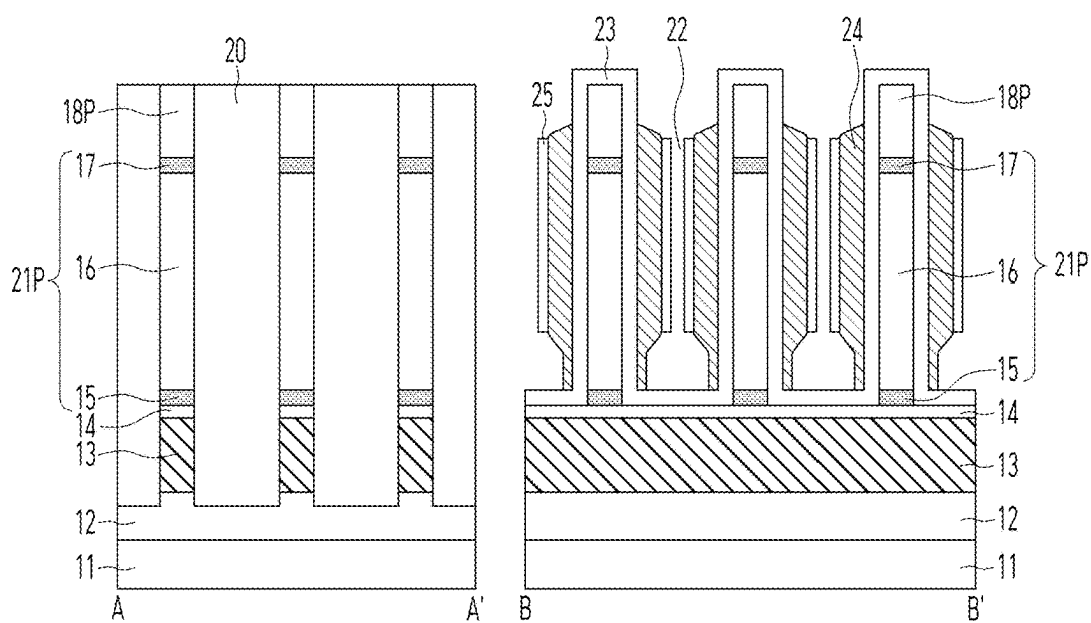


FIG. 2G

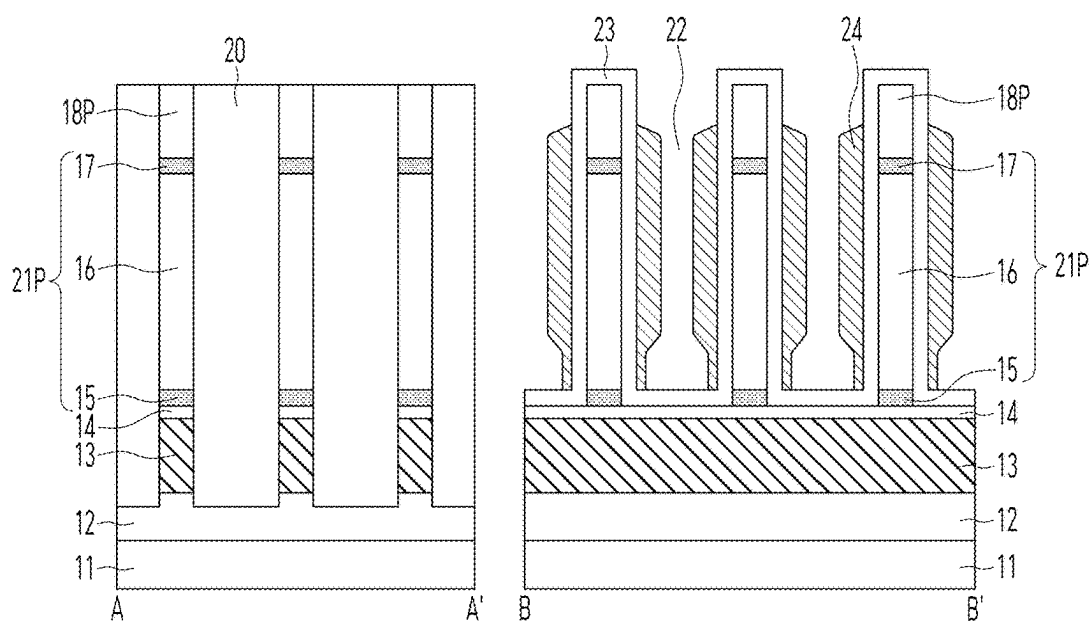


FIG. 2H

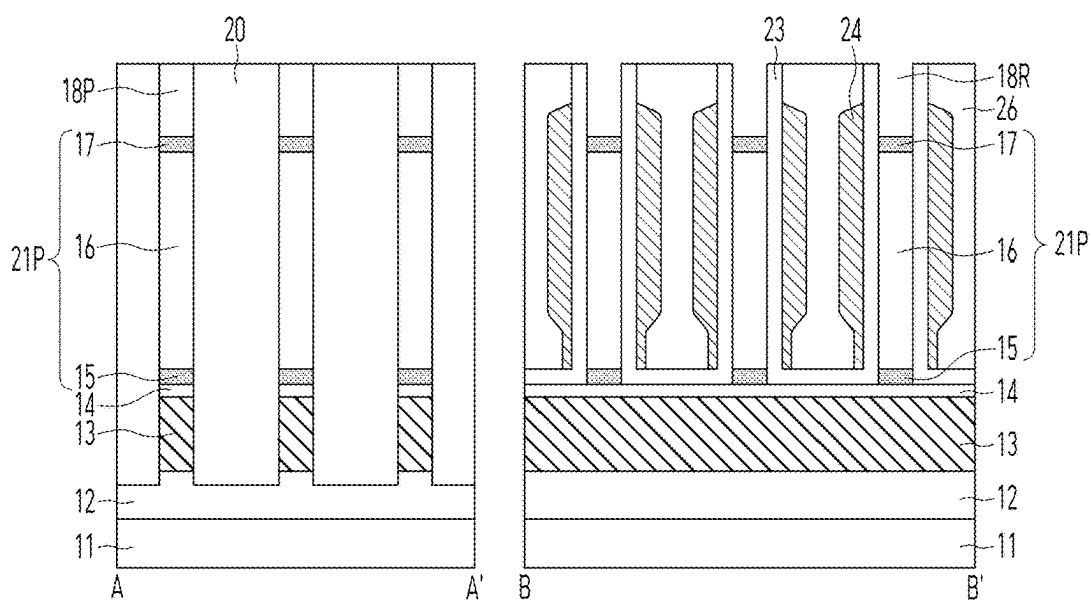


FIG. 2I

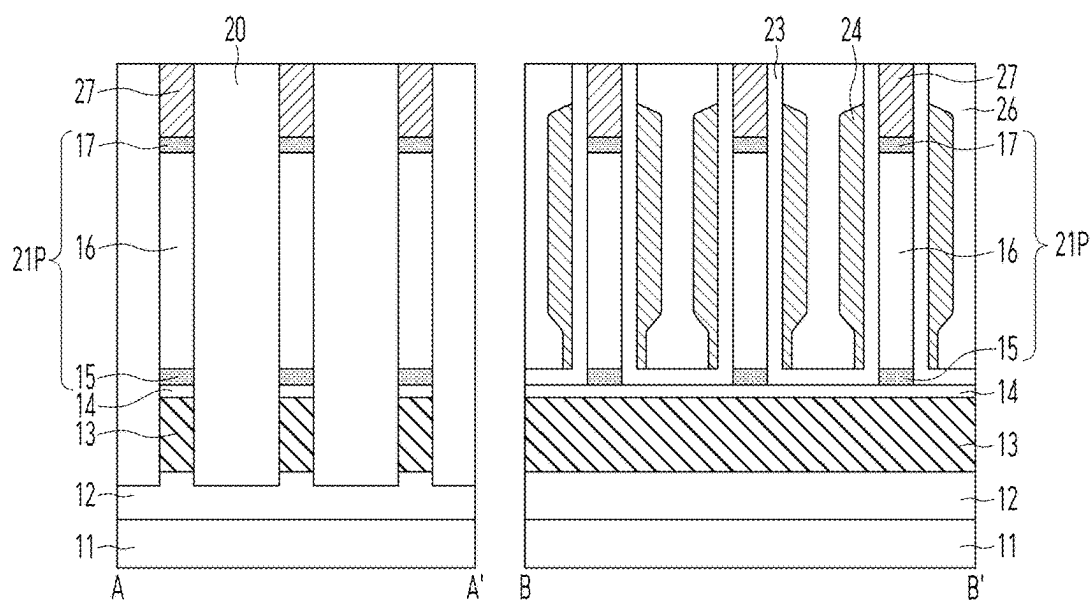


FIG. 2J

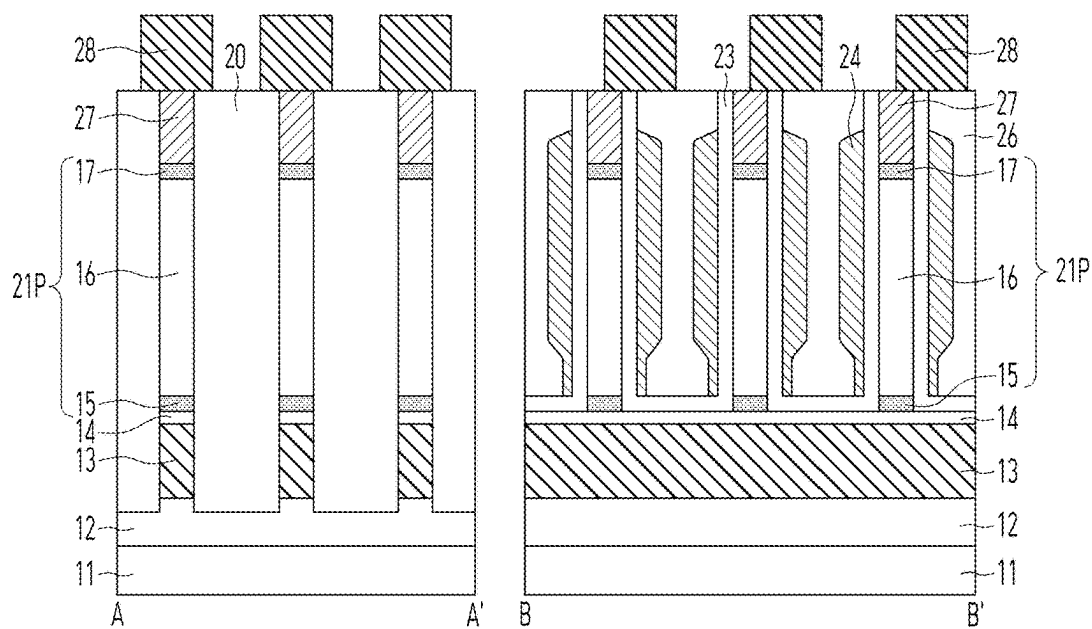


FIG. 2K

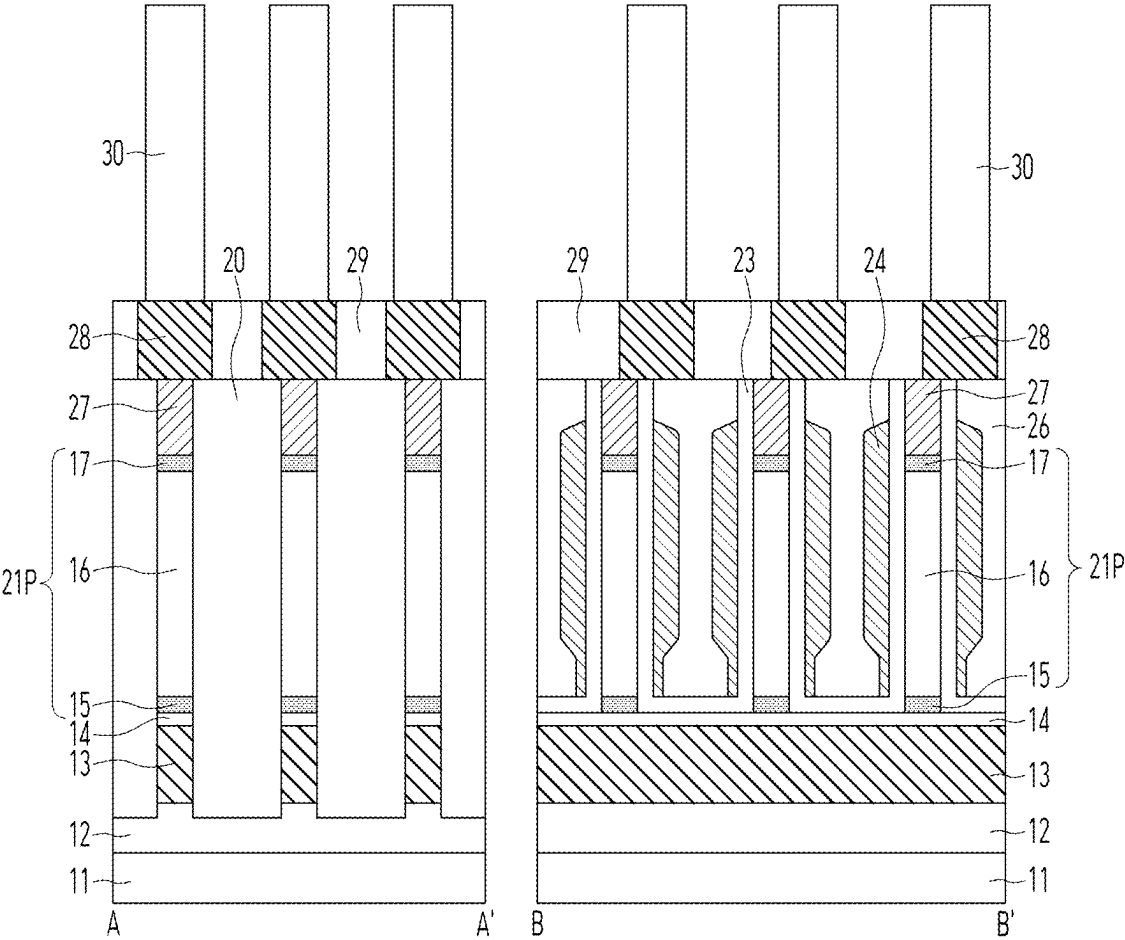


FIG. 3A

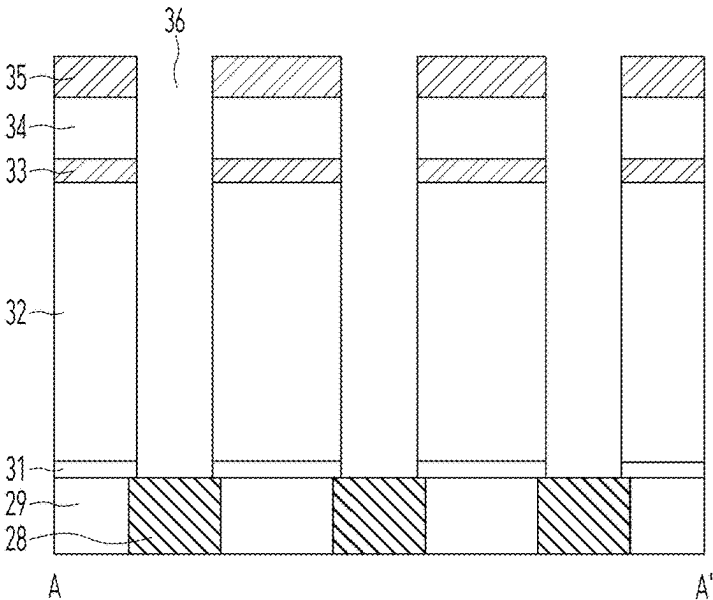


FIG. 3B

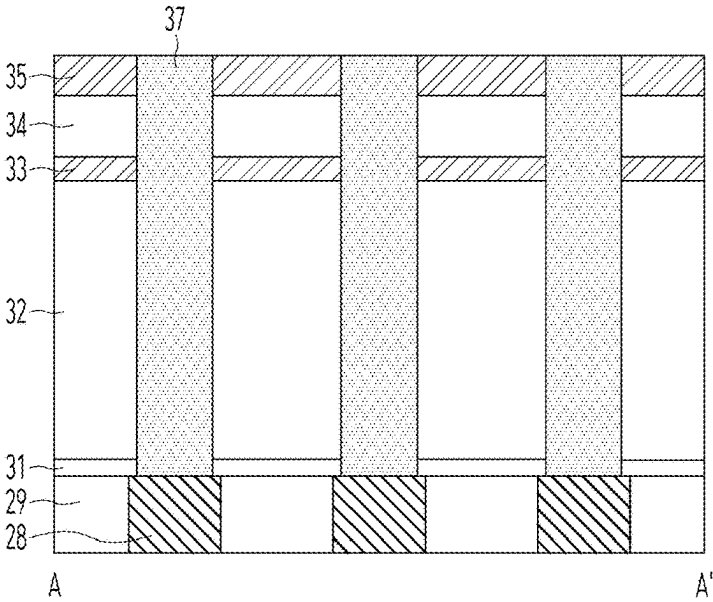


FIG. 3C

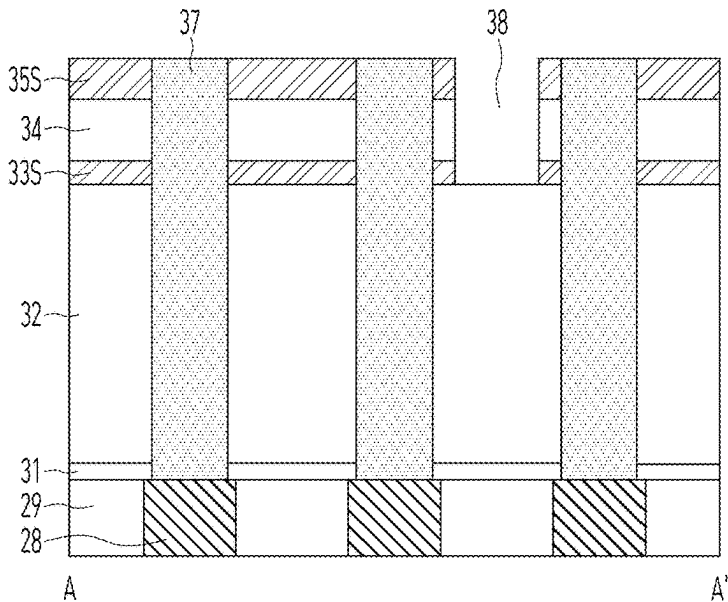


FIG. 3D

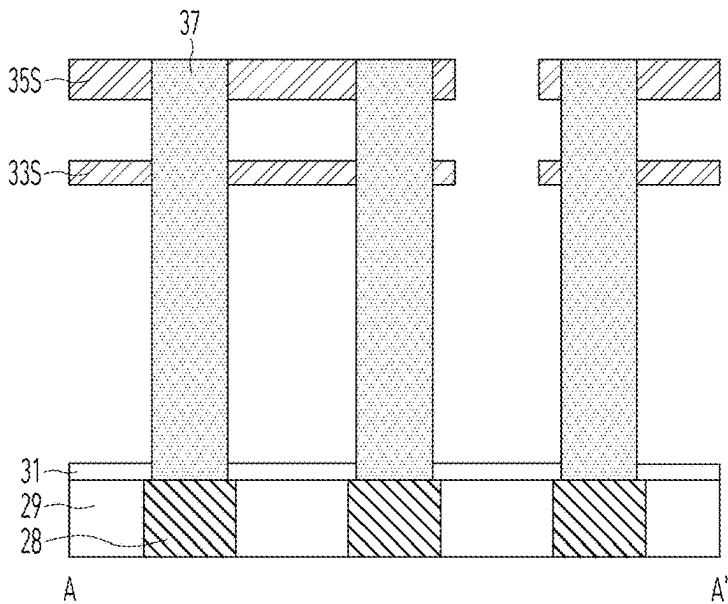


FIG. 3E

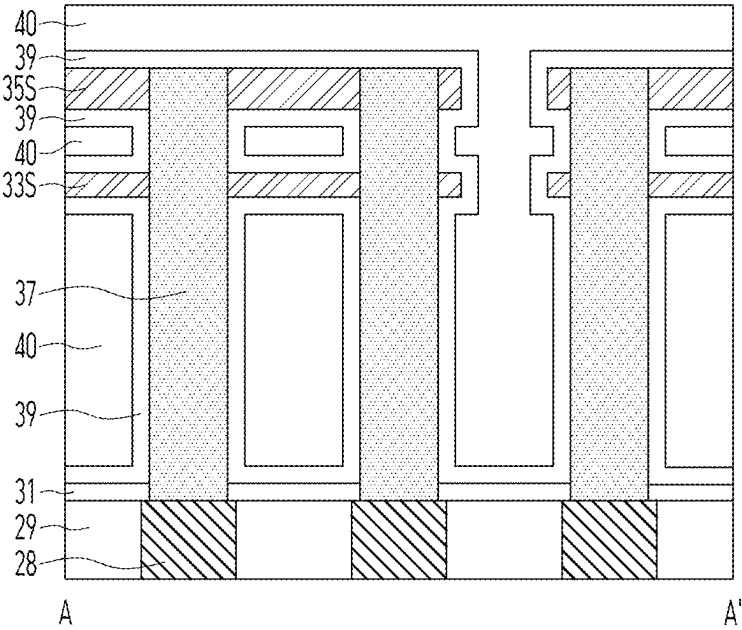


FIG. 4A

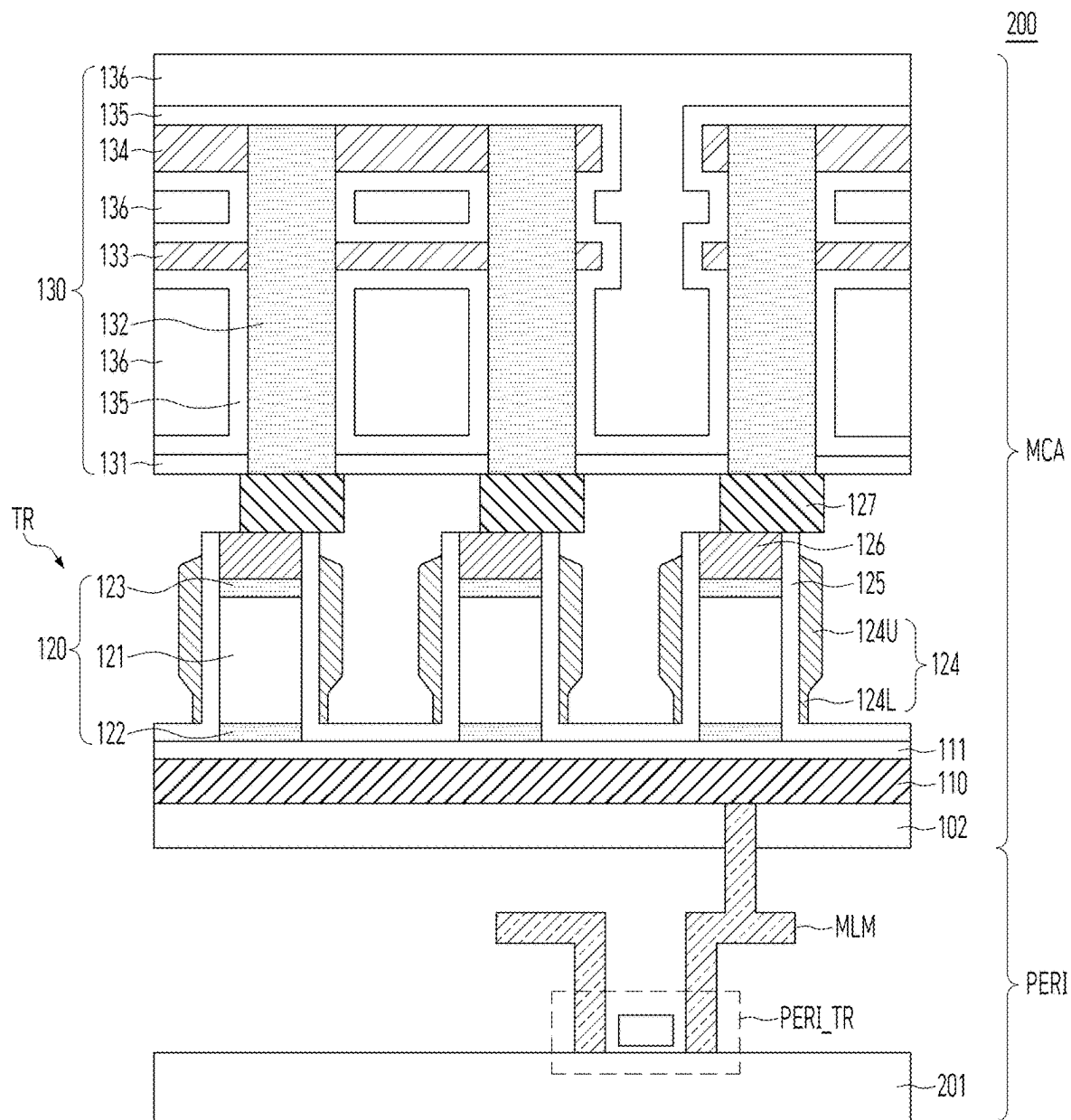




FIG. 4B

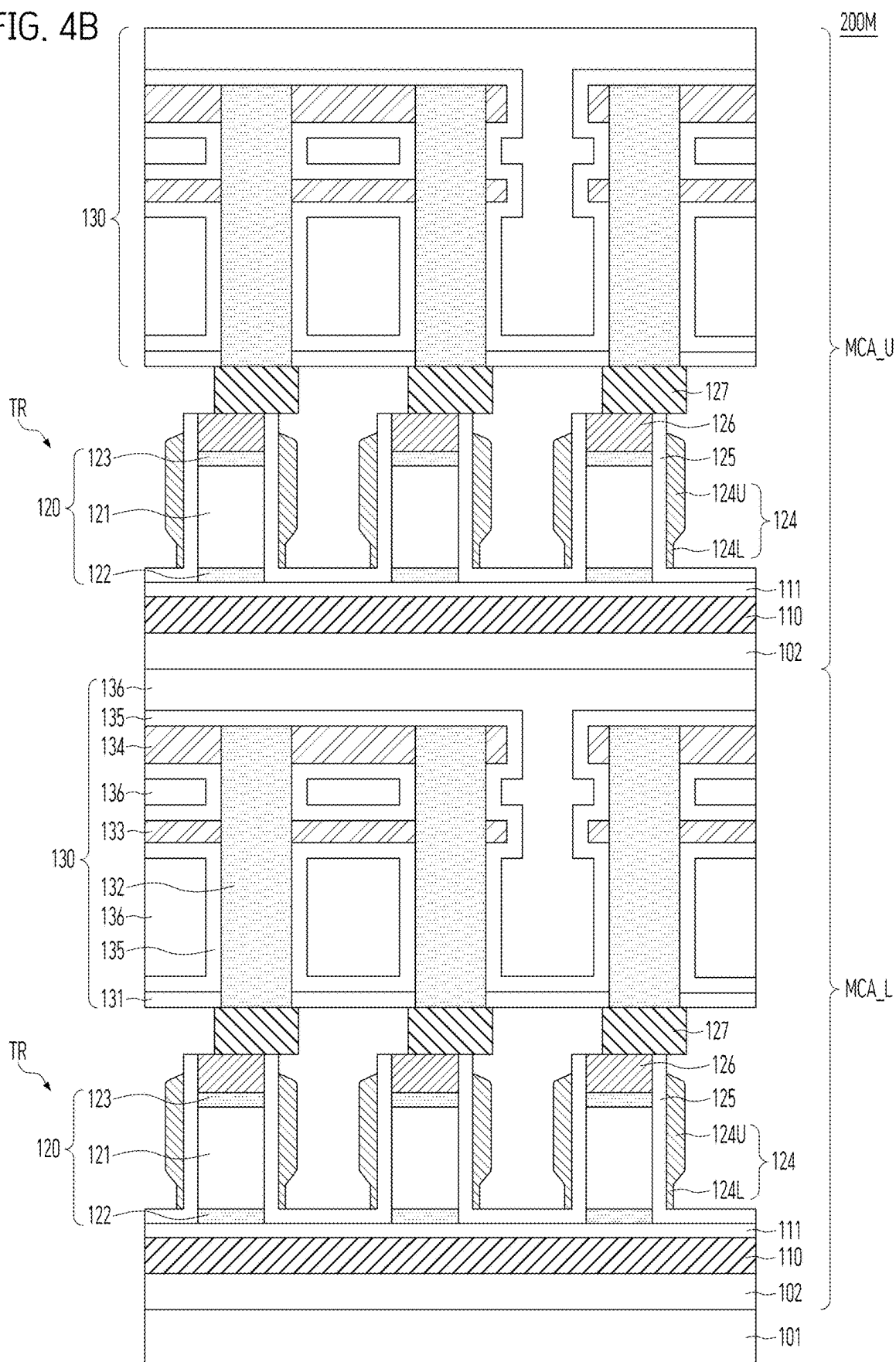


FIG. 5

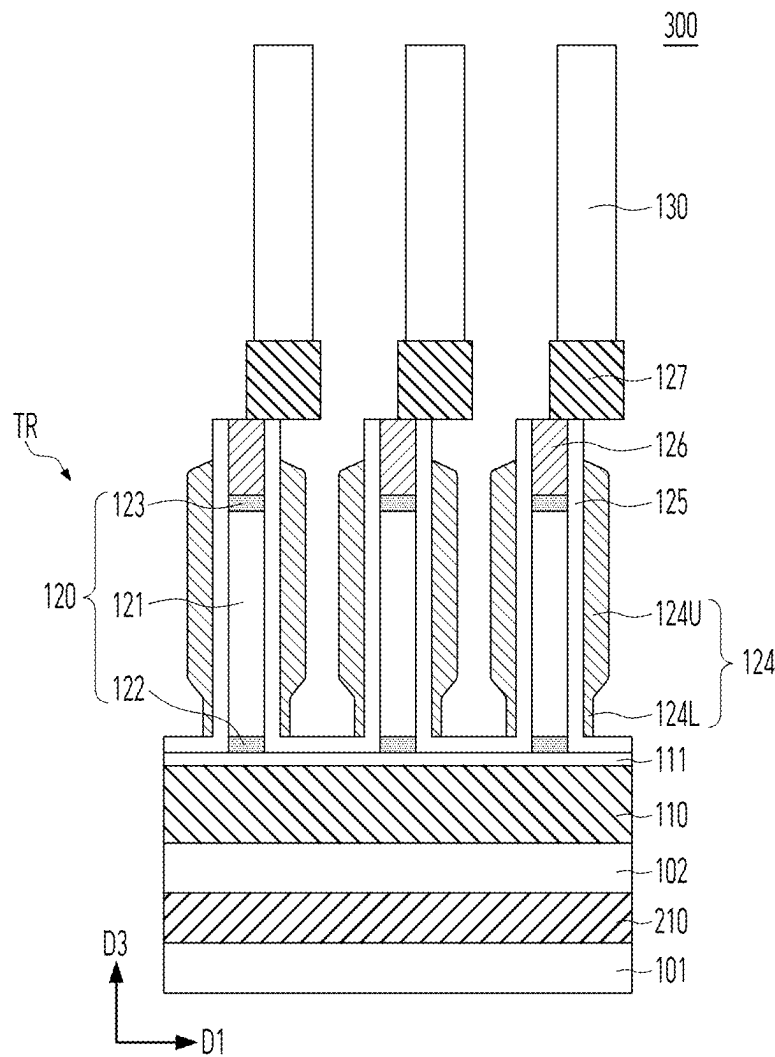


FIG. 6A

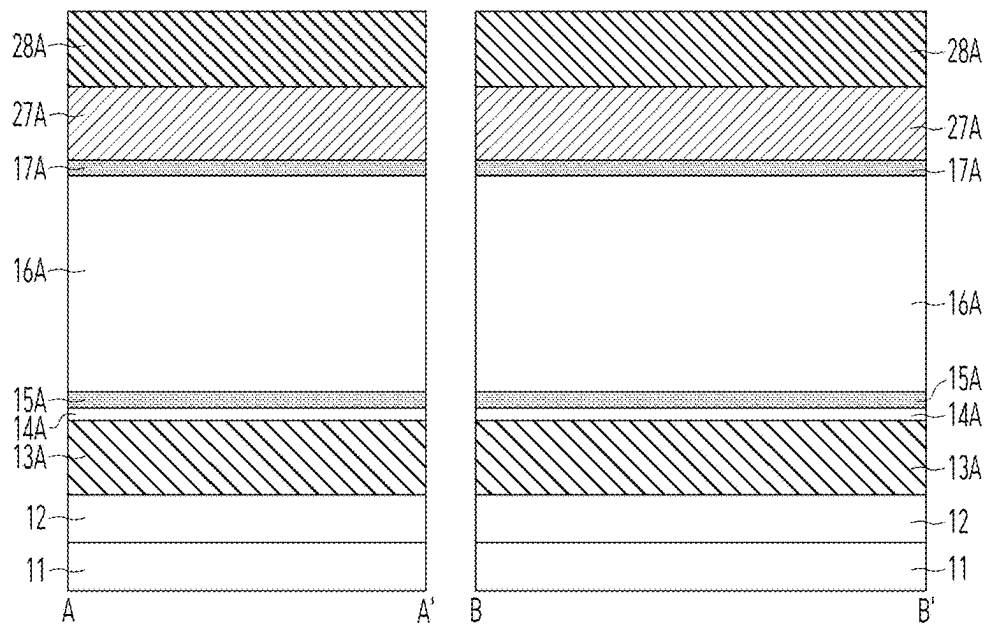


FIG. 6B

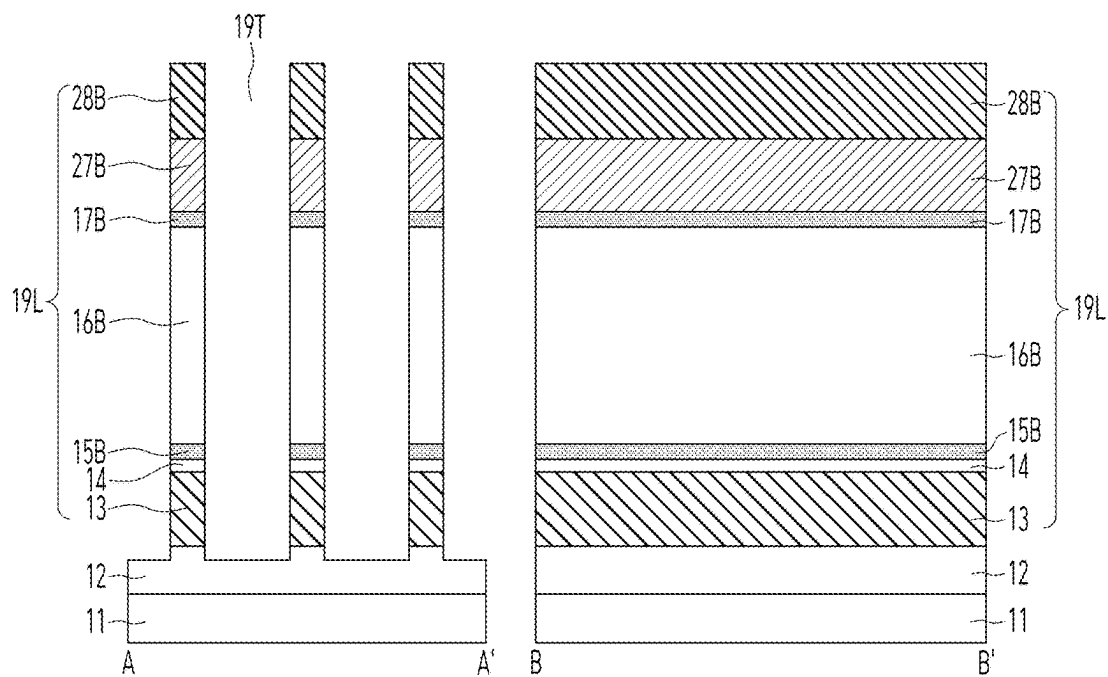




FIG. 7A

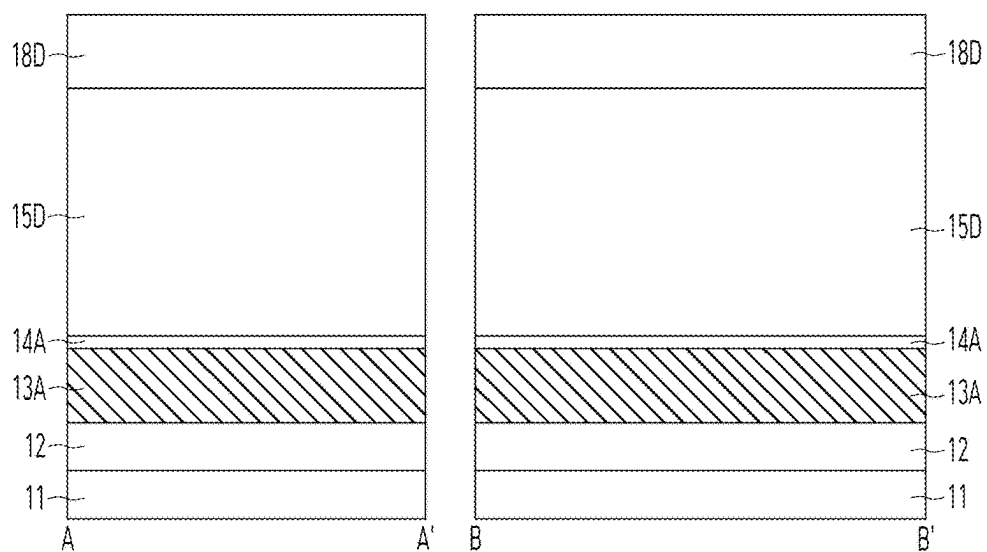


FIG. 7B

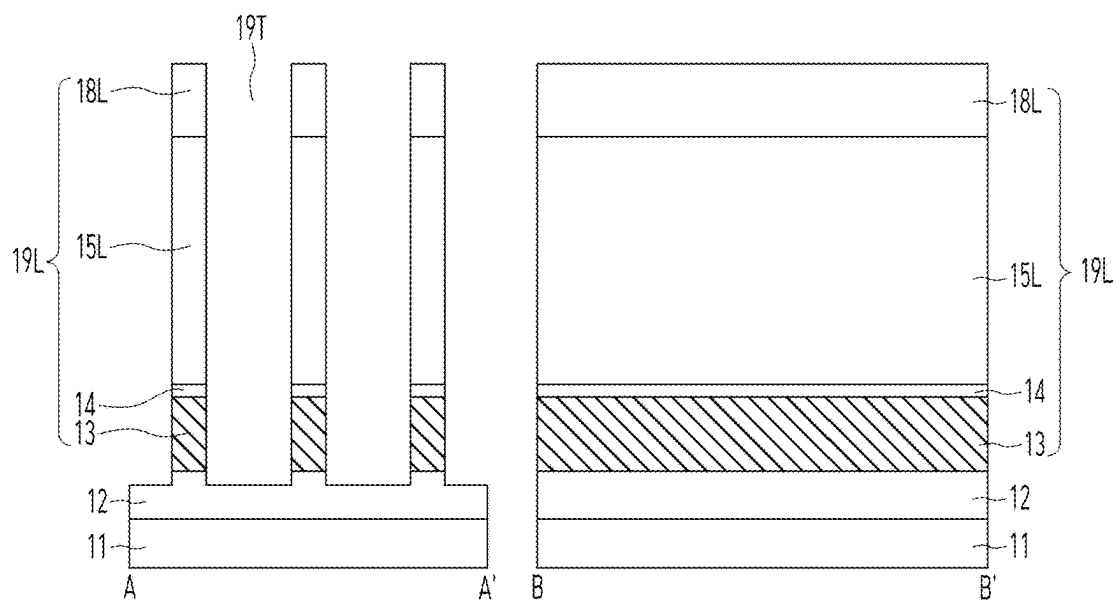


FIG. 7C

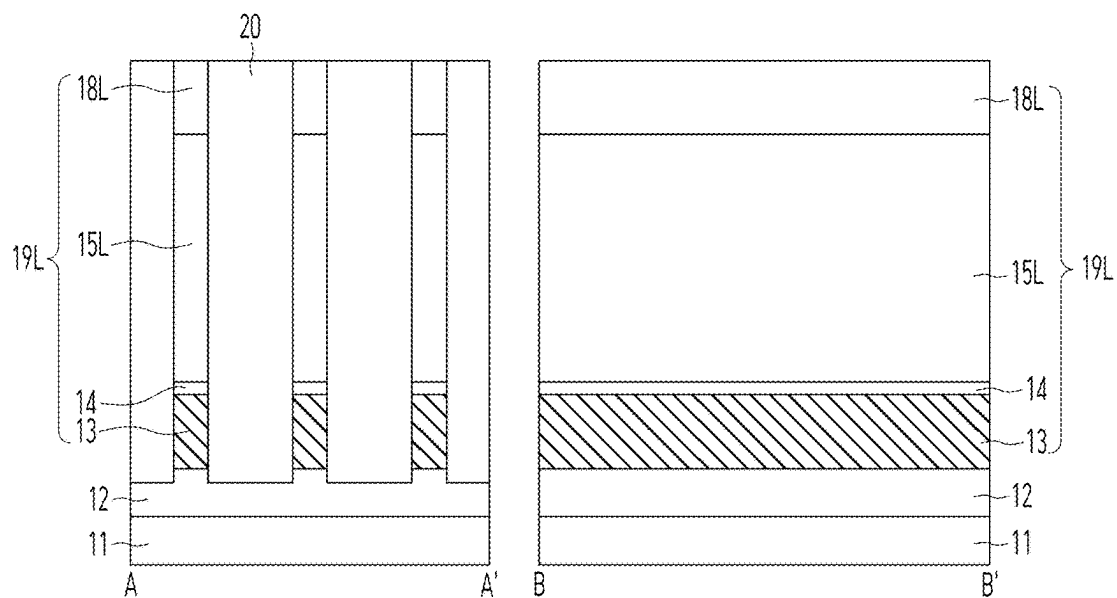


FIG. 7D

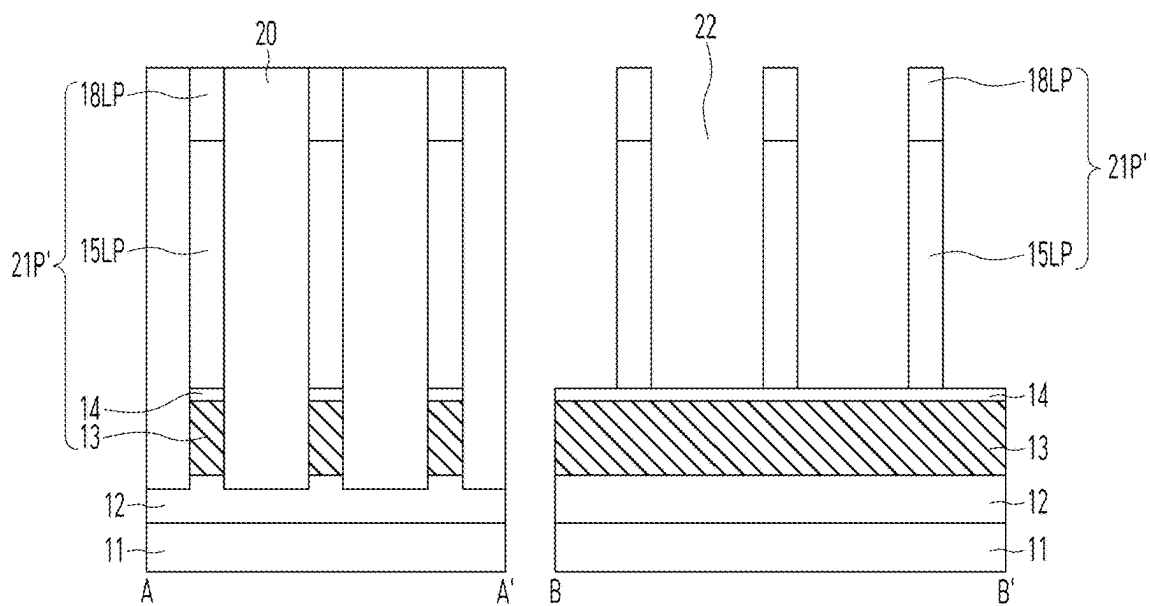


FIG. 7E

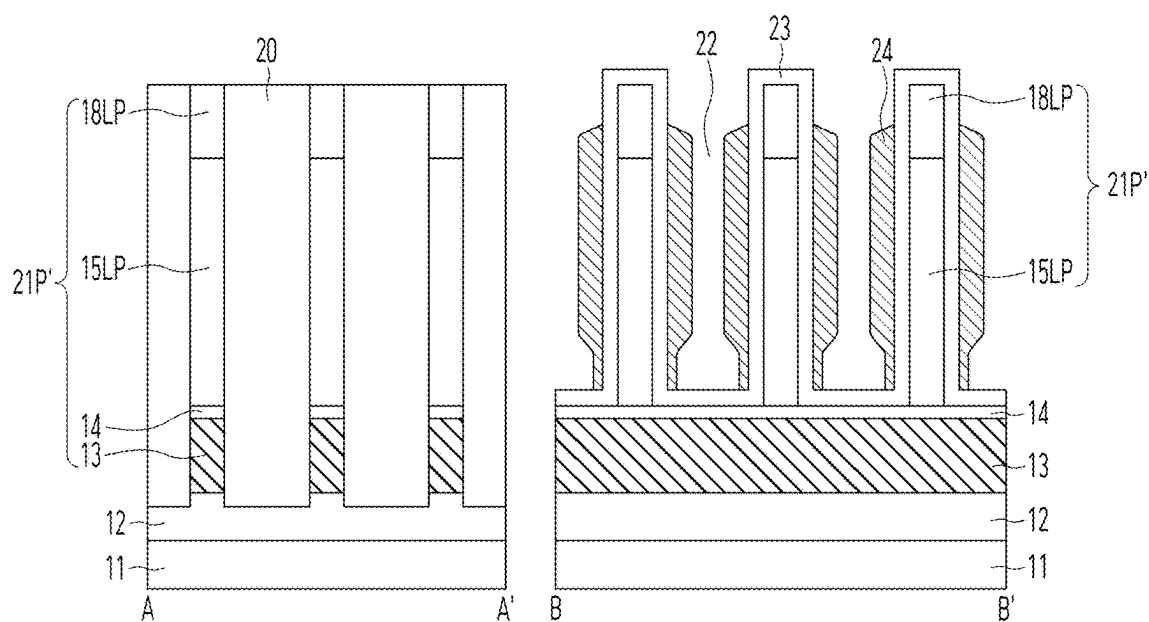


FIG. 7F

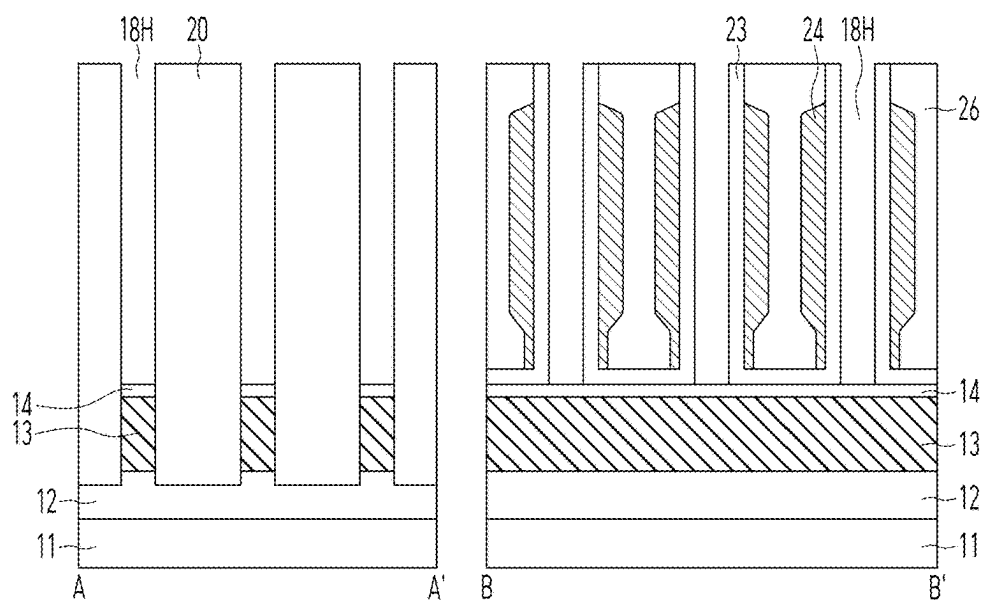


FIG. 7G

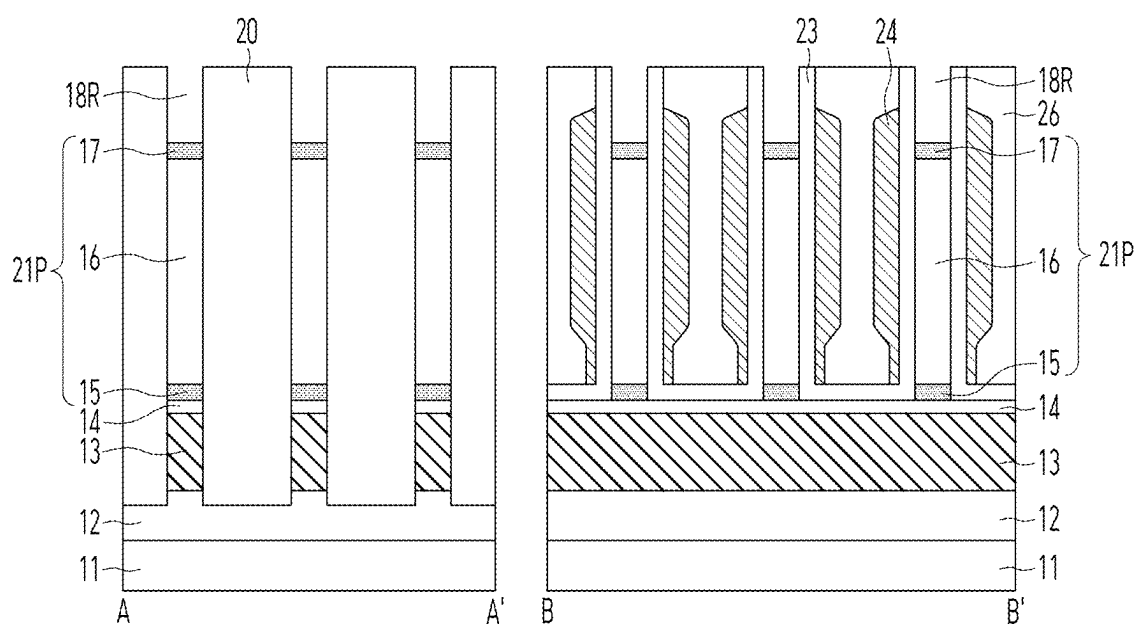




FIG. 8

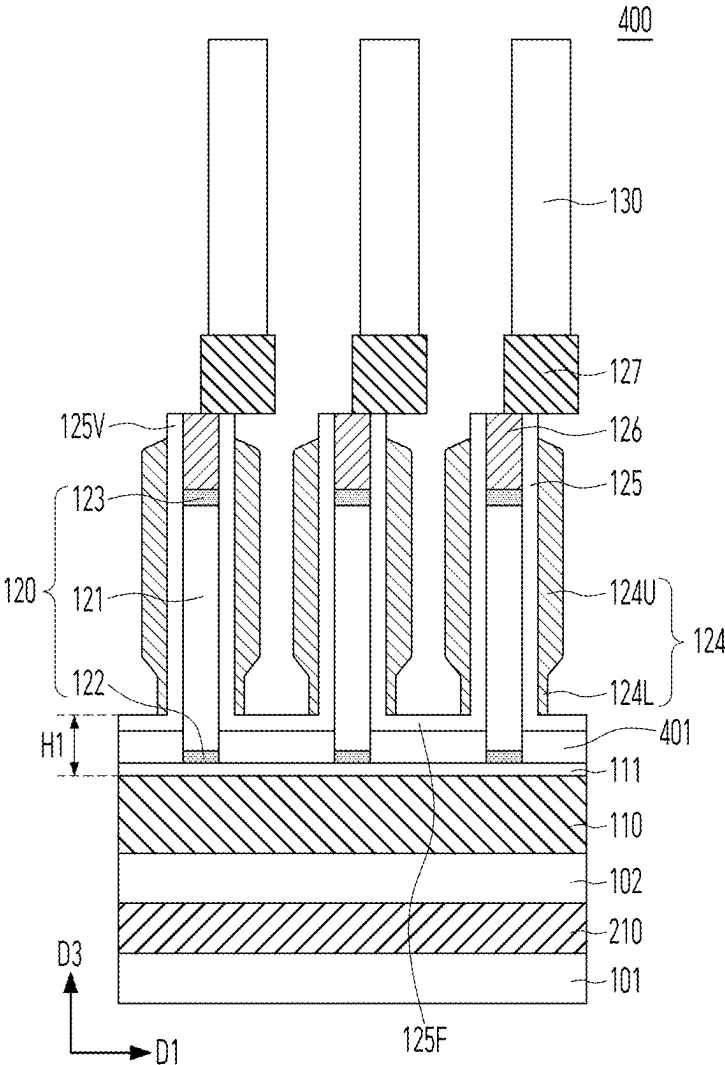


FIG. 9

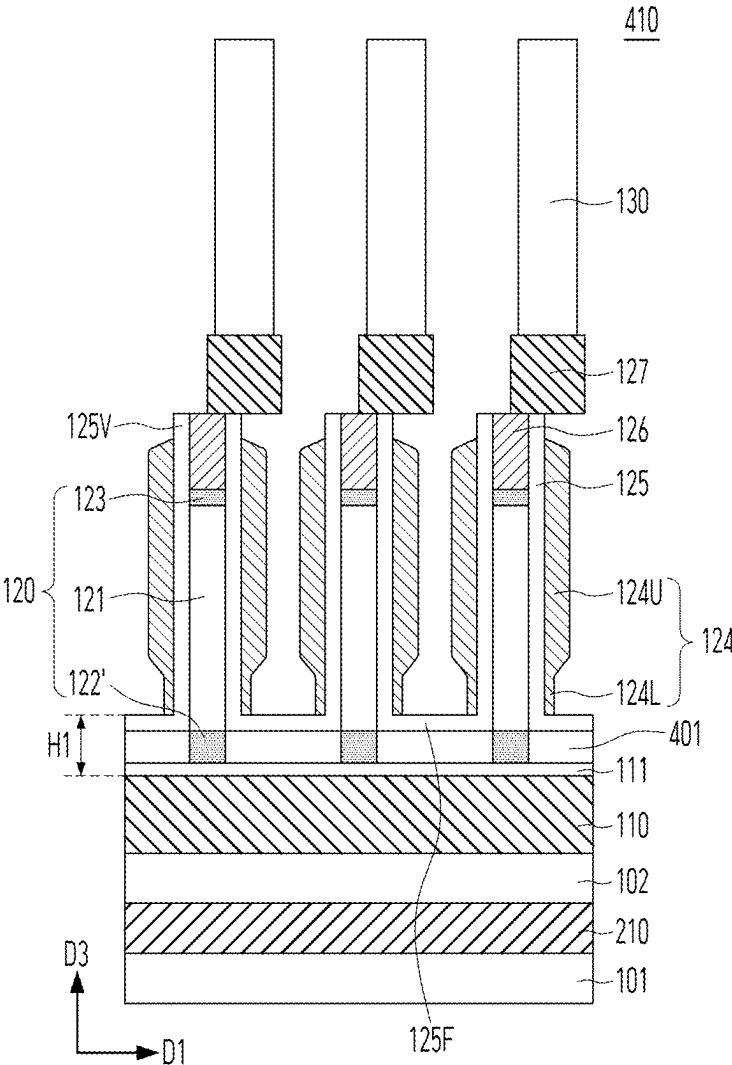


FIG. 10

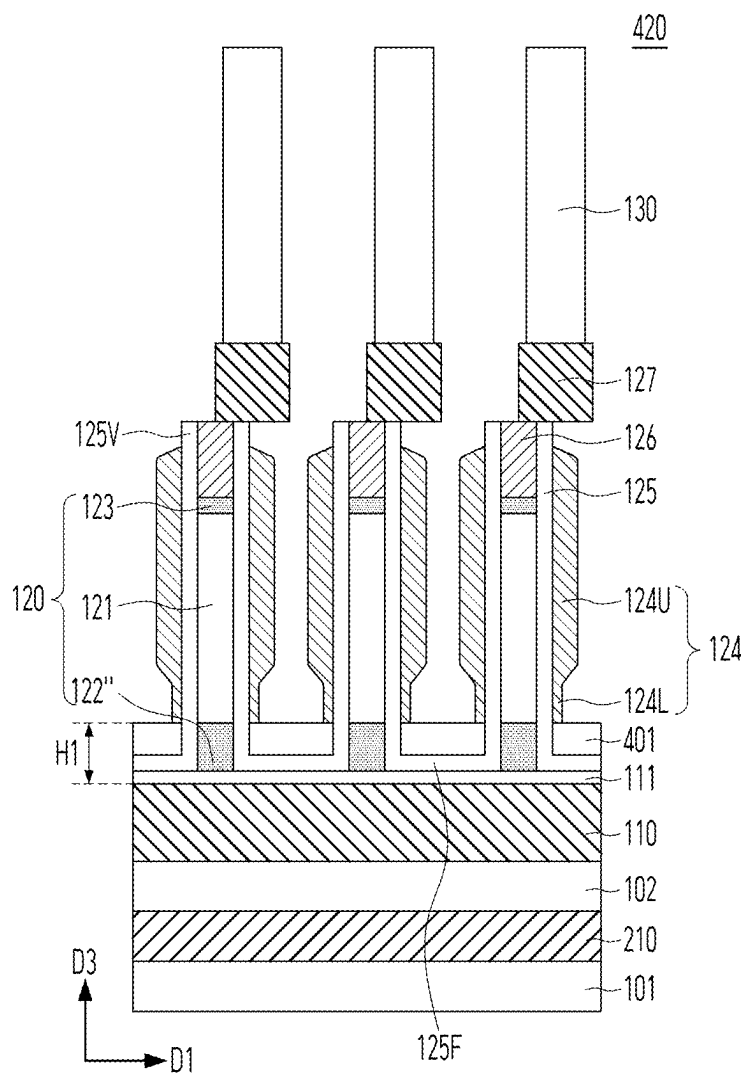


FIG. 11A

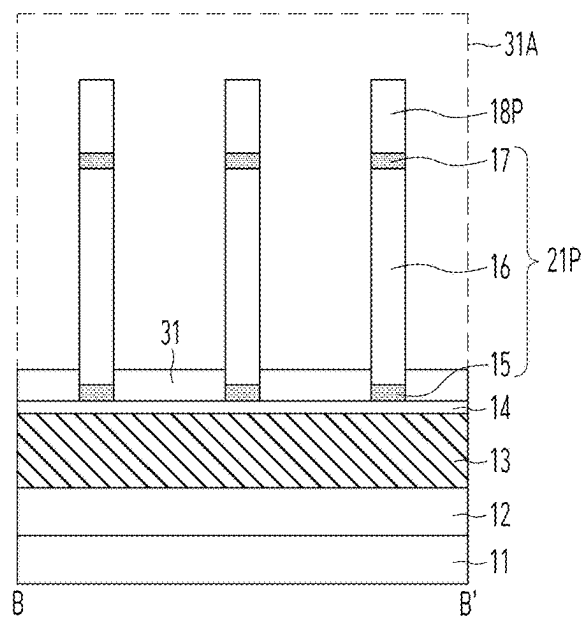


FIG. 11B

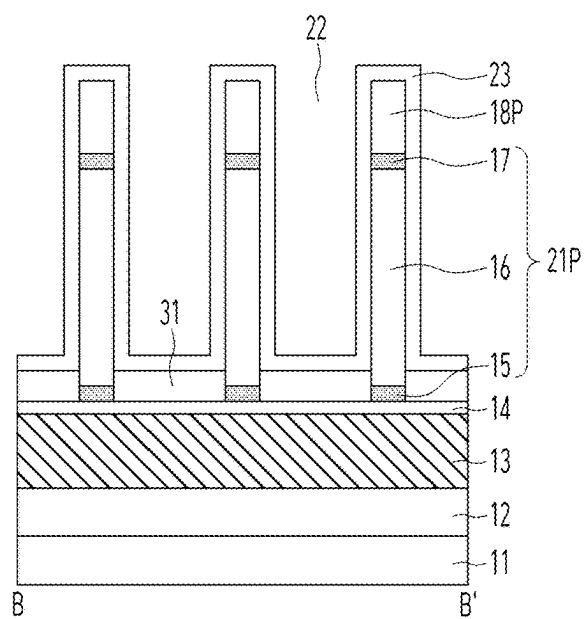


FIG. 11C

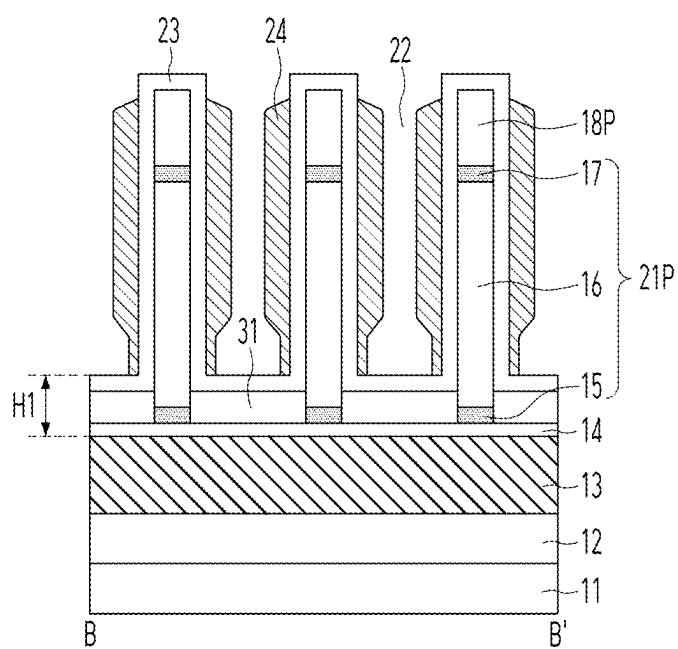


FIG. 12A

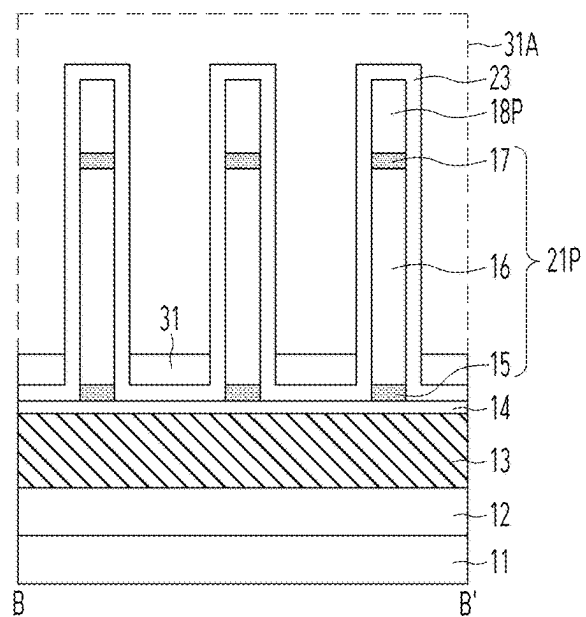


FIG. 12B

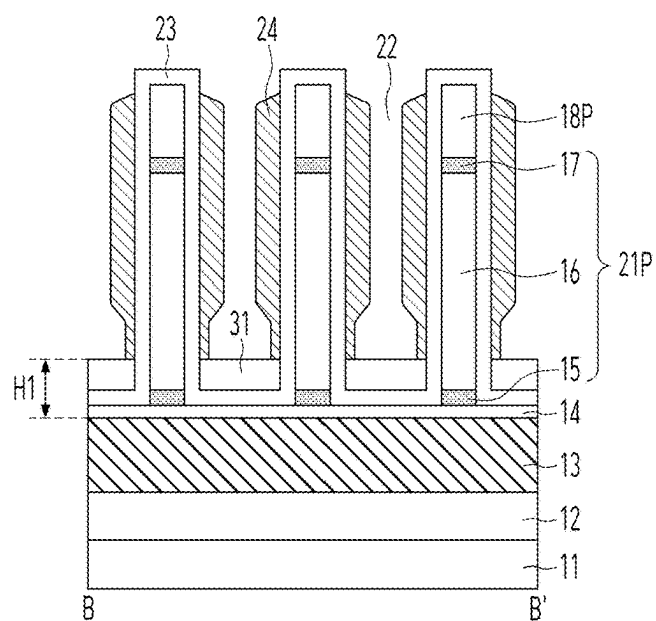


FIG. 13A

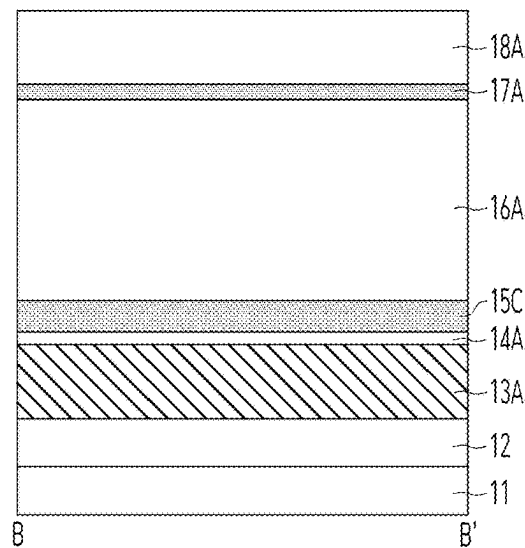


FIG. 13B

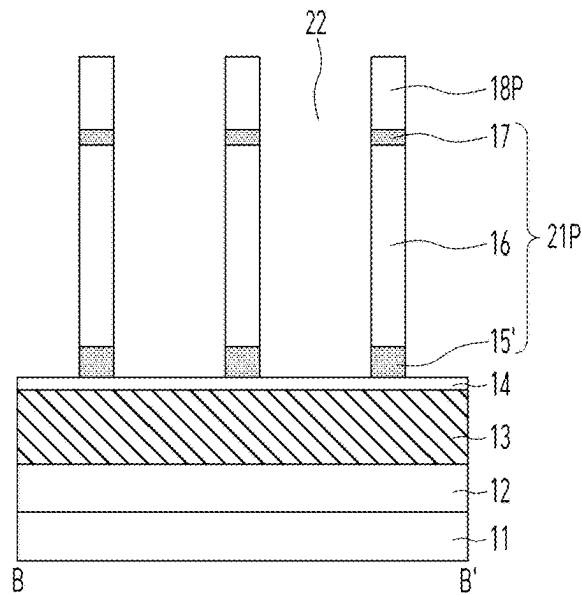


FIG. 13C

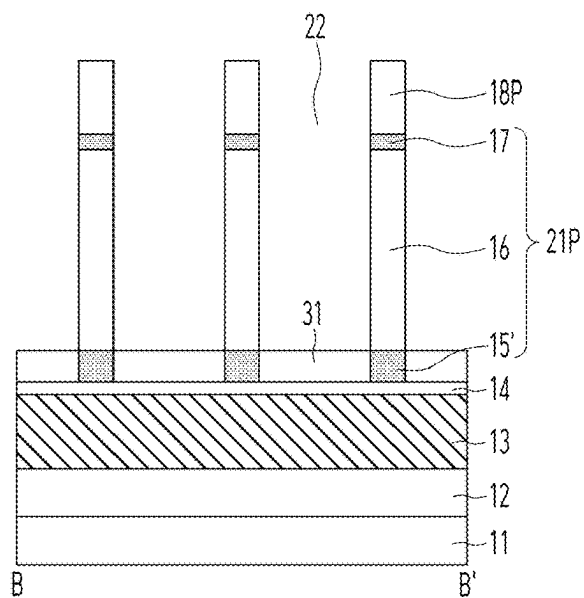


FIG. 13D

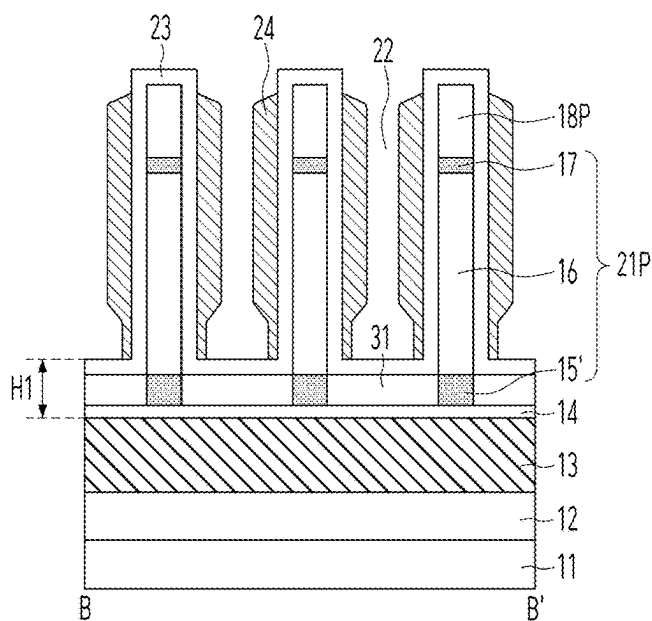




FIG. 14A

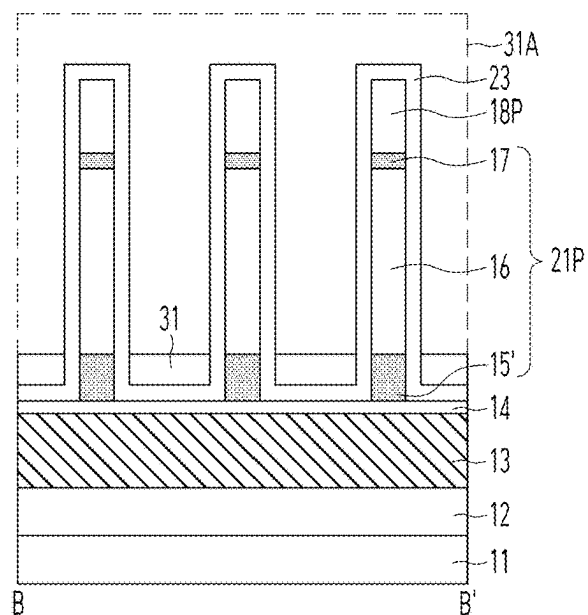


FIG. 14B

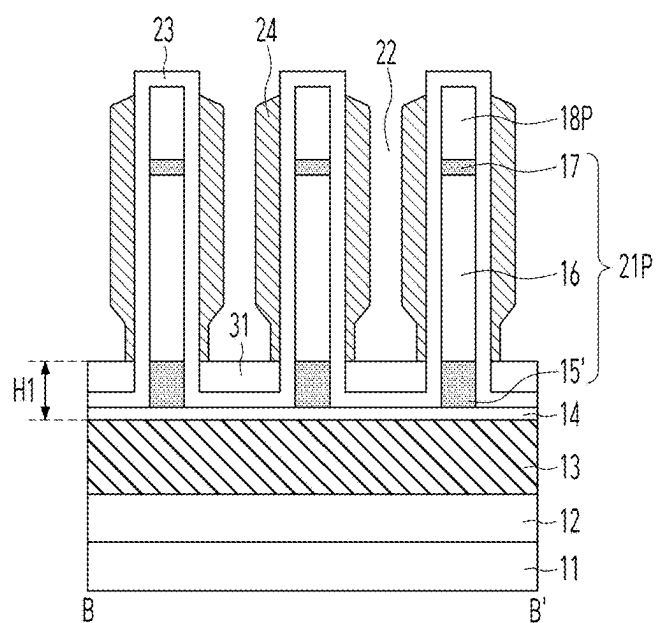


FIG. 15A

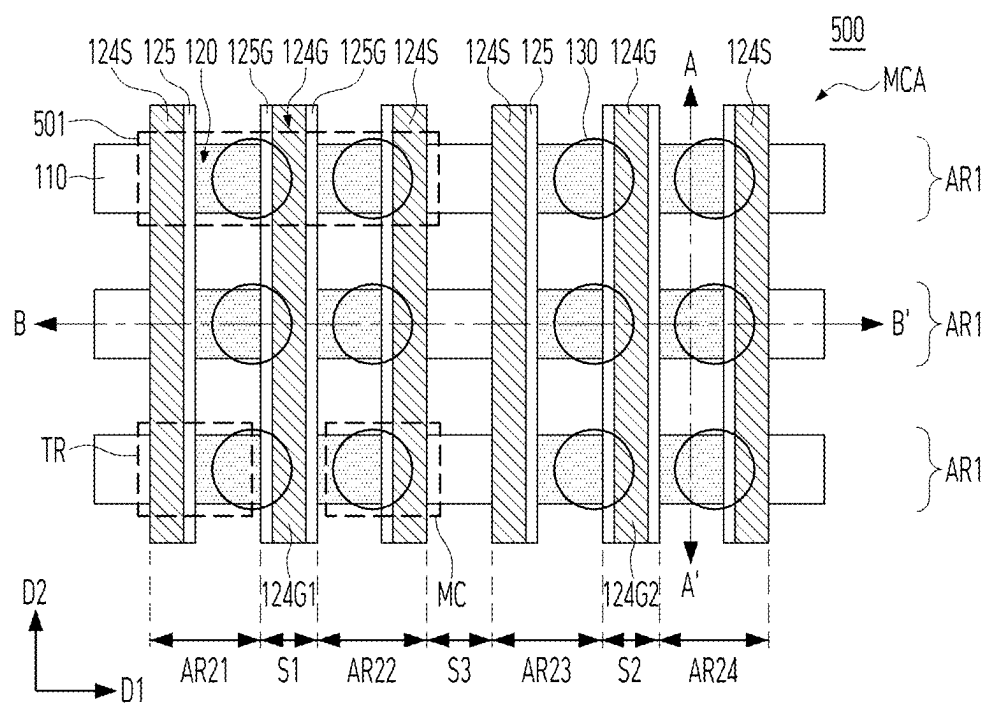




FIG. 15C

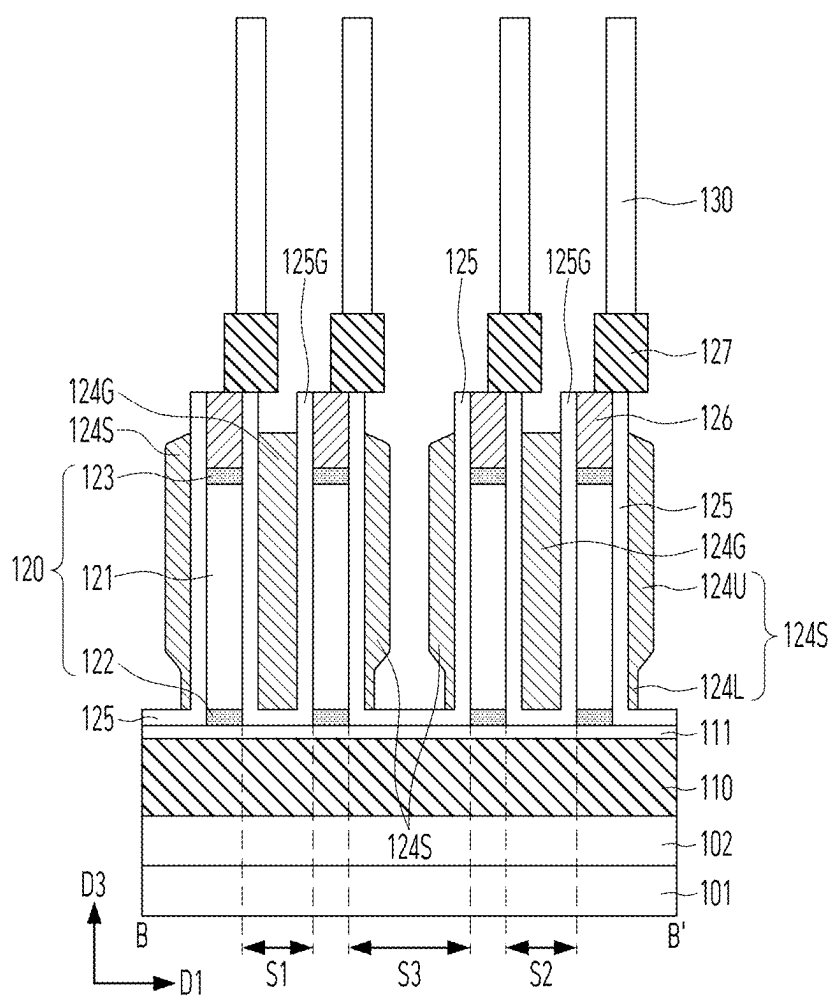


FIG. 15D

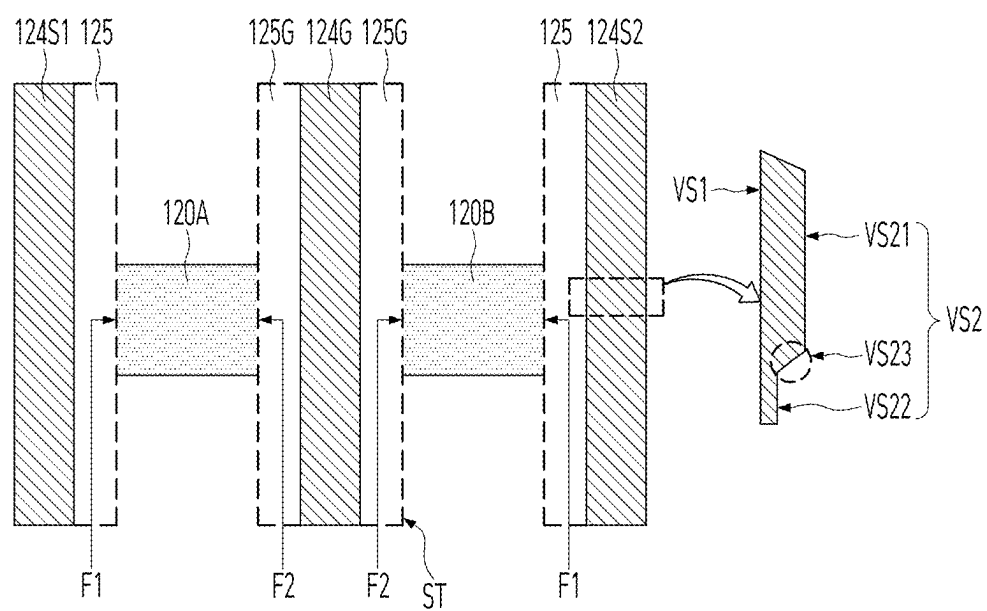


FIG. 16A

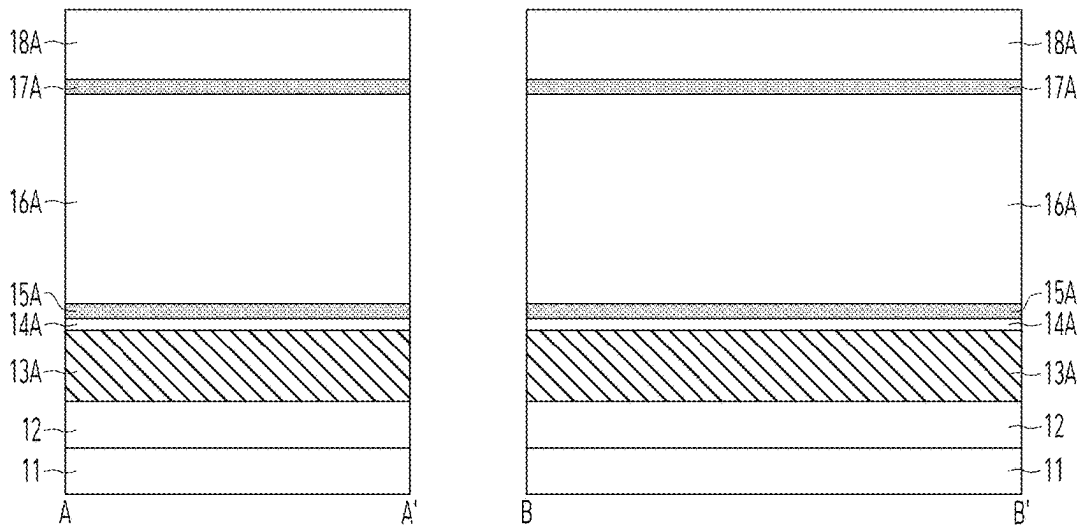


FIG. 16B

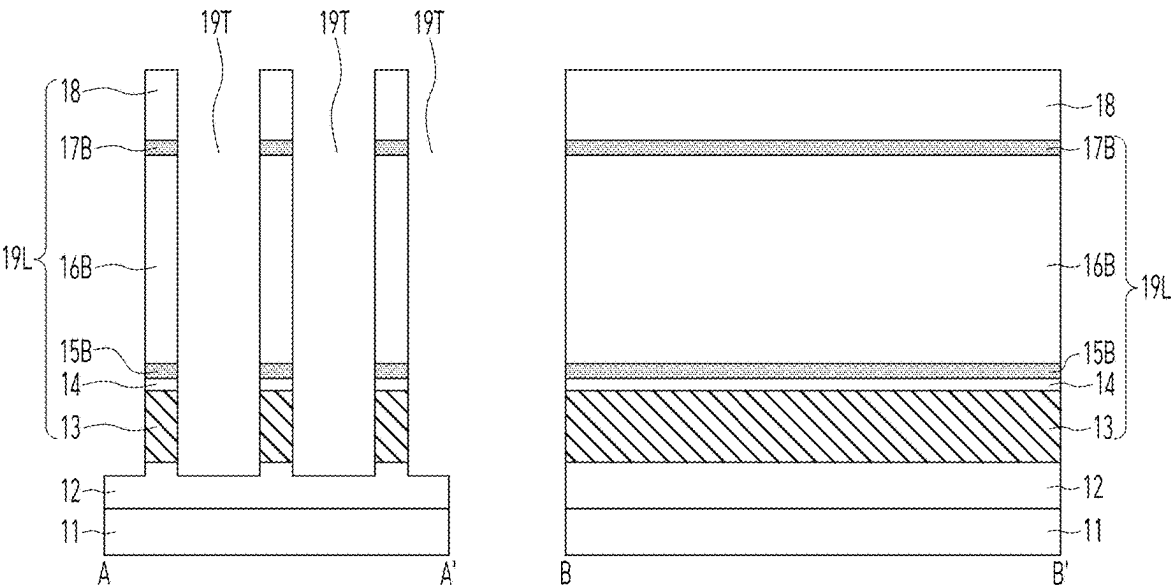


FIG. 16C

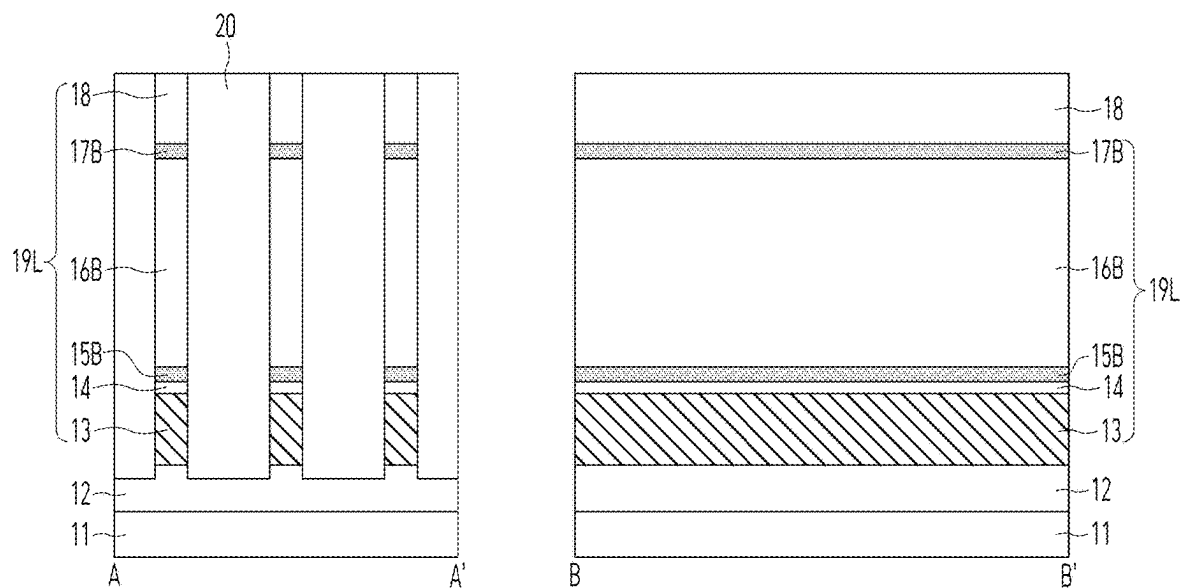


FIG. 16D

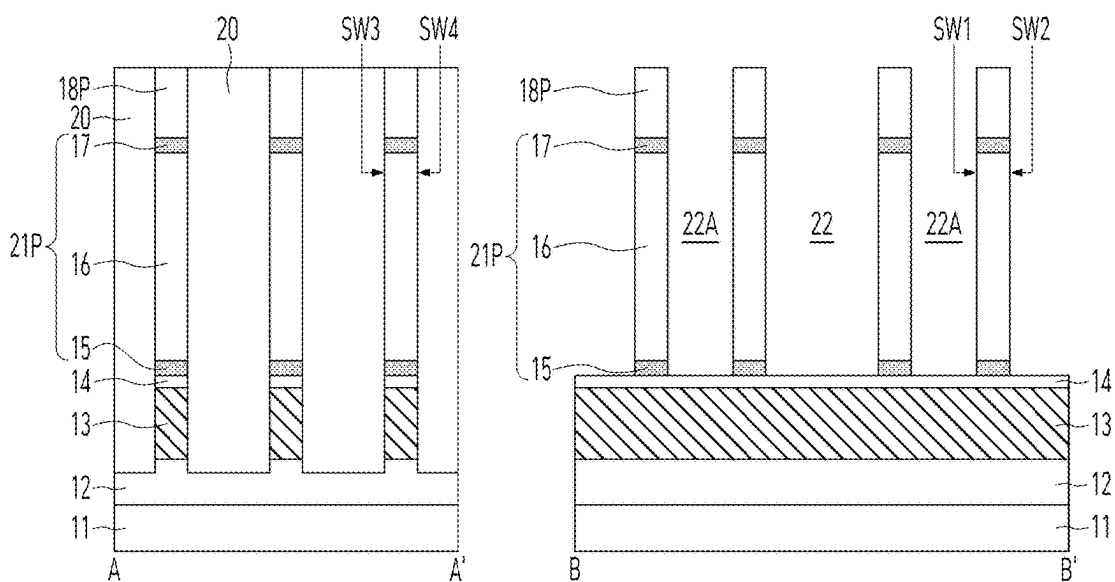


FIG. 16E

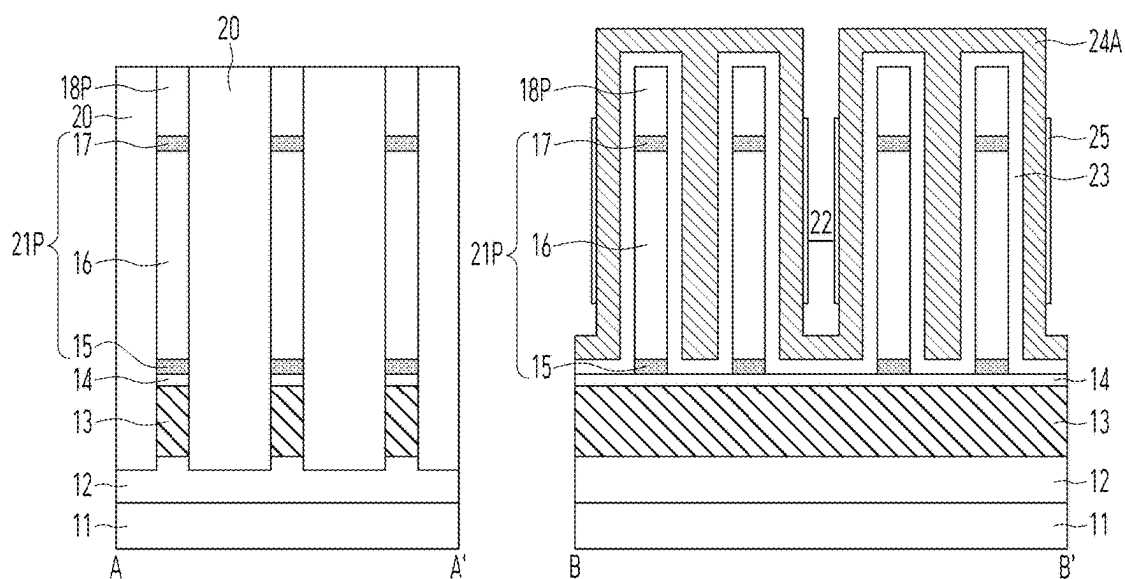


FIG. 16F

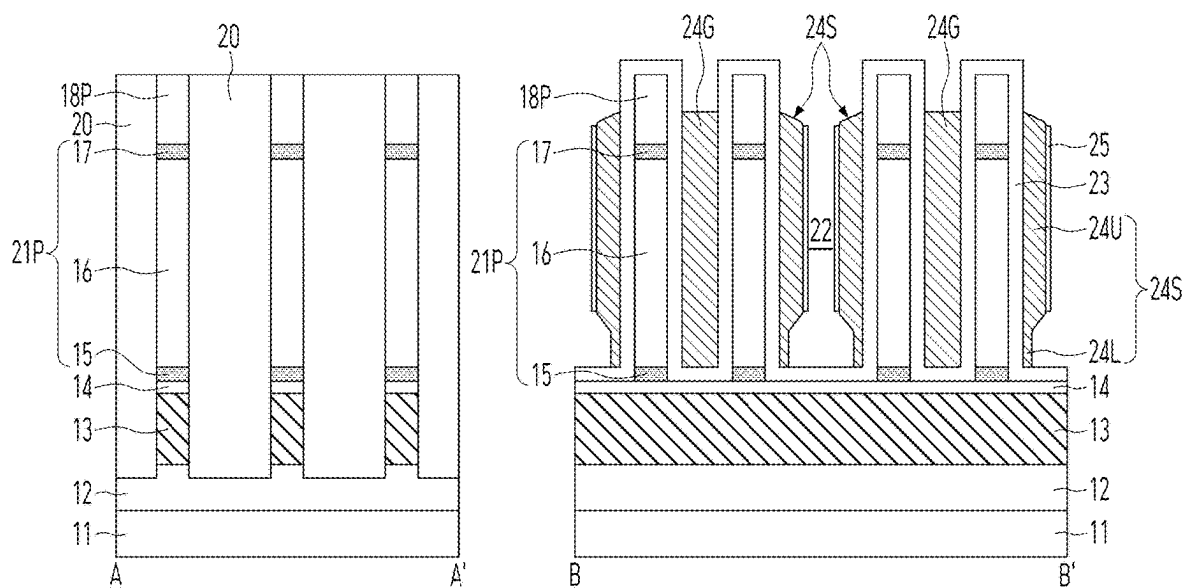




FIG. 16G

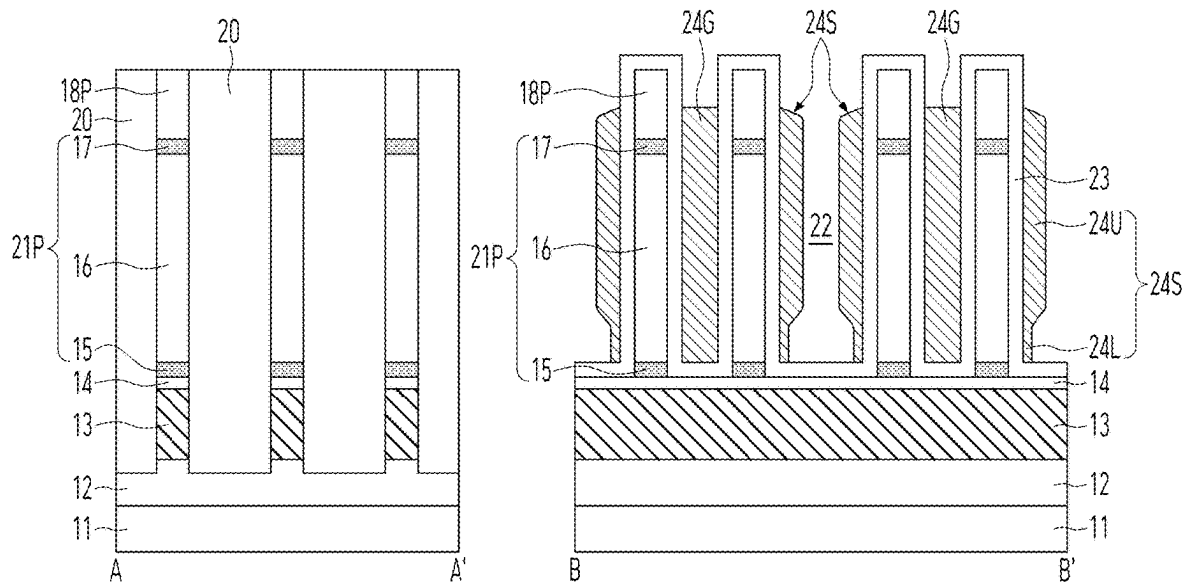


FIG. 16H

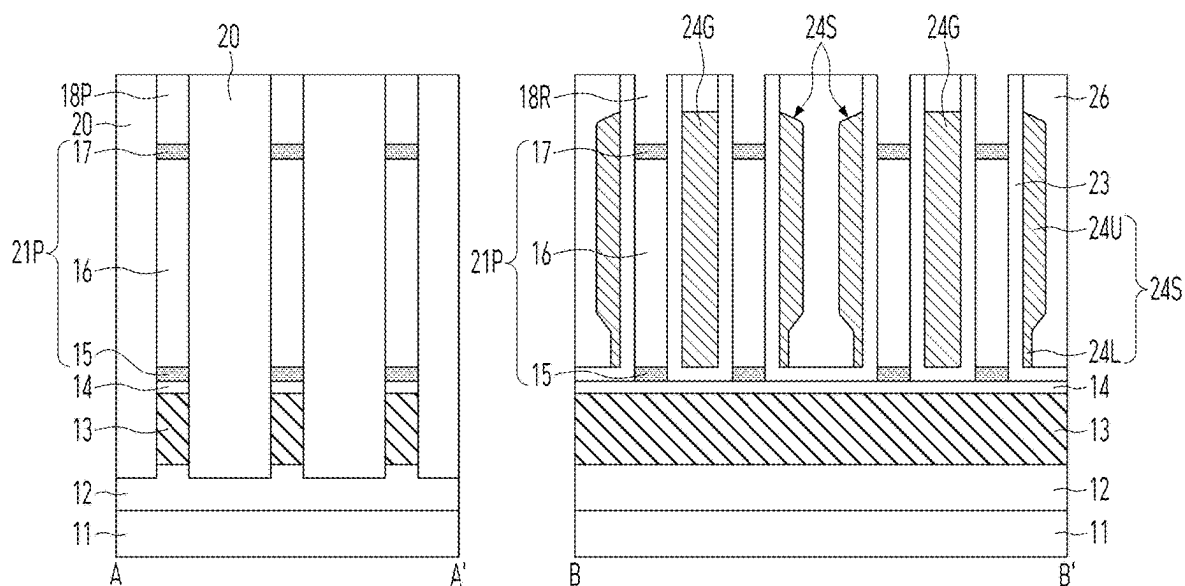
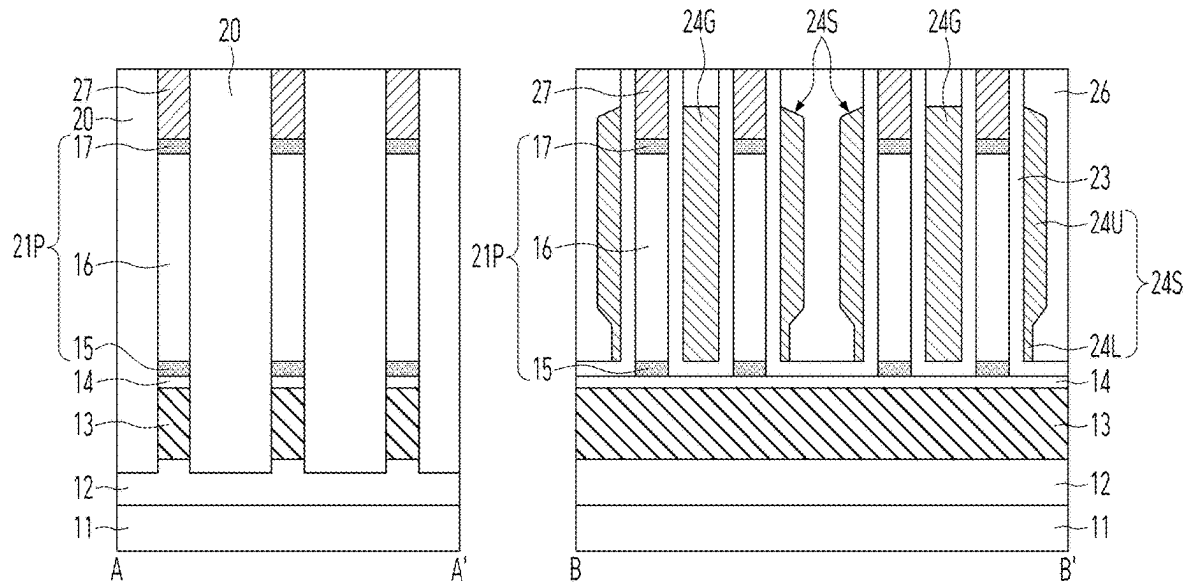


FIG. 16I



## SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE SAME

### CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation-in-part application of U.S. patent application Ser. No. 17/844,941, filed on Jun. 21, 2022. The present application claims priority of Korean Patent Application No. 10-2021-0160342, filed on Nov. 19, 2021, which is incorporated herein by reference in its entirety.

### BACKGROUND

#### 1. Field

[0002] Embodiments of the present disclosure relate to a semiconductor device, and more particularly, to a semiconductor device including a bit line and a method for fabricating the semiconductor device.

#### 2. Description of the Related Art

[0003] To fabricate highly integrated semiconductor devices, a transistor having a vertical channel has been proposed.

[0004] A vertical channel transistor using monocrystalline silicon as a channel has limitations in the degree of integration. Also, its electrical characteristics deteriorate due to gate-induced drain leakage (GIDL) occurring in an overlapping region between a source/drain and a gate and junction leakage of a PN junction.

### SUMMARY

[0005] Embodiments of the present disclosure are directed to a highly integrated semiconductor device, and a method for fabricating the semiconductor device.

[0006] In accordance with an embodiment of the present disclosure, a semiconductor device includes a bit line; an oxide semiconductor pillar extending vertically from the bit line; a capacitor disposed over the oxide semiconductor pillar; and a word line disposed over a sidewall of the oxide semiconductor pillar, wherein the oxide semiconductor pillar includes a lower oxide semiconductor interface layer coupled to the bit line; an upper oxide semiconductor interface layer coupled to the capacitor; and an oxide semiconductor channel layer disposed between the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer.

[0007] In accordance with another embodiment of the present disclosure, a semiconductor device includes a substrate; a peripheral circuit portion disposed over the substrate; and a memory cell array including a bit line, a transistor, and a memory element that are vertically stacked over the peripheral circuit portion, wherein the transistor includes an oxide semiconductor channel layer disposed between the bit line and the memory element; a tapered vertical word line disposed over a sidewall of the oxide semiconductor channel layer; a lower oxide semiconductor interface layer between the bit line and the oxide semiconductor channel layer; and an upper oxide semiconductor interface layer disposed between the capacitor and the oxide semiconductor channel layer.

[0008] In accordance with yet another embodiment of the present disclosure, a method for fabricating a semiconductor

device includes forming a bit line over a substrate; forming an oxide semiconductor pillar by stacking a lower interface layer, an oxide semiconductor channel layer, and an upper interface layer over the bit line in the recited order; forming a word line on a sidewall of the oxide semiconductor pillar; and forming a capacitor over the oxide semiconductor pillar.

[0009] In accordance with still another embodiment of the present disclosure, a semiconductor device includes a first conductive line; an oxide semiconductor pillar extending vertically from the first conductive line; a memory element disposed over the oxide semiconductor pillar; and a second conductive line disposed over a sidewall of the oxide semiconductor pillar, wherein the oxide semiconductor pillar includes a lower oxide semiconductor interface layer coupled to the first conductive line; an upper oxide semiconductor interface layer coupled to the memory element; and an oxide semiconductor channel layer disposed between the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer.

[0010] In accordance with another embodiment of the present disclosure, a vertical channel transistor includes a lower oxide semiconductor interface layer; an upper oxide semiconductor interface layer; an oxide semiconductor channel layer extending vertically between the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer; and a tapered vertical gate disposed over a sidewall of the oxide semiconductor channel layer. The oxide semiconductor channel layer may include IGZO, ITZO or ZTO, and the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer may include indium-rich IGZO.

[0011] In accordance with another embodiment of the present disclosure, a semiconductor device includes a first pillar and a second pillar spaced apart from each other through a shield trench, each pillar including an inner side that defines inner walls of the shield trench and an outer side that faces the inner side; a shield gate formed in the shield trench; a first main gate formed on the outer side of the first pillar; a second main gate formed on the outer side of the second pillar; a bit line formed on lower portions of the first and second pillars; and a capacitor formed on an upper portion of each of the first and second pillars.

[0012] In accordance with another embodiment of the present disclosure, a semiconductor device includes a trench; a first row array of first oxide semiconductor pillars and a second row array of second oxide semiconductor pillars spaced apart from each other by the trench, each pillar including an inner side that defines inner sidewalls of the trench and an outer side that faces the inner side; a common gate extending along the first row array and the second row array, and formed in the trench while being formed on the inner sides of the first and second oxide semiconductor pillars; a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and a second gate extending along the outer sides of the second oxide semiconductor pillars of the second row array.

[0013] In accordance with another embodiment of the present disclosure, a method for fabricating a semiconductor device includes forming a bit line; forming a first row array of first oxide semiconductor pillars and a second row array of second oxide semiconductor pillars spaced apart from each other by a trench, each pillar including an inner side that defines inner sidewalls of the trench and an outer side that faces the inner side over the bit line; forming a common

gate extending along the first row array and the second row array on the inner sides of the first and second oxide semiconductor pillars; forming a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and forming a second gate extending along the outer sides of the second oxide semiconductor pillars of the second row array.

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1A is a schematic perspective view illustrating a semiconductor device.

[0015] FIG. 1B is a schematic plan view illustrating the semiconductor device of FIG. 1A.

[0016] FIG. 1C is a cross-sectional view taken along a line A-A' shown in FIG. 1B.

[0017] FIG. 1D is a cross-sectional view taken along a line B-B' shown in FIG. 1B.

[0018] FIG. 1E is a detailed cross-sectional view illustrating a capacitor.

[0019] FIGS. 2A to 2K are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with an embodiment of the present disclosure.

[0020] FIGS. 3A to 3E are cross-sectional views illustrating a method for forming a capacitor.

[0021] FIG. 4A is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure.

[0022] FIG. 4B is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure.

[0023] FIG. 5 is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure.

[0024] FIGS. 6A to 6C are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0025] FIGS. 7A to 7G are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0026] FIGS. 8 to 10 are cross-sectional views illustrating semiconductor devices in accordance with other embodiments of the present disclosure.

[0027] FIGS. 11A to 11C are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0028] FIGS. 12A and 12B are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0029] FIGS. 13A to 13D are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0030] FIGS. 14A and 14B are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0031] FIG. 15A is a schematic plan view illustrating a semiconductor device in accordance with an embodiment of the present disclosure.

[0032] FIG. 15B is a cross-sectional view taken along a line A-A' shown in FIG. 15A.

[0033] FIG. 15C is a cross-sectional view taken along a line B-B' shown in FIG. 15A.

[0034] FIG. 15D is an enlarged view illustrating a portion 501 of FIG. 15A.

[0035] FIGS. 16A to 16I illustrate a method for fabricating a semiconductor device in accordance with an embodiment of the present disclosure.

#### DETAILED DESCRIPTION

[0036] Embodiments of the present disclosure will be described below in more detail with reference to the accompanying drawings. The embodiments may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art. Throughout this disclosure, like reference numerals refer to like parts throughout the various figures and embodiments of the present disclosure.

[0037] The drawings are not necessarily to scale and in some instances, proportions may have been exaggerated to clearly illustrate features of the embodiments. When a first layer is referred to as being “on” a second layer or “on” a substrate, it not only refers to a case where the first layer is formed directly on the second layer or the substrate but also a case where a third layer exists between the first layer and the second layer or the substrate.

[0038] FIG. 1A is a schematic perspective view illustrating a semiconductor device. FIG. 1B is a schematic plan view illustrating the semiconductor device. FIG. 1C is a cross-sectional view taken along a line A-A' shown in FIG. 1B. FIG. 1D is a cross-sectional view taken along a line B-B' shown in FIG. 1B. FIG. 1E is a detailed cross-sectional view illustrating a capacitor 130.

[0039] Referring to FIGS. 1A to 1E, the semiconductor device 100 may include a plurality of first conductive lines 110, a plurality of oxide semiconductor pillars 120 vertically extending from the first conductive lines 110, a plurality of memory elements 130 formed over the oxide semiconductor pillars 120, and a plurality of second conductive lines 124 disposed over the respective sidewalls of the oxide semiconductor pillars 120. Each oxide semiconductor pillar 120 may include a lower interface layer 122 coupled to the first conductive line 110, an upper interface layer 123 coupled to the memory element 130, and an oxide semiconductor channel layer 121 disposed between the lower interface layer 122 and the upper interface layer 123. The lower interface layer 122 and the upper interface layer 123 may include an oxide semiconductor layer. The lower interface layer 122 and the upper interface layer 123 may be referred to as a lower oxide semiconductor interface layer and an upper oxide semiconductor interface layer, respectively.

[0040] The semiconductor device 100 may include a Dynamic Random Access Memory (DRAM), and the first conductive lines 110 and the second conductive lines 124 may correspond to bit lines and word lines, respectively. The memory elements 130 may correspond to capacitors. Hereinafter, the first conductive lines 110 and the second conductive lines 124 may be simply referred to as bit lines 110 and word lines 124, respectively, and the memory elements 130 may be simply referred to as capacitors 130.

[0041] The semiconductor device 100 may include an oxide semiconductor channel layer 121, a bit line 110 disposed at a lower level than the oxide semiconductor channel layer 121, a capacitor 130 disposed at a higher level than the oxide semiconductor channel layer 121, a tapered vertical word line 124 disposed over a sidewall of the oxide semiconductor channel layer 121, a lower interface layer 122 between the bit line 110 and the oxide semiconductor channel layer 121, and an upper interface layer 123 between the capacitor 130 and the oxide semiconductor channel layer 121.

[0042] The semiconductor device 100 may be described in detail as follows.

[0043] The semiconductor device 100 may include a substrate 101, a buffer layer 102 over the substrate 101, a bit line 110 over the buffer layer 102, a vertical channel transistor TR over the bit line 110, and a capacitor 130 over the vertical channel transistor TR. The vertical channel transistor TR may include an oxide semiconductor pillar 120, a tapered vertical word line 124 disposed over both sidewalls of the oxide semiconductor pillar 120, and a gate dielectric layer 125 between the oxide semiconductor pillar 120 and the tapered vertical word line 124. The oxide semiconductor pillar 120 may include the oxide semiconductor channel layer 121, the lower interface layer 122 between the oxide semiconductor channel layer 121 and the bit line 110, and the upper interface layer 123 between the oxide semiconductor channel layer 121 and the capacitor 130. A barrier layer 111 may be disposed between the bit line 110 and the lower interface layer 122.

[0044] The semiconductor device 100 may further include first and second contact plugs 126 and 127 between the upper interface layer 123 and the capacitor 130.

[0045] The substrate 101 may be a material suitable for semiconductor processing. The substrate 101 may include a semiconductor substrate. The substrate 101 may be formed of a silicon-containing material. The substrate 101 may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate 101 may include other semiconductor materials such as germanium. The substrate 101 may include a III/V group semiconductor substrate, for example, a compound semiconductor substrate, such as GaAs. The substrate 101 may include a Silicon-On-Insulator (SOI) substrate.

[0046] The buffer layer 102 may include silicon oxide, silicon nitride, or a combination thereof. To reduce parasitic capacitance, the buffer layer 102 may be formed of silicon oxide. For example, the buffer layer 102 may include tetra ethyl ortho silicate (TEOS).

[0047] The bit line 110 may extend in a first direction D1 over the buffer layer 102. The bit line 110 may include a metal-based material. The bit line 110 may include a metal, a metal nitride, a metal silicide, or a combination thereof. The bit line 110 may have a thickness of approximately 100 to 400 Å. The bit line 110 may include a tungsten layer.

[0048] The barrier layer 111 may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier layer 111 may include titanium nitride, molybdenum, or ruthenium. The barrier layer 111 may have a thickness of approximately 10 to 50 Å. For example, the barrier layer 111 may include a titanium nitride layer.

[0049] The lower interface layer 122 and the upper interface layer 123 may include an oxide semiconductor material having a lower resistance than the oxide semiconductor channel layer 121. The lower interface layer 122 and the upper interface layer 123 may include a metallic-rich oxide semiconductor material, and the oxide semiconductor channel layer 121 may include an oxygen-rich oxide semiconductor material. For example, the oxide semiconductor channel layer 121 may include indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO), or zinc tin oxide (ZTO), and the lower interface layer 122 and the upper interface layer 123 may include indium-rich IGZO. Indium-rich IGZO may refer to a material with a higher indium content than gallium (Ga) and zinc (Zinc) in IGZO, for example, the content of indium may be approximately 40% or more.

[0050] The oxide semiconductor channel layer 121 may include an oxide semiconductor material. The oxide semiconductor channel layer 121 may contain indium. The oxide semiconductor channel layer 121 may include IGZO. The oxide semiconductor channel layer 121 may be formed to have a thickness of approximately 200 to 1000 Å.

[0051] The lower interface layer 122 may include an oxide semiconductor material. The lower interface layer 122 may contain indium. The lower interface layer 122 may include an indium-rich oxide semiconductor material. For example, the lower interface layer 122 may include indium-rich IGZO. The lower interface layer 122 may be formed to have a thickness of approximately 10 to 50 Å.

[0052] The upper interface layer 123 may include an oxide semiconductor material. The upper interface layer 123 may contain indium. The upper interface layer 123 may include an indium-rich oxide semiconductor material. For example, the upper interface layer 123 may include indium-rich IGZO. The upper interface layer 123 may be formed to have a thickness of approximately 10 to 50 Å.

[0053] As described above, the oxide semiconductor pillar 120 may extend vertically in a third direction D3 over the bit line 110. The oxide semiconductor pillar 120 may be vertically stacked over the bit line 110 in an order of the lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include an oxide semiconductor material. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include IGZO, but the lower interface layer 122 and the upper interface layer 123 may have a higher indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor channel layer 121 may be IGZO, and the lower interface layer 122 and the upper interface layer 123 may be indium-rich IGZO. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may be referred to as an active pillar.

[0054] A tapered vertical word line 124 having a double structure may be disposed over a sidewall of the oxide semiconductor channel layer 121. The tapered vertical word line 124 and the bit line 110 may extend in directions crossing each other. The tapered vertical word line 124 may have a reverse tapered shape. The reverse tapered shape may refer to a shape the width of the bottom portion of which becomes gradually narrower than the width of the upper portion. For example, the tapered vertical word line 124 may

include a lower level portion **124L** which is adjacent to the bit line **110** and an upper level portion **124U** which is adjacent to the capacitor **130**. The thickness of the lower level portion **124L** in the first direction **D1** may be thinner than the thickness of the upper level portion **124U**. The lower level portion **124L** may be adjacent to the lower interface layer **122**, and the upper level portion **124U** may be adjacent to the upper interface layer **123**. The lower level portion **124L** and the upper level portion **124U** may be formed of the same material. The lower level portion **124L** and the bit line structure **BL** may be spaced apart from each other with a space. The lower level portion **124L** of the tapered vertical word line **124** and the bit line structure **BL** may not contact each other.

**[0055]** The tapered vertical word line **124** may include a metal-based material. The tapered vertical word line **124** may include a metal, a metal nitride, or a combination thereof. The tapered vertical word line **124** may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

**[0056]** A gate dielectric layer **125** may be formed between the tapered vertical word line **124** and the oxide semiconductor pillar **120**. The gate dielectric layer **125** may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The gate dielectric layer **125** may be disposed between the oxide semiconductor pillar **120** and the tapered vertical word line **124**. The gate dielectric layer **125** may include a horizontal portion that extends to be disposed between the lower level portion **124L** of the tapered vertical word line **124** and the bit line **110**, and the horizontal portion of the gate dielectric layer **125** may directly contact the barrier layer **111**. The oxide semiconductor pillar **120**, the gate dielectric layer **125**, and the tapered vertical word line **124** may form the vertical channel transistor **TR**. The tapered vertical word line **124** may be referred to as a tapered vertical gate.

**[0057]** The contact plugs **126** and **127** may include a first contact plug **126** and a second contact plug **127** over the first contact plug **126**. The first contact plug **126** may directly contact the upper interface layer **123**, and the second contact plug **127** may directly contact the capacitor **130**. The first contact plug **126** and the second contact plug **127** may vertically overlap with each other. The first contact plug **126** and the second contact plug **127** may include a metal-based material. The first contact plug **126** and the second contact plug **127** may include a metal, a metal nitride, or a combination thereof. The first contact plug **126** and the second contact plug **127** may be formed of the same metal-based material. According to another embodiment of the present disclosure, the first contact plug **126** and the second contact plug **127** may be formed of different metal-based materials. Sidewalls of the first contact plugs **126** may be surrounded by the gate dielectric layer **125**.

**[0058]** The capacitor **130** may include a lower electrode **132**, a dielectric layer **135**, and an upper electrode **136**. The lower electrode **132** may be formed over the second contact plug **127**. The lower electrode **132** may have a pillar shape. The lower electrodes **132** may be supported by the supporters **133** and **134**. A sidewall of the bottom portion of the lower electrode **132** may contact the etch stop layer **131**. According to another embodiment of the present disclosure, the lower electrodes **132** may have a cylindrical shape.

**[0059]** Although not illustrated, dielectric pillars may be disposed between the oxide semiconductor pillars **120** in the direction of the A-A'.

**[0060]** FIGS. 2A to 2K are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with an embodiment of the present disclosure.

**[0061]** Referring to FIG. 2A, a buffer layer **12** may be formed over a substrate **11**. The substrate **11** may be a material appropriate for semiconductor processing. The substrate **11** may include a semiconductor substrate. The substrate **11** may be formed of a silicon-containing material. The substrate **11** may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate **11** may include other semiconductor materials, such as germanium. The substrate **11** may include a III/V-group semiconductor substrate, for example, a compound semiconductor substrate, such as GaAs. The substrate **11** may include a Silicon-On-Insulator (SOI) substrate. The buffer layer **12** may include a dielectric material. The buffer layer **12** may include silicon oxide, silicon nitride, or a combination thereof.

**[0062]** A conductive layer **13A** may be formed over the buffer layer **12**. The conductive layer **13A** may include a metal-based material. The conductive layer **13A** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The conductive layer **13A** may be formed by atomic layer deposition (ALD), physical vapor deposition (PVD), or chemical vapor deposition (CVD). The conductive layer **13A** may be formed to have a thickness of approximately 100 to 400 Å. For example, as for the conductive layer **13A**, a tungsten layer may be deposited to have a thickness of approximately 200 Å by physical vapor deposition (PVD). To reduce parasitic capacitance between the substrate **11** and the conductive layer **13A**, the buffer layer **12** may be formed of silicon oxide. For example, the buffer layer **12** may include tetra ethyl ortho silicate (TEOS).

**[0063]** A barrier material layer **14A** may be formed over the conductive layer **13A**. The barrier material layer **14A** may include a metal-based material. The barrier material layer **14A** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier material layer **14A** may include titanium nitride, molybdenum, or ruthenium. The barrier material layer **14A** may be formed by atomic layer deposition (ALD), physical vapor deposition (PVD), or chemical vapor deposition (CVD). The barrier material layer **14A** may be formed to have a thickness of approximately 10 to 50 Å. For example, as the barrier material layer **14A**, titanium nitride may be deposited to have a thickness of approximately 20 Å by physical vapor deposition (PVD).

**[0064]** An oxide semiconductor stack may be formed over the barrier material layer **14A**. For example, the oxide semiconductor stack may include a lower interface material layer **15A**, a channel material layer **16A**, and an upper interface material layer **17A**.

**[0065]** A lower interface material layer **15A** may be formed over the barrier material layer **14A**. The lower interface material layer **15A** may include a conductive material. The lower interface material layer **15A** may include an oxide semiconductor material. The lower interface material layer **15A** may contain indium. The lower interface material layer **15A** may include an indium-rich

oxide semiconductor material. For example, the lower interface material layer 15A may include indium-rich IGZO. The lower interface material layer 15A may be formed to have a thickness of approximately 10-50 Å.

[0066] A channel material layer 16A may be formed over the lower interface material layer 15A. The channel material layer 16A may include a conductive material. The channel material layer 16A may include an oxide semiconductor material. The channel material layer 16A may contain indium. The channel material layer 16A may include IGZO. The channel material layer 16A may be formed to have a thickness of approximately 200 to 1000 Å.

[0067] An upper interface material layer 17A may be formed over the channel material layer 16A. The upper interface material layer 17A may include a conductive material. The upper interface material layer 17A may include an oxide semiconductor material. The upper interface material layer 17A may contain indium. The upper interface material layer 17A may include an indium-rich oxide semiconductor material. For example, the upper interface material layer 17A may include indium-rich IGZO. The upper interface material layer 17A may be formed to have a thickness of approximately 10 to 50 Å.

[0068] As described above, the lower interface material layer 15A, the channel material layer 16A, and the upper interface material layer 17A may be vertically stacked over the barrier material layer 14A. The lower interface material layer 15A, the channel material layer 16A, and the upper interface material layer 17A may all include an oxide semiconductor material. The lower interface material layer 15A, the channel material layer 16A, and the upper interface material layer 17A may all include IGZO, but the lower interface material layer 15A and the upper interface material layer 17A may have a higher indium concentration than the channel material layer 16A. The channel material layer 16A may be IGZO, and the lower interface material layer 15A and the upper interface material layer 17A may be indium-rich IGZO. Since the lower interface material layer 15A and the upper interface material layer 17A contain a high concentration of indium, its resistance may be reduced lower than that of the channel material layer 16A. Also, channel seamless interconnection of the channel material layer 16A may be possible.

[0069] Subsequently, a sacrificial layer 18A may be formed over the upper interface material layer 17A. The sacrificial layer 18A may include a stack of different materials. The sacrificial layer 18A may include silicon nitride.

[0070] Referring to FIG. 2B, sacrificial lines 18 may be formed over the upper interface material layer 17A. The sacrificial lines 18 may be formed by etching the sacrificial layer 18A. The sacrificial lines 18 may serve to protect the lower interface material layer 15A, the channel material layer 16A, and the upper interface material layer 17A from the subsequent processes. For example, the sacrificial lines 18 may be used as an etch barrier during an etching process of the lower interface material layer 15A, the channel material layer 16A, and the upper interface material layer 17A.

[0071] Each of the sacrificial lines 18 may include a stack of different materials. The sacrificial lines 18 may include silicon nitride. The etching process of the sacrificial layer 18A to form the sacrificial lines 18 may include a double patterning process.

[0072] Subsequently, the upper interface material layer 17A, the channel material layer 16A, and the lower interface material layer 15A may be etched by using the sacrificial lines 18 as an etch barrier, and then the barrier material layer 14A and the conductive layer 13A may be etched.

[0073] A plurality of line structures 19L and first trenches 19T may be formed over the buffer layer 12 by a series of the etching processes described above. Each of the line structures 19L may include a stack of a bit line 13, a bit line barrier layer 14, a lower interface layer 15B, a channel material 16B, an upper interface layer 17B, and a sacrificial line 18. The bit line barrier layer 14 may be formed by etching the barrier material layer 14A, and the bit line 13 may be formed by etching the conductive layer 13A. The lower interface layer 15B, the channel material 16B, and the upper interface layer 17B may be formed by etching the lower interface material layer 15A, the channel material layer 16A, and the upper interface material layer 17A, respectively. The first trenches 19T may be disposed between the line structures 19L. The stack of the lower interface layer 15B, the channel material 16B, and the upper interface layer 17B may be referred to as an 'oxide semiconductor line'.

[0074] Referring to FIG. 2C, dielectric lines 20 may be formed between the line structures 19L. The dielectric lines 20 may include a dielectric material. For example, the dielectric lines 20 may include silicon oxide, silicon nitride, silicon carbon oxide (SiCO), spin-on-dielectric, or a combination thereof. After a dielectric material is deposited to fill the first trenches 19T between the line structures 19L to form the dielectric lines 20, a planarization process of the dielectric material may be performed. The dielectric lines 20 may fill the first trenches 19T, respectively. To form the dielectric lines 20, a planarization process may be performed after a silicon nitride, silicon carbon oxide, and a spin-on dielectric layer are sequentially formed.

[0075] Referring to FIG. 2D, a portion of the line structures 19L may be selectively etched to form the sacrificial pillars 18P and the oxide semiconductor pillars 21P. A bit line barrier layer 14 and a bit line 13 may be disposed below the oxide semiconductor pillars 21P.

[0076] Each of the oxide semiconductor pillars 21P may include a stack of a lower interface layer 15, a channel layer 16, and an upper interface layer 17. The lower interface layer 15, the channel layer 16, and the upper interface layer 17 may be formed by etching the lower interface layer 15B, the channel material 16B, and the upper interface layer 17B, respectively. A sacrificial pillar 18P may be formed over the upper interface layer 17.

[0077] Second trenches 22 may be formed between the oxide semiconductor pillars 21P. The bottom surfaces of the second trenches 22 may expose the top surface of the bit line barrier layer 14. The first trenches 19T and the second trenches 22 may intersect with each other. The first trenches 19T may be deeper than the second trenches 22. In the direction of the line B-B', the dielectric lines 20 may be cut by the second trenches 22. Hereinafter, the dielectric lines 20 may be simply referred to as 'dielectric pillars 20'. The dielectric pillars 20 may be disposed between the oxide semiconductor pillars 21P in the direction of the line A-A'.

[0078] Each of the oxide semiconductor pillars 21P may include first to fourth sidewalls SW1 to SW4. A first sidewall SW1 and a second sidewall SW2 of the individual oxide semiconductor pillar 21P may be exposed by the second

trenches 22, and a third sidewall SW3 and a fourth sidewall SW4 of the individual oxide semiconductor pillar 21P may not be exposed by the dielectric pillars 20. The sacrificial pillar 18P may also include exposed sidewalls and non-exposed sidewalls just as the oxide semiconductor pillar 21P.

[0079] Referring to FIG. 2E, a gate dielectric layer 23 may be formed over the exposed first and second sidewalls SW1 and SW2 of the oxide semiconductor pillars 21P. The gate dielectric layer 23 may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The high-k material may include a material having a higher dielectric constant than that of silicon oxide. For example, the high-k material may include a material having a dielectric constant which is greater than approximately 3.9. In another embodiment, the high-k material may include a material having a dielectric constant which is greater than approximately 10. In another embodiment, the high-k material may include a material having a dielectric constant of approximately 10 to 30. The high-k material may include at least one metallic element. The high-k material may include a hafnium-containing material. The hafnium-containing material may include hafnium oxide, hafnium silicon oxide, hafnium silicon oxynitride, or a combination thereof. According to another embodiment of the present disclosure, the high-k material may include lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, zirconium silicon oxynitride, aluminum oxide, or a combination thereof. As for the high-k material, other known high-k materials may be selectively used. The gate dielectric layer 23 may include a metal oxide. The gate dielectric layer 23 may be formed conformally over the first and second sidewalls SW1 and SW2 of the oxide semiconductor pillar 21P. The gate dielectric layer 23 may conformally cover the exposed sidewalls and the top surface of the sacrificial pillar 18P. The gate dielectric layer 23 and the lower interface layer 15 may have the same thickness. According to another embodiment of the present disclosure, the gate dielectric layer 23 may be thinner than the lower interface layer 15.

[0080] A word line conductive layer 24A may be formed over the gate dielectric layer 23. The word line conductive layer 24A may be conformally formed over the gate dielectric layer 23. The word line conductive layer 24A may include a metal-based material. The word line conductive layer 24A may include a metal, a metal nitride, or a combination thereof. The word line conductive layer 24A may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0081] Spacers 25 may be formed over the word line conductive layer 24A. The spacers 25 may include an oxide. To form the spacers 25, an etch-back process may be performed after depositing silicon oxide over the word line conductive layer 24A. The upper surface of the spacers 25 may be disposed at a lower level than the upper surface of the sacrificial pillars 18P.

[0082] Referring to FIG. 2F, tapered vertical word lines 24 may be formed. To form the tapered vertical word lines 24, the word line conductive layer 24A may be selectively etched by using the spacers 25 as an etch barrier. The spacers 25 may serve to protect the tapered vertical word lines 24 during an etching process of the word line conductive layer 24A. In the etching process of the word line conductive layer

24A, an etch-back process and a wet etching process may be sequentially performed. Non-tapered vertical word lines may be formed by an etch-back process, and the tapered vertical word lines 24 may be formed by the subsequent wet etching process. The bottom portion of the tapered vertical word lines 24 may become thin by the wet etching process. The tapered vertical word lines 24 may correspond to the tapered vertical word lines 124 of FIGS. 1A, 1B, and 1D, and the bottom portion of the tapered vertical word lines 24 may correspond to the lower level portion 124L.

[0083] Referring to FIG. 2G, the spacers 25 may be removed. Each of the tapered vertical word lines 24 may have a double structure and the tapered vertical word lines 24 may be respectively formed on the first and second sidewalls of the oxide semiconductor pillars 21P. Referring to FIGS. 1A, 1B, and 1D, the tapered vertical word lines 24 may extend in the second direction D2, and the bit lines 13 may extend in the first direction D1. The oxide semiconductor pillars 21P and the dielectric pillars 20 may be alternately disposed in the second direction D2, and the tapered vertical word lines 24 may extend along the exposed sidewalls of the oxide semiconductor pillars 21P and the dielectric pillar 20.

[0084] Referring to FIG. 2H, an inter-layer dielectric layer 26 may be formed over the tapered vertical word line 24. The inter-layer dielectric layer 26 may be planarized to expose the upper surfaces of the sacrificial pillars 18P. The inter-layer dielectric layer 26 may include silicon oxide, such as a spin-on dielectric layer (SOD). The sacrificial pillars 18P may serve as an etch stop layer during a planarization process of the inter-layer dielectric layer 26.

[0085] Subsequently, the sacrificial pillars 18P may be selectively removed. Accordingly, hole-shaped recesses 18R may be formed. The hole-shaped recesses 18R may selectively expose the surfaces of the upper interface layers 17. The sacrificial pillars 18P may be removed by using a wet etching process.

[0086] Referring to FIG. 2I, first contact plugs 27 filling the hole-shaped recesses 18R may be formed. The first contact plugs 27 may directly contact the upper interface layers 17. The first contact plugs 27 may include a metal, a metal nitride, or a combination thereof. For example, to form the first contact plugs 27, titanium nitride may be deposited to fill the hole-shaped recesses 18R and then titanium nitride may be planarized to expose the surface of the inter-layer dielectric layer 26.

[0087] Referring to FIG. 2J, second contact plugs 28 may be formed over the first contact plugs 27. The first contact plugs 27 and the second contact plugs 28 may partially overlap with each other. The second contact plugs 28 may include a metal-based material. The second contact plugs 28 may include a metal, a metal nitride, or a combination thereof. For example, the second contact plugs 28 may include tungsten. According to another embodiment of the present disclosure, the first contact plugs 27 and the second contact plugs 28 may be formed of the same metal-based material.

[0088] A metal-based material may be deposited and etched to form the second contact plugs 28.

[0089] Referring to FIG. 2K, a spacer material layer 29 may be formed over the second contact plugs 28. The spacer material layer 29 may include silicon nitride. The spacer material layer 29 may be formed between the neighboring second contact plugs 28.



[0090] A capacitor 30 may be formed over the second contact plug 28.

[0091] FIGS. 3A to 3E are cross-sectional views illustrating a method for forming the capacitor 30. Hereinafter, the structures formed before the formation of the second contact plugs 28 are omitted.

[0092] Referring to FIG. 3A, an etch stop layer 31 may be formed over the second contact plugs 28 and the spacer material layer 29. A first mold layer 32, a first supporter layer 33, a second mold layer 34, and a second supporter layer 35 may be sequentially formed over the etch stop layer 31.

[0093] The etch stop layer 31 may be formed of a material having an etch selectivity with respect to the first mold layer 32. The etch stop layer 31 may include silicon nitride or silicon oxynitride. The first mold layer 32 may include a dielectric material. The first mold layer 32 may be formed of silicon oxide (SiO<sub>2</sub>). The first mold layer 32 may be formed to be thicker than the first supporter layer 33. The first mold layer 32 may be formed by using a deposition process, such as chemical vapor deposition (CVD), atomic layer deposition (ALD), or physical vapor deposition (PVD). The first mold layer 32 may include silicon oxide doped with phosphorus or silicon oxide doped with boron. The first mold layer 32 may include USG (Undoped Silicate Glass), PSG (Phosphorous Silicate Glass), BSG (Boron Silicate Glass), BPSG (Boron Phosphorous Silicate Glass), FSG (Fluorine Silicate Glass), or a combination thereof. Phosphorus-doped silicon oxide and boron-doped silicon oxide may be readily removed during the subsequent process because the etching rate with respect to an etching solution is high.

[0094] The first supporter layer 33 may be formed of a material having an etch selectivity with respect to the first mold layer 32 and the second mold layer 34. The first supporter layer 33 may include silicon nitride or silicon carbon nitride (SiCN).

[0095] The second mold layer 34 may include a dielectric material. The second mold layer 34 may be formed of silicon oxide (SiO<sub>2</sub>). The second mold layer 34 may be formed to be thicker than the first supporter layer 33. The second mold layer 34 may be formed by a deposition process, such as chemical vapor deposition (CVD), atomic layer deposition (ALD), or physical vapor deposition (PVD). The second mold layer 34 may include phosphorus-doped silicon oxide or boron-doped silicon oxide. The second mold layer 34 may include USG, PSG, BSG, BPSG, FSG, or a combination thereof. The first mold layer 32 and the second mold layer 34 may be formed of the same material or different materials.

[0096] According to another embodiment of the present disclosure, the first mold layer 32 and the second mold layer 34 may be formed of a silicon material, such as amorphous silicon or polysilicon.

[0097] The second supporter layer 35 may be formed of a material having an etch selectivity with respect to the second mold layer 34. The second supporter layer 35 may include silicon nitride or silicon carbon nitride (SiCN).

[0098] The first supporter layer 33 and the second supporter layer 35 may be formed of the same material or different materials. Both of the first supporter layer 33 and the second supporter layer 35 may be formed of silicon nitride. According to another embodiment of the present disclosure, the first supporter layer 33 may be formed of silicon nitride, and the second supporter layer 35 may be

formed of silicon carbon nitride. The second supporter layer 35 may be thicker than the first supporter layer 33.

[0099] According to another embodiment of the present disclosure, another supporter layer may be further formed. For example, the supporter structure may be a multi-level supporter layer structure.

[0100] Subsequently, an opening 36 may be formed. To form the opening 36, the second supporter layer 35, the second mold layer 34, the first supporter layer 33, and the first mold layer 32 may be sequentially etched by using a mask layer (not shown) as an etch barrier. An etching process for forming the opening 36 may stop at the etch stop layer 31. To form the opening 36, a dry etching process, a wet etching process, or a combination thereof may be used. The opening 36 may be referred to as a hole in which a lower electrode (or a storage node) is to be formed.

[0101] Subsequently, the etch stop layer 31 may be etched to expose the upper surface of the second contact plug 38 below the opening 36.

[0102] Referring to FIG. 3B, a lower electrode 37 may be formed in the opening 36. The lower electrode 37 may fill the inside of the opening 36. The lower electrode 37 may include polysilicon, a metal, a metal nitride, a conductive metal oxide, a metal silicide, a noble metal, or a combination thereof. The lower electrode 37 may include at least one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta<sub>2</sub>N), titanium aluminum nitride (TiAlN), tungsten (W), tungsten nitride (WN), ruthenium (Ru), ruthenium oxide (RuO<sub>2</sub>), iridium (Ir), iridium oxide (IrO<sub>2</sub>), platinum (Pt), and a combination thereof. The lower electrode 37 may include titanium nitride (TiN). The lower electrode 37 may include titanium nitride (ALD-TiN) which is formed by atomic layer deposition (ALD). According to another embodiment of the present disclosure, the lower electrode 37 may include a hybrid structure of a titanium nitride cylinder and a polysilicon pillar.

[0103] Referring to FIG. 3C, the second supporter layer 35, the second mold layer 34, and the first supporter layer 33 may be sequentially etched. As a result, a supporter opening 38 exposing the first mold layer 32 may be formed, and an upper-level supporter 35S and a lower-level supporter 33S may be formed.

[0104] The upper level supporter 35S and the lower level supporter 33S may contact the outer wall of the lower electrode 37. The upper-level supporter 35S and the lower-level supporter 33S may prevent the lower electrodes 37 from collapsing in the subsequent process of removing the second mold layer 34 and the first mold layer 32.

[0105] Referring to FIG. 3D, the second mold layer 34 and the first mold layer 32 may be removed through the supporter opening 38. The first and second mold layers 32 and 34 may be removed by a wet dip-out process. The wet dip-out process for removing the first and second mold layers 32 and 34 may be performed by using an etching solution capable of selectively removing the first and second mold layers 32 and 34. When the first and second mold layers 32 and 34 include silicon oxide, the first and second mold layers 32 and 34 may be removed by a wet etching process using hydrofluoric acid (HF).

[0106] Referring to FIG. 3E, a dielectric layer 39 may be formed over the lower electrode 37 and the lower and upper level supporters 33S and 35S. The dielectric layer 39 may include a high-k material having a higher dielectric constant than silicon oxide. The high-k material may include hafnium

oxide ( $\text{HfO}_2$ ), zirconium oxide ( $\text{ZrO}_2$ ), aluminum oxide ( $\text{Al}_2\text{O}_3$ ), titanium oxide ( $\text{TiO}_2$ ), tantalum oxide ( $\text{Ta}_2\text{O}_5$ ), niobium oxide ( $\text{Nb}_2\text{O}_5$ ) or strontium titanium oxide ( $\text{SrTiO}_3$ ). According to another embodiment of the present disclosure, the dielectric layer 39 may be formed of a composite layer including two or more layers of the above-mentioned high-k materials. According to the embodiment of the present disclosure, the dielectric layer 39 may be formed of a zirconium oxide-based material having excellent leakage current characteristics while sufficiently lowering the equivalent oxide thickness (EOT). For example, it may include a ZAZ ( $\text{ZrO}_2/\text{Al}_2\text{O}_3/\text{ZrO}_2$ ) stack. According to another embodiment of the present disclosure, the dielectric layer 27 may include a  $\text{TiO}_2/\text{ZrO}_2/\text{Al}_2\text{O}_3/\text{ZrO}_2$  stack, a  $\text{TiO}_2/\text{HfO}_2/\text{Al}_2\text{O}_3/\text{HfO}_2$  stack, a  $\text{Ta}_2\text{O}_5/\text{ZrO}_2/\text{Al}_2\text{O}_3/\text{ZrO}_2$  stack, or a  $\text{Ta}_2\text{O}_5/\text{HfO}_2/\text{Al}_2\text{O}_3/\text{HfO}_2$  stack.

[0107] Subsequently, an upper electrode 40 may be formed over the dielectric layer 39. The upper electrode 40 may fill the space between the neighboring lower electrodes 37. The upper electrode 40 may extend to cover the upper portions of the lower electrodes 37. The upper electrode 40 may include a conductive material. The upper electrode 40 may be stacked (reference numerals omitted) in the order of a liner electrode, a gap-fill electrode, and a low-resistance electrode. The liner electrode of the upper electrode 40 may include titanium nitride, and the gap-fill electrode of the upper electrode 40 may include silicon germanium. The low resistance electrode of the upper electrode 40 may include tungsten or tungsten nitride.

[0108] FIG. 4A is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure. The semiconductor device 200 of FIG. 4A may be similar to the semiconductor device 100 shown in FIGS. 1A to 1E. Hereinafter, for the detailed descriptions of the constituent elements which also appear in FIGS. 1A to 1E, the description of FIGS. 1A to 1E may be referred to.

[0109] Referring to FIG. 4A, the semiconductor device 200 may include a peripheral circuit portion PERI including a substrate 201, and a memory cell array MCA over the peripheral circuit portion PERI. The memory cell array MCA may include a bit line 110, a transistor TR, and a capacitor 130. The transistor TR may include a vertical channel transistor. The memory cell array MCA may include a plurality of memory cells sharing the bit line 110.

[0110] The transistor TR may include an oxide semiconductor pillar 120 which is disposed between the bit line 110 and the capacitor 130, a tapered vertical word line 124 which is disposed over a sidewall of the oxide semiconductor pillar 120, and a gate dielectric layer 125 disposed between the oxide semiconductor pillar 120 and the tapered vertical word line 124. The oxide semiconductor pillar 120 may include a lower interface layer 122, an oxide semiconductor channel layer 121, and an upper interface layer 123. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include an oxide semiconductor material. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include IGZO, but the lower interface layer 122 and the upper interface layer 123 may have a greater indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor

channel layer 121 may be IGZO, and the lower interface layer 122 and the upper interface layer 123 may be indium-rich IGZO.

[0111] The tapered vertical word line 124 may include a lower level portion 124L and an upper level portion 124U.

[0112] The bit line 110 and a peripheral transistor PERI\_TR of the peripheral circuit portion PERI may be coupled to each other through a metal interconnection MLM. The uppermost layer of the metal interconnection MLM may pass through a buffer layer 102 to be coupled to the bit line 110.

[0113] The semiconductor device 200 may include a peripheral-under-cell (PUC) structure in which the memory cell array MCA is formed over the peripheral circuit portion PERI.

[0114] FIG. 4B is a cross-sectional view illustrating a semiconductor device 200M in accordance with another embodiment of the present disclosure. The semiconductor device 200M of FIG. 4B may be similar to the semiconductor device 100 shown in FIGS. 1A to 1E. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 1A to 1E, descriptions of FIGS. 1A to 1E may be referred to.

[0115] Referring to FIG. 4B, the semiconductor device 200M may be an array of memory cells including a bit line 110, an oxide semiconductor pillar 120, and a capacitor 130. For example, the lower level memory cell array MCA\_L and the upper level memory cell array MCA\_U may be vertically stacked. Each of the memory cells of the lower level memory cell array MCA\_L and the upper level memory cell array MCA\_U may include a bit line 110, a transistor TR, and a capacitor 130. The transistor TR may include a vertical channel transistor. The transistor TR may include an oxide semiconductor pillar 120 which is disposed between the bit line 110 and the capacitor 130, a tapered vertical word line 124 which is disposed over a sidewall of the oxide semiconductor pillar 120, and a gate dielectric layer 125 disposed between the oxide semiconductor pillar 120 and the tapered vertical word line 124. The oxide semiconductor pillar 120 may include a lower interface layer 122, an oxide semiconductor channel layer 121, and an upper interface layer 123. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include an oxide semiconductor material. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include IGZO, but the lower interface layer 122 and the upper interface layer 123 may have a higher indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor channel layer 121 may be IGZO, and the lower interface layer 122 and the upper interface layer 123 may be indium-rich IGZO. The tapered vertical word line 124 may include a lower level portion 124L and an upper level portion 124U.

[0116] The upper electrode 136 of the lower level memory cell array MCA\_L may directly contact the buffer layer 102 of the upper level memory cell array MCA\_U.

[0117] The lower level memory cell array MCA\_L and the upper level memory cell array MCA\_U may be stacked without wafer bonding.

[0118] FIG. 5 is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure. The semiconductor device 300 of FIG. 5 may be similar to the semiconductor device 100 of FIGS. 1A to 1E. Hereinafter, for the detailed descriptions of

the constituent elements also appearing in FIGS. 1A to 1E, descriptions of FIGS. 1A to 1E may be referred to.

[0119] Referring to FIG. 5, the semiconductor device 300 may include a bit line 110, a vertical channel transistor TR, and a capacitor 130 that are disposed vertically in the third direction D3. The vertical channel transistor TR may include an oxide semiconductor pillar 120 disposed between the bit line 110 and the capacitor 130, a tapered vertical word line 124 disposed over a sidewall of the oxide semiconductor pillar 120, and a gate dielectric layer 125 disposed between the oxide semiconductor pillar 120 and the tapered vertical word line 124. The oxide semiconductor pillar 120 may include a lower interface layer 122, an oxide semiconductor channel layer 121, and an upper interface layer 123. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include an oxide semiconductor material. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include IGZO, but the lower interface layer 122 and the upper interface layer 123 may have a higher indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor channel layer 121 may be IGZO, and the lower interface layer 122 and the upper interface layer 123 may be indium-rich IGZO.

[0120] The semiconductor device 300 may further include a dummy plate 210 below the bit line 110. The dummy plate 210 may include a metal-based material. The buffer layer 102 may be disposed between the dummy plate 210 and the bit line 110. Although parasitic capacitance may be increased due to the dummy plate 210, coupling noise may be improved because the coupling capacitance ratio between the bit lines 110 is decreased. Referring to FIGS. 1B and 5, the coupling capacitance ratio between the bit lines 110 that are spaced apart from each other in the second direction D2 may decrease by the dummy plate 210. The optimal ratio of the parasitic capacitance may be controlled by adjusting the thickness of the buffer layer 102 between the dummy plate 210 and the bit line 110.

[0121] FIGS. 6A to 6C are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method of fabricating the semiconductor device of FIGS. 6A to 6C may be similar to that of FIGS. 2A to 2K. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 2A to 2K, descriptions of FIGS. 2A to 2K may be referred to.

[0122] Referring to FIG. 6A, a buffer layer 12 may be formed over the substrate 11. A conductive layer 13A, a barrier material layer 14A, a lower interface material layer 15A, a channel material layer 16A, and an upper interface material layer 17A may be sequentially formed over the buffer layer 12.

[0123] Subsequently, a first contact layer 27A and a second contact layer 28A may be stacked over the upper interface material layer 17A. The first contact layer 27A may include titanium nitride, and the second contact layer 28A may include tungsten.

[0124] Referring to FIG. 6B, line structures 19L may be formed. Each of the line structures 19L may include a stack of a bit line 13, a bit line barrier layer 14, a lower interface layer 15B, a channel material 16B, an upper interface layer

17B, a first contact layer 27B, and the second contact layer 28B. First trenches 19T may be formed between the line structures 19L.

[0125] Referring to FIG. 6C, a portion of the line structures 19L may be selectively etched. As a result, oxide semiconductor pillars 21P may be formed over the bit line barrier layer 14. Each of the oxide semiconductor pillars 21P may include a stack of a lower interface layer 15, a channel layer 16, and an upper interface layer 17. The lower interface layer 15, the channel layer 16, and the upper interface layer 17 may be formed by etching the lower interface layer 15B, the channel material 16B, and the upper interface layer 17B, respectively. A stack of first contact plugs 27 and second contact plugs 28 may be formed over the upper interface layers 17. The first contact plugs 27 may be formed by etching the first contact layer 27B, and the second contact plugs 28 may be formed by etching the second contact layer 28B.

[0126] Trenches 22 may be formed between the oxide semiconductor pillars 21P. The first contact plugs 27 and the second contact plugs 28 may be cut by the trenches 22, and accordingly, each of the first contact plugs 27 and the second contact plugs 28 may have a pillar shape.

[0127] Subsequently, a gate dielectric layer 23 may be formed on the sides of the oxide semiconductor pillars 21P. The gate dielectric layer 23 may cover the sidewalls of the first contact plugs 27 and the second contact plugs 28.

[0128] Subsequently, tapered vertical word lines 24 may be formed over the gate dielectric layer 23.

[0129] Subsequently, as illustrated in FIG. 2K, a capacitor 30 may be formed over the second contact plugs 28.

[0130] FIGS. 7A to 7G are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method of fabricating the semiconductor device of FIGS. 7A to 7G may be similar to that of FIGS. 2A to 2K. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 2A to 2K, descriptions of FIGS. 2A to 2K may be referred to.

[0131] Referring to FIG. 7A, a buffer layer 12 may be formed over the substrate 11. A conductive layer 13A and a barrier material layer 14A may be formed over the buffer layer 12.

[0132] Subsequently, a first sacrificial layer 15D and a second sacrificial layer 18D may be sequentially formed over the barrier material layer 14A. The first sacrificial layer 15D may include polysilicon, and the second sacrificial layer 18D may include silicon nitride.

[0133] Referring to FIG. 7B, second sacrificial layer lines 18L may be formed. The second sacrificial layer lines 18L may be formed by etching the second sacrificial layer 18D. The second sacrificial layer lines 18L may be formed by a double patterning process.

[0134] Subsequently, a first sacrificial layer 15D may be etched by using the second sacrificial layer lines 18L as an etch barrier, and the barrier material layer 14A and the conductive layer 13A may be etched continuously.

[0135] A plurality of line structures 19L may be formed over the buffer layer 12 by a series of the etching processes described above. Each of the line structures 19L may include a stack of a bit line 13, a bit line barrier layer 14, a first sacrificial layer line 15L, and a second sacrificial layer line 18L. The bit line barrier layer 14 may be formed by etching the barrier material layer 14A, and the bit line 13 may be

formed by etching the conductive layer 13A. The first sacrificial layer line 15L may be formed by etching the first sacrificial layer 15D. First trenches 19T may be formed between the line structures 19L.

[0136] Referring to FIG. 7C, dielectric lines 20 may be formed between the line structures 19L. The dielectric lines 20 may include a dielectric material. For example, the dielectric lines 20 may include silicon oxide, silicon nitride, silicon carbon oxide (SiCO), a spin-on-dielectric layer, or a combination thereof. The dielectric lines 20 may be formed by depositing a dielectric material to fill the first trenches 19T between the line structures 19L and performing a planarization process of a dielectric material. The dielectric lines 20 may be formed by sequentially forming a silicon nitride, a silicon carbon oxide, and a spin-on dielectric layer and performing a planarization process.

[0137] Referring to FIG. 7D, a portion of the line structures 19L may be selectively etched to form sacrificial pillars 21P'. The bit line barrier layer 14 and the bit line 13 may be disposed below the sacrificial pillars 21P'.

[0138] Each of the sacrificial pillars 21P' may include a stack of a first sacrificial pillar 15LP and a second sacrificial pillar 18LP. The first sacrificial pillar 15LP may be formed by etching the first sacrificial layer line 15L, and the second sacrificial pillar 18LP may be formed by etching the second sacrificial layer line 18L.

[0139] Second trenches 22 may be formed between the sacrificial pillars 21P'.

[0140] Subsequently, the gate dielectric layer 23 and the tapered vertical word line 24 may be formed by a series of processes as illustrated in FIGS. 2E and 2F. Referring to FIG. 7E, the gate dielectric layer 23 may be formed on the sides of the sacrificial pillars 21P'. The tapered vertical word line 24 may be formed over the gate dielectric layer 23. The bottom portion of the tapered vertical word line 24 may have a thinner thickness than the other portions.

[0141] Referring to FIG. 7F, an inter-layer dielectric layer 26 may be formed over the tapered vertical word line 24. The inter-layer dielectric layer 26 may be planarized to expose the upper surfaces of the second sacrificial pillars 18LP.

[0142] Subsequently, the second sacrificial pillars 18LP and the first sacrificial pillars 15LP may be selectively removed. Accordingly, pillar-shaped openings 18H may be formed. The bottom surface of the pillar-shaped openings 18H may expose the surface of the barrier layer 14.

[0143] Referring to FIG. 7G, oxide semiconductor pillars 21P filling the pillar-shaped openings 18H may be formed. Each of the oxide semiconductor pillars 21P may include a stack of the lower interface layer 15, the channel layer 16, and the upper interface layer 17. The lower interface layer 15, the channel layer 16, and the upper interface layer 17 may be formed by epitaxial growth, individually.

[0144] The lower interface layer 15 may contain indium. The lower interface layer 15 may include an indium-rich oxide semiconductor material. For example, the lower interface layer 15 may include indium-rich IGZO. The lower interface layer 15 may be formed to have a thickness of approximately 10 to 50 Å.

[0145] A channel layer 16 may be formed on the lower interface layer 15. The channel layer 16 may contain indium. The channel layer 16 may include IGZO. The channel layer 16 may be formed to have a thickness of 200 to 1000 Å.

[0146] An upper interface layer 17 may be formed over the channel layer 16. The upper interface layer 17 may contain indium. The upper interface layer 17 may include an indium-rich oxide semiconductor material. For example, the upper interface layer 17 may include indium-rich IGZO. The upper interface layer 17 may be formed to have a thickness of approximately 10 to 50 Å.

[0147] The oxide semiconductor pillars 21P may partially fill the pillar-shaped openings 18H, and thus hole-shaped recesses 18R may be defined by the oxide semiconductor pillars 21P.

[0148] Subsequently, a series of the processes as described in FIGS. 2I to 2K may be performed.

[0149] FIGS. 8 to 10 are cross-sectional views illustrating semiconductor devices in accordance with other embodiments of the present disclosure. FIGS. 8 to 10 may be similar to the semiconductor device 100 shown in FIGS. 1A to 1E, and also may be similar to the semiconductor device 300 shown in FIG. 5. Hereinafter, as for the detailed descriptions on the constituent elements also appearing in FIGS. 1A to 1E and FIG. 5, descriptions of FIGS. 1A to 1E and FIG. 5 may be referred to.

[0150] Referring to FIG. 8, the semiconductor device 400 may include a substrate 101, a buffer layer 102, a bit line 110, a barrier layer 111, an oxide semiconductor pillar 120, a tapered vertical word line 124, a gate dielectric layer 125, and a capacitor 130. The semiconductor device 400 may further include a first contact plug 126 and a second contact plug 127 between the oxide semiconductor pillar 120 and the capacitor 130. The oxide semiconductor pillar 120 may include a lower interface layer 122, an oxide semiconductor channel layer 121, and an upper interface layer 123. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include an oxide semiconductor material. The lower interface layer 122, the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include IGZO, but the lower interface layer 122 and the upper interface layer 123 may have a higher indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor channel layer 121 may be IGZO, and the lower interface layer 122 and the upper interface layer 123 may be indium-rich IGZO.

[0151] The semiconductor device 400 may further include a dummy plate 210 below the bit line 110. The buffer layer 102 may be disposed between the dummy plate 210 and the bit line 110. According to another embodiment of the present disclosure, the dummy plate 210 may be omitted.

[0152] The semiconductor device 400 may further include an isolating dielectric layer 401 which is disposed between the tapered vertical word line 124 and the barrier layer 111. The isolating dielectric layer 401 may be disposed below the gate dielectric layer 125. The isolating dielectric layer 401 may further increase the distance between the lower level portion 124L of the tapered vertical word line 124 and the bit line 110 (refer to a reference numeral H1).

[0153] Referring to FIG. 9, the semiconductor device 410 may include a substrate 101, a buffer layer 102, a bit line 110, a barrier layer 111, an oxide semiconductor pillar 120, a tapered vertical word line 124, a gate dielectric layer 125, and a capacitor 130. The semiconductor device 410 may further include a first contact plug 126 and a second contact plug 127 between the oxide semiconductor pillar 120 and the capacitor 130. The oxide semiconductor pillar 120 may

include a lower interface layer 122', an oxide semiconductor channel layer 121, and an upper interface layer 123. The lower interface layer 122', the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include IGZO, but the lower interface layer 122' and the upper interface layer 123 may have a higher indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor channel layer 121 may be IGZO, and the lower interface layer 122' and the upper interface layer 123 may be indium-rich IGZO.

[0154] The semiconductor device 410 may further include a dummy plate 210 below the bit line 110. The buffer layer 102 may be disposed between the dummy plate 210 and the bit line 110. According to another embodiment of the present disclosure, the dummy plate 210 may be omitted.

[0155] The semiconductor device 410 may further include an isolating dielectric layer 401 which is disposed between the tapered vertical word line 124 and the barrier layer 111. The isolating dielectric layer 401 may further increase the distance between the lower level portion 124L of the tapered vertical word line 124 and the bit line 110 (refer to a reference numeral H1).

[0156] The height of the lower interface layer 122' of the oxide semiconductor pillar 120 may be higher than the height of the upper interface layer 123. Also, the height of the lower interface layer 122' may be higher than the height of the lower interface layer 122 of FIG. 8. The lower interface layer 122' and the isolating dielectric layer 401 may have the same height.

[0157] Referring to FIG. 10, the semiconductor device 420 may include a substrate 101, a buffer layer 102, a bit line 110, a barrier layer 111, an oxide semiconductor pillar 120, a tapered vertical word line 124, a gate dielectric layer 125, and a capacitor 130. The semiconductor device 420 may further include a first contact plug 126 and a second contact plug 127 between the oxide semiconductor pillar 120 and the capacitor 130. The oxide semiconductor pillar 120 may include a lower interface layer 122'', an oxide semiconductor channel layer 121, and an upper interface layer 123. The lower interface layer 122'', the oxide semiconductor channel layer 121, and the upper interface layer 123 may all include IGZO, but the lower interface layer 122'' and the upper interface layer 123 may have a higher indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor channel layer 121 may be IGZO, and the lower interface layer 122'' and the upper interface layer 123 may be indium-rich IGZO.

[0158] The semiconductor device 420 may further include a dummy plate 210 below the bit line 110. The buffer layer 102 may be disposed between the dummy plate 210 and the bit line 110. According to another embodiment of the present disclosure, the dummy plate 210 may be omitted.

[0159] The semiconductor device 420 may further include an isolating dielectric layer 401 which is disposed between the tapered vertical word line 124 and the gate dielectric layer 125. The isolating dielectric layer 401 may be disposed over the gate dielectric layer 125. The isolating dielectric layer 401 may further increase the distance between the lower level portion 124L of the tapered vertical word line 124 and the bit line 110 (refer to a reference numeral H1).

[0160] The height of the lower interface layer 122'' of the oxide semiconductor pillar 120 may be greater than the height of the upper interface layer 123. The height of the lower interface layer 122'' may be higher than the height of

the lower interface layer 122' of FIG. 9. According to another embodiment of the present disclosure, the height of the lower interface layer 122'' may be the same as the height of the lower interface layer 122' of FIG. 9. According to another embodiment of the present disclosure, the height of the lower interface layer 122'' may be the same as the height of the lower interface layer 122 of FIG. 8.

[0161] In the semiconductor devices 400, 410, and 420 shown in FIGS. 8 to 10, the tapered vertical word line 124 and the bit line 110 may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer 401 and the gate dielectric layer 125 (refer to a reference numeral H1). The isolating dielectric layer 401 may include silicon oxide.

[0162] The gate dielectric layer 125 of the semiconductor devices 400, 410, and 420 shown in FIGS. 8 to 10 may include a vertical portion 125V which is disposed between the oxide semiconductor pillar 120 and the tapered vertical word line 124, and a horizontal portion 125F which extends from the vertical portion 125V and is disposed between the bit line 110 and the tapered vertical word line 124. The isolating dielectric layer 401 may be disposed at a lower level than the horizontal portion 125F of the gate dielectric layer 125 (refer to FIGS. 8 and 9) or may be disposed at a higher level than the horizontal portion 125F of the gate dielectric layer 125 (refer to FIG. 10).

[0163] FIGS. 11A to 14B are embodiments of a method for fabricating the semiconductor devices in accordance with FIGS. 8 to 10.

[0164] FIGS. 11A to 11C are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. 11A to 11C may be similar to that of FIGS. 2A to 2K. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 2A to 2K, descriptions of FIGS. 2A to 2K may be referred to. FIGS. 11A to 11C illustrate the fabrication method according to the line B-B' shown in FIG. 1A.

[0165] First, referring to FIGS. 2A to 2D, a buffer layer 12 may be formed over the substrate 11, and a bit line 13 and a barrier layer 14 may be formed over the buffer layer 12. Subsequently, oxide semiconductor pillars 21P including a lower interface layer 15, a channel layer 16, and an upper interface layer 17 may be formed over the barrier layer 14. Sacrificial pillars 18P may be disposed over the upper interface layer 17. Second trenches 22 may be formed between the oxide semiconductor pillars 21P.

[0166] Subsequently, referring to FIG. 11A, an isolating dielectric layer 31 may be formed over the barrier layer 14. The isolating dielectric layer 31 may be formed by depositing a dielectric layer 31A over the oxide semiconductor pillars 21P and then performing an etch-back process onto the dielectric layer 31A. The isolating dielectric layer 31 may include silicon oxide. The height of the isolating dielectric layer 31 may be higher than the height of the lower interface layer 15.

[0167] Referring to FIG. 11B, a gate dielectric layer 23 may be formed over the isolating dielectric layer 31. The gate dielectric layer 23 may be formed on the exposed sidewalls of the oxide semiconductor pillars 21P and the sacrificial pillars 18P. The isolating dielectric layer 31 may be disposed between the gate dielectric layer 23 and the bit

line 13. The isolating dielectric layer 31 may be disposed at a lower level than the gate dielectric layer 23.

[0168] Referring to FIG. 11C, the tapered vertical word lines 24 may be formed. Tapered vertical word lines 24 may be formed with reference to the methods illustrated in FIGS. 2E to 2G. The gate dielectric layer 23, the isolating dielectric layer 31, and the barrier layer 14 may be disposed between the bottom portion of the tapered vertical word lines 24 and the bit line 13. The tapered vertical word lines 24 and the bit line 13 may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer 31 and the gate dielectric layer 23 (refer to a reference numeral H1).

[0169] FIGS. 12A and 12B are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. 12A and 12B may be similar to that of FIGS. 2A to 2K. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 2A to 2K, descriptions of FIGS. 2A to 2K may be referred to. FIGS. 12A and 12B may be the fabrication method according to the line B-B' shown in FIG. 1A.

[0170] First, referring to FIGS. 2A to 2D, a buffer layer 12 may be formed over the substrate 11, and a bit line 13 and a barrier layer 14 may be formed over the buffer layer 12. Subsequently, oxide semiconductor pillars 21P including a lower interface layer 15, a channel layer 16, and an upper interface layer 17 may be formed over the barrier layer 14. Sacrificial pillars 18P may be disposed over the upper interface layer 17.

[0171] Subsequently, referring to FIG. 12A, a gate dielectric layer 23 may be formed over the barrier layer 14. The gate dielectric layer 23 may be formed on the exposed sidewalls of the oxide semiconductor pillars 21P and the sacrificial pillars 18P.

[0172] Subsequently, an isolating dielectric layer 31 may be formed over the gate dielectric layer 23. The isolating dielectric layer 31 may be formed by depositing a dielectric layer 31A over the gate dielectric layer 23 and performing an etch-back process onto the dielectric layer 31A. The isolating dielectric layer 31 may include silicon oxide. The height of the isolating dielectric layer 31 may be higher than the height of the lower interface layer 15. A gate dielectric layer 23 may be disposed between the isolating dielectric layer 31 and the bit line 13.

[0173] As described above, the isolating dielectric layer 31 may be formed after the gate dielectric layer 23 is formed.

[0174] Referring to FIG. 12B, tapered vertical word lines 24 may be formed. The tapered vertical word lines 24 may be formed with reference to the methods illustrated in FIGS. 2E to 2G. A gate dielectric layer 23, an isolating dielectric layer 31, and a barrier layer 14 may be disposed between the bottom portion of the tapered vertical word lines 24 and the bit line 13. The tapered vertical word lines 24 and the bit line 13 may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer 31 and the gate dielectric layer 23 (refer to a reference numeral H1).

[0175] FIGS. 13A to 13D are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. 13A to 13D may be similar to that of FIGS. 2A to 2K. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 2A to 2K, descriptions

of FIGS. 2A to 2K may be referred to. FIGS. 13A to 13D may be the fabrication method according to the line B-B' shown in FIG. 1A.

[0176] Referring to FIGS. 2A and 13A, a buffer layer 12 may be formed over a substrate 11, and a conductive layer 13A and a barrier material layer 14A may be sequentially formed over the buffer layer 12.

[0177] A lower interface layer 15C may be formed over the barrier material layer 14A. The lower interface layer 15C may include a conductive material. The lower interface layer 15C may include an oxide semiconductor material. The lower interface layer 15C may contain indium. The lower interface layer 15C may include an indium-rich oxide semiconductor material. For example, the lower interface layer 15C may include indium-rich IGZO. The lower interface layer 15C may be formed to have a thickness of approximately 10 to 50 Å. The lower interface layer 15C may be thicker than the lower interface material layer 15A shown in FIG. 2A.

[0178] A channel material layer 16A may be formed over the lower interface layer 15C. The channel material layer 16A may include a conductive material. The channel material layer 16A may include an oxide semiconductor material. The channel material layer 16A may contain indium. The channel material layer 16A may include IGZO. The channel material layer 16A may be formed to have a thickness of approximately 200 to 1000 Å.

[0179] An upper interface material layer 17A may be formed over the channel material layer 16A. The upper interface material layer 17A may include a conductive material. The upper interface material layer 17A may include an oxide semiconductor material. The upper interface material layer 17A may contain indium. The upper interface material layer 17A may include an indium-rich oxide semiconductor material. For example, the upper interface material layer 17A may include indium-rich IGZO. The upper interface material layer 17A may be formed to have a thickness of approximately 10 to 50 Å. The upper interface material layer 17A may be thinner than the lower interface layer 15C.

[0180] As described above, the lower interface layer 15C, the channel material layer 16A, and the upper interface material layer 17A may be vertically stacked over the barrier material layer 14A. The lower interface layer 15C, the channel material layer 16A, and the upper interface material layer 17A may all include an oxide semiconductor material. The lower interface layer 15C, the channel material layer 16A, and the upper interface material layer 17A may all include IGZO, but the lower interface layer 15C and the upper interface material layer 17A may have a higher indium concentration than the channel material layer 16A. The channel material layer 16A may be IGZO, and the lower interface material layer 15A and the upper interface material layer 17A may be indium-rich IGZO. Since the lower interface layer 15C and the upper interface material layer 17A contain a high concentration of indium, the resistance may be reduced lower than that of the channel material layer 16A. Also, channel seamless interconnection of the channel material layer 16A is possible.

[0181] Subsequently, a sacrificial layer 18A may be formed over the upper interface material layer 17A. The sacrificial layer 18A may include a stack of different materials. The sacrificial layer 18A may include silicon nitride.

[0182] Referring to FIG. 13B, a bit line 13 and a barrier layer 14 may be formed over the buffer layer 12 by a series of the processes as illustrated in FIGS. 2B to 2D. Subsequently, oxide semiconductor pillars 21P including a lower interface layer 15', a channel layer 16, and an upper interface layer 17 may be formed over the barrier layer 14. Sacrificial pillars 18P may be disposed over the upper interface layer 17. Second trenches 22 may be formed between the oxide semiconductor pillars 21P.

[0183] Referring to FIG. 13C, an isolating dielectric layer 31 may be formed over the barrier layer 14. A method of forming the isolating dielectric layer 31 may be the same as that of FIG. 11A. The height of the isolating dielectric layer 31 and the lower interface layer 15' may be the same as each other.

[0184] Referring to FIG. 13D, a gate dielectric layer 23 may be formed over the isolating dielectric layer 31, and tapered vertical word lines 24 may be formed over the gate dielectric layer 23. The tapered vertical word lines 24 may be formed with reference to the methods illustrated in FIGS. 2E to 2G. A gate dielectric layer 23, an isolating dielectric layer 31, and a barrier layer 14 may be disposed between the bottom portion of the tapered vertical word lines 24 and the bit line 13. The tapered vertical word lines 24 and the bit line 13 may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer 31 and the gate dielectric layer 23 (refer to a reference numeral H1).

[0185] FIGS. 14A and 14B are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. 14A and 14B may be similar to that of FIGS. 2A through 2K. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 2A to 2K, descriptions of FIGS. 2A to 2K may be referred to. FIGS. 14A and 14B may be the fabrication method according to the line B-B' shown in FIG. 1A.

[0186] First, as illustrated in FIGS. 2A to 2D and 13A and 13B, a buffer layer 12 may be formed over the substrate 11, and a bit line 13 and a barrier layer 14 may be formed over the buffer layer 12. Subsequently, oxide semiconductor pillars 21P including a lower interface layer 15', a channel layer 16, and an upper interface layer 17 may be formed over the barrier layer 14. Sacrificial pillars 18P may be disposed over the upper interface layer 17.

[0187] Subsequently, referring to FIG. 14A, a gate dielectric layer 23 may be formed over the barrier layer 14. The gate dielectric layer 23 may be formed on the exposed sidewalls of the oxide semiconductor pillars 21P and the sacrificial pillars 18P.

[0188] Subsequently, an isolating dielectric layer 31 may be formed over the gate dielectric layer 23. The isolating dielectric layer 31 may be formed by depositing the dielectric layer 31A over the gate dielectric layer 23 and performing an etch-back process onto the dielectric layer 31A. The isolating dielectric layer 31 may include silicon oxide. The gate dielectric layer 23 may be disposed between the isolating dielectric layer 31 and the bit line 13.

[0189] As described above, the isolating dielectric layer 31 may be formed after the gate dielectric layer 23 is formed.

[0190] Referring to FIG. 14B, tapered vertical word lines 24 may be formed. The tapered vertical word lines 24 may be formed with reference to the methods shown in FIGS. 2E to 2G. The gate dielectric layer 23, the isolating dielectric

layer 31, and the barrier layer 14 may be disposed between the bottom portion of the tapered vertical word lines 24 and the bit line 13. The tapered vertical word lines 24 and the bit line 13 may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer 31 and the gate dielectric layer 23 (refer to a reference numeral H1).

[0191] FIG. 15A is a schematic plan view illustrating a semiconductor device 500 in accordance with an embodiment of the present disclosure. FIG. 15B is a cross-sectional view taken along a line A-A' shown in FIG. 15A. FIG. 15C is a cross-sectional view taken along a line B-B' shown in FIG. 15A. FIG. 15D is an enlarged view illustrating a portion 501 of FIG. 15A. The semiconductor device 500 of FIGS. 15A to 15D may be similar to the semiconductor device 100 of FIGS. 1A to 1E. Hereinafter, for the detailed description of the constituent elements of the semiconductor device 500 also appearing in FIGS. 1A to 1E, FIGS. 1A to 1E may be referred to.

[0192] Referring to FIGS. 15A to 15D, the semiconductor device 500 may include a lower horizontal conductive line, an upper horizontal conductive line, a shield line, a vertically oriented pillar, and a data storage element. The lower horizontal conductive line may include a bit line 110. The upper horizontal conductive line may include a main gate 124S. The vertically oriented pillar may include an oxide semiconductor pillar 120. The data storage element may include a capacitor 130. The shield line may include a shield gate 124G.

[0193] The semiconductor device 500 may include the bit line 110, the oxide semiconductor pillar 120 extending vertically from the bit line 110, the capacitor 130 over the oxide semiconductor pillar 120, and the main gate 124S disposed on a sidewall of the oxide semiconductor pillar 120.

[0194] Referring back to FIG. 15B, the oxide semiconductor pillar 120 may include a first oxide semiconductor interface layer 122 coupled to the bit line 110, a second oxide semiconductor interface layer 123 coupled to the capacitor 130, and an oxide semiconductor channel layer 121 between the first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123. Each of the first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123 may include an oxide semiconductor layer. The first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123 may be referred to as a lower oxide semiconductor interface layer and an upper oxide semiconductor interface layer, respectively. The semiconductor device 500 may include a dynamic random access memory (DRAM). As shown in FIG. 15A, the bit line 110 may extend in the first direction D1, and the main gate 124S may extend in the second direction D2. The first direction D1 and the second direction D2 may intersect with each other perpendicularly.

[0195] The bit line 110 may be disposed at a lower level than the oxide semiconductor channel layer 121. The capacitor 130 may be disposed at a higher level than the oxide semiconductor channel layer 121. As shown in FIG. 15C, the main gate 124S may be disposed on the sidewall of a first side of the oxide semiconductor channel layer 121. The first oxide semiconductor interface layer 122 may be disposed between the bit line 110 and the oxide semiconductor channel layer 121. The second oxide semiconductor interface layer 123 may be disposed between the capacitor 130

and the oxide semiconductor channel layer **121**. The bit line **110**, the first oxide semiconductor interface layer **122**, the oxide semiconductor channel layer **121**, the second oxide semiconductor interface layer **123**, and the capacitor **130** may be disposed vertically in the third direction **D3**.

[0196] The semiconductor device **500** may be described in detail as follows.

[0197] Referring back to FIGS. **15A**, **15B** and **15C**, a buffer layer **102** may be formed over a substrate **101**, and the bit line **110** may be formed over the buffer layer **102**. A vertical channel transistor TR may be formed over the bit line **110**, and the capacitor **130** may be formed over the vertical channel transistor TR. The vertical channel transistor TR may include the oxide semiconductor pillar **120**, the main gate **124S** disposed on the sidewall of the first side of the oxide semiconductor pillar **120**, and a gate dielectric layer **125** between the oxide semiconductor pillar **120** and the main gate **124S**. The oxide semiconductor pillar **120** may include the oxide semiconductor channel layer **121**, the first oxide semiconductor interface layer **122** between the oxide semiconductor channel layer **121** and the bit line **110**, and the second oxide semiconductor interface layer **123** between the oxide semiconductor channel layer **121** and the capacitor **130**. A barrier layer **111** may be disposed between the bit line **110** and the first oxide semiconductor interface layer **122**.

[0198] One bit line **110**, one vertical channel transistor TR, and one capacitor **130** may form a memory cell MC. The memory cell MC may have an one transistor, one capacitor (1T-1C or 1T1C) structure. The semiconductor device **500** may include a memory cell array MCA, that is, an array of memory cells MC. The memory cell array MCA may include a plurality of first arrays AR1 and a plurality of second arrays AR21 to AR24. The first arrays AR1 may be disposed spaced apart from each other in the second direction **D2**. The second arrays AR21 to AR24 may be disposed spaced apart from each other in the first direction **D1**. The first arrays AR1 and the second arrays AR21 to AR24 may intersect with each other. The first arrays AR1 may be column arrays, while the second arrays AR21 to AR24 may be row arrays. The first arrays AR1 may include a plurality of memory cells MC that are disposed in the first direction **D1**. The second arrays AR21 to AR24 may include a plurality of memory cells MC that are disposed in the second direction **D2**. In each first array AR1, a plurality of memory cells MC may share one bit line **110**. In each second array AR21 to AR24, a plurality of memory cells MC may share one main gate **124S**. The second arrays AR21 to AR24 may include a first sub-array AR21, a second sub-array AR22, a third sub-array AR23, and a fourth sub-array AR24. A first space **S1** between the first sub-array AR21 and the second sub-array AR22 may be the same as a second space **S2** between the third sub-array AR23 and the fourth sub-array AR24. A third space **S3** between the second sub-array AR22 and the third sub-array AR23 may be greater than each of the first and second spaces **S1** and **S2**. According to another embodiment of the present disclosure, the number of the sub-arrays may be four or more. The first and second spaces **S1** and **S2** may be referred to as shield trenches.

[0199] The semiconductor device **500** may further include shield gates **124G**. The shield gates **124G** may be disposed between the main gates **124S**. The shield gates **124G** may be disposed between the neighboring vertical channel transistors TR. The shield gates **124G** may be disposed between the neighboring memory cells MC.

[0200] Referring back to FIG. **15A**, the shield gates **124G** may be disposed between the neighboring second arrays AR21 to AR24. The shield gates **124G** may include a first shield gate **124G1** and a second shield gate **124G2**.

[0201] For example, the first shield gate **124G1** may be disposed between the first sub-array AR21 and the second sub-array AR22. To be specific, the first shield gate **124G1** may be disposed between the vertical channel transistors TR of the first sub-array AR21 and the vertical channel transistors TR of the second sub-array AR22. Since the first shield gate **124G1** is disposed between the first sub-array AR21 and the second sub-array AR22, the interference between the vertical channel transistors TR of the first sub-array AR21 and the vertical channel transistors TR of the second sub-array AR22 may be blocked off. The first sub-array AR21 and the second sub-array AR22 may share the first shield gate **124G1**.

[0202] The second shield gate **124G2** may be disposed between the third sub-array AR23 and the fourth sub-array AR24. To be specific, the second shield gate **124G2** may be disposed between the vertical channel transistors TR of the third sub-array AR23 and the vertical channel transistors TR of the fourth sub-array AR24. Since the second shield gate **124G2** is disposed between the third sub-array AR23 and the fourth sub-array AR24, the interference between the vertical channel transistors TR of the third sub-array AR23 and the vertical channel transistors TR of the fourth sub-array AR24 may be blocked off. The third sub-array AR23 and the fourth sub-array AR24 may share the second shield gate **124G2**.

[0203] Referring back to FIGS. **15B** and **15C**, the semiconductor device **500** may further include first and second contact plugs **126** and **127** between the second oxide semiconductor interface layer **123** and the capacitor **130**.

[0204] The substrate **101** may be a material appropriate for semiconductor processing. The substrate **101** may include a semiconductor substrate. The substrate **101** may be formed of a silicon-containing material. The substrate **101** may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate **101** may also include another semiconductor material, such as germanium. The substrate **101** may include a III/V-group semiconductor substrate, for example, a compound semiconductor substrate, such as gallium arsenide (GaAs). The substrate **101** may also include a Silicon-On-Insulator (SOI) substrate.

[0205] The buffer layer **102** may include silicon oxide, silicon nitride, or a combination thereof. The buffer layer **102** may be formed of silicon oxide to reduce parasitic capacitance. For example, the buffer layer **102** may include Tetraethyl Orthosilicate (TEOS).

[0206] The bit line **110** may extend in the first direction **D1** over the buffer layer **102**. The bit line **110** may include a metal-based material. The bit line **110** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The bit line **110** may have a thickness of approximately 100 to 400 Å. The bit line **110** may include a tungsten layer.

[0207] The barrier layer **111** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier layer **111** may include titanium nitride, molybdenum, or ruthenium. The barrier layer **111** may have a thickness of approximately 10 to 50 Å. For example, the barrier layer **111** may include a titanium nitride layer.



[0208] Each of the first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123 may include an oxide semiconductor material having a lower resistance than the oxide semiconductor channel layer 121. Each of the first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123 may include a metal-rich oxide semiconductor material. The oxide semiconductor channel layer 121 may include an oxygen-rich oxide semiconductor material. For example, the oxide semiconductor channel layer 121 may include indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO) or zinc tin oxide (ZTO), and each of the first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123 may include indium-rich IGZO. Indium-rich IGZO may refer to IGZO whose indium content is greater than gallium (Ga) and zinc (Zinc). For example, the content of indium may be approximately 40% or more.

[0209] The oxide semiconductor channel layer 121 may include an oxide semiconductor material. The oxide semiconductor channel layer 121 may contain indium. The oxide semiconductor channel layer 121 may include IGZO. The oxide semiconductor channel layer 121 may be formed to have a thickness of approximately 200 to 1,000 Å.

[0210] The first oxide semiconductor interface layer 122 may include an oxide semiconductor material. The first oxide semiconductor interface layer 122 may contain indium. The first oxide semiconductor interface layer 122 may include an indium-rich oxide semiconductor material. For example, the first oxide semiconductor interface layer 122 may include indium-rich IGZO. The first oxide semiconductor interface layer 122 may be formed to have a thickness of approximately 10 to 50 Å.

[0211] The second oxide semiconductor interface layer 123 may include an oxide semiconductor material. The second oxide semiconductor interface layer 123 may contain indium. The second oxide semiconductor interface layer 123 may include an indium-rich oxide semiconductor material. For example, the second oxide semiconductor interface layer 123 may include indium-rich IGZO. The second oxide semiconductor interface layer 123 may be formed to have a thickness of approximately 10 to 50 Å.

[0212] As described above, the oxide semiconductor pillar 120 may extend vertically in the third direction D3 over the bit line 110. The oxide semiconductor pillar 120 may include the first oxide semiconductor interface layer 122, the oxide semiconductor channel layer 121, and the second oxide semiconductor interface layer 123 that are vertically stacked over the bit line 110 in the mentioned order. The first oxide semiconductor interface layer 122, the oxide semiconductor channel layer 121, and the second oxide semiconductor interface layer 123 may all include an oxide semiconductor material. The first oxide semiconductor interface layer 122, the oxide semiconductor channel layer 121, and the second oxide semiconductor interface layer 123 may all include IGZO (indium gallium zinc oxide). The first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123 may have a higher indium concentration than the oxide semiconductor channel layer 121. The oxide semiconductor channel layer 121 may include IGZO, and each of the first oxide semiconductor interface layer 122 and the second oxide semiconductor interface layer 123 may include indium-rich IGZO. The first oxide semiconductor interface layer 122, the oxide semi-

conductor channel layer 121, and the second oxide semiconductor interface layer 123 may be referred to as an active pillar.

[0213] The main gate 124S of a single structure may be disposed on a sidewall of the oxide semiconductor channel layer 121. The main gate 124S may be referred to as a tapered vertical word line. The main gate 124S and the bit line 110 may extend in the directions intersecting with each other. The main gate 124S may have a reverse tapered shape in the third direction D3. The reverse tapered shape may refer to a shape whose bottom portion has a narrower width than the upper portion. For example, the main gate 124S may include a lower level portion 124L disposed adjacent to the bit line 110 and an upper level portion 124U disposed adjacent to the capacitor 130. The thickness of the lower level portion 124L in the first direction D1 may be thinner than the thickness of the upper level portion 124U. The lower level portion 124L may be disposed adjacent to the first oxide semiconductor interface layer 122. The upper level portion 124U may be disposed adjacent to the second oxide semiconductor interface layer 123. The lower level portion 124L and the upper level portion 124U may be formed of the same material. The lower level portion 124L and the bit line 110 may be disposed spaced apart from each other with a gap. The lower level portion 124L of the main gate 124S may not contact the bit line 110 and the barrier layer 111.

[0214] The main gate 124S may include a metal-based material. The main gate 124S may include a metal, a metal nitride, or a combination thereof. The main gate 124S may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0215] The gate dielectric layer 125 may be formed between the main gate 124S and the oxide semiconductor pillar 120. The gate dielectric layer 125 may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The gate dielectric layer 125 may be disposed between the oxide semiconductor pillar 120 and the main gate 124S. The gate dielectric layer 125 may include a horizontal portion that extends to be disposed between the lower level portion 124L of the main gate 124S and the bit line 110, and the horizontal portion of the gate dielectric layer 125 may directly contact the barrier layer 111. The oxide semiconductor pillar 120, the gate dielectric layer 125, and the main gate 124S may form a vertical channel transistor TR. The main gate 124S may be referred to as a tapered vertical gate.

[0216] The shield gate 124G may be disposed between the neighboring oxide semiconductor pillars 120. The shield gate 124G may have the same width at the top and bottom portions. The gate dielectric layer 125 may be disposed between the shield gate 124G and the oxide semiconductor pillars 120. The shield gate 124G may include a metal-based material. The shield gate 124G may include a metal, a metal nitride, or a combination thereof. The main gate 124S may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0217] The main gate 124S may be referred to as a front gate, and the shield gate 124G may be referred to as a common shield gate.

[0218] The main gate 124S may function to control the vertical channel transistor TR, and the shield gate 124G may function to block off the interference between the neighboring vertical channel transistors TR. The shield gate 124G

may be referred to as a back gate, a shield word line, a common gate, or a shield line.

[0219] Referring back to FIGS. 15B and 15C, the contact plugs 126 and 127 may include a first contact plug 126 and a second contact plug 127 over the first contact plug 126. The first contact plug 126 may directly contact the second oxide semiconductor interface layer 123. The second contact plug 127 may directly contact the capacitor 130. As shown in FIG. 15B, the first contact plug 126 and the second contact plug 127 may vertically overlap with each other in the A-A' direction. As shown in FIG. 15C, the first contact plug 126 and the second contact plug 127 may vertically partially overlap with each other in the B-B' direction. The first contact plug 126 and the second contact plug 127 may include a metal-based material. Each of the first contact plug 126 and the second contact plug 127 may include a metal, a metal nitride, or a combination thereof. The first contact plug 126 and the second contact plug 127 may be formed of the same metal-based material. According to another embodiment of the present disclosure, the first contact plug 126 and the second contact plug 127 may be formed of different metal-based materials. The sidewalls of the first contact plug 126 may be surrounded by the gate dielectric layer 125.

[0220] Referring back to FIG. 1E, the capacitor 130 may include the lower electrode 132, the dielectric layer 135, and the upper electrode 136. The lower electrode 132 may be formed over the second contact plug 127. The lower electrode 132 may have a pillar shape. The lower electrodes 132 may be supported by supporters 133 and 134. The sidewalls of the bottom portions of the lower electrodes 132 may contact the etch stop layer 131. According to another embodiment of the present disclosure, the lower electrode 132 may have a cylindrical shape.

[0221] In some embodiments, dielectric pillars may be disposed between the oxide semiconductor pillars 120 in the A-A' direction.

[0222] Referring to FIG. 15D, two neighboring oxide semiconductor pillars 120A and 120B will be described as follows. From the perspective of a top view, each of the first and second oxide semiconductor pillars 120A and 120B may include an outer side F1 and an inner side F2 which faces the outer side F1. The outer sides F1 may be disposed adjacent to first and second main gates 124S1 and 124S2. The inner sides F2 may be disposed adjacent to the shield gate 124G. The first and second main gates 124S1 and 124S2 may have a tapered shape. Each of the first and second main gates 124S1 and 124S2 may include a first sidewall VS1 having a straight profile, and a second sidewall VS2 facing the first sidewall VS1 and having a tapered profile. The second sidewall VS2 may include a first segment VS21 of a straight profile, a second segment VS22 of a straight profile, and a third segment VS23 of a tapered profile between the first segment VS21 and the second segment VS22. In this way, the second sidewall VS2 may have a straight profile and a tapered profile merged therein. The first sidewalls VS1 of the first and second main gates 124S1 and 124S2 may be close to the outer sides F1 of the first and second oxide semiconductor pillars 120A and 120B. The gate dielectric layer 125 may be disposed between the outer sides F1 of the first and second oxide semiconductor pillars 120A and 120B and the first and second main gates 124S1 and 124S2. A shield gate dielectric layer 125G may be disposed between the shield gate 124G and the inner sides F2 of the first and second

oxide semiconductor pillars 120A and 120B. The first and second oxide semiconductor pillars 120A and 120B may be disposed adjacent to each other with a shield trench ST therebetween. The shield trench ST may correspond to the first and second spaces S1 and S2 illustrated in FIG. 15A. Each of the first and second oxide semiconductor pillars 120A and 120B may include an inner side F2 that defines the inner sidewalls of the shield trench ST and an outer side F1 that faces the inner side F2. The shield gate 124G may be disposed between the first oxide semiconductor pillar 120A and the second oxide semiconductor pillar 120B. The first main gate 124S1 may be close to the outer side F1 of the first oxide semiconductor pillar 120A. The second main gate 124S2 may be close to the outer side F1 of the second oxide semiconductor pillar 120B. The shield gate dielectric layer 125G may be referred to as a 'common gate dielectric layer', and the shield gate 124G may be referred to as a 'common gate'.

[0223] Referring to FIGS. 15A to 15D, the semiconductor device 500 may include first and second pillars 120 disposed adjacent to each other with a shield trench ST therebetween and each including an inner side F2 that defines the inner sidewalls of the shield trench ST and an outer side F1 that faces the inner side F2, the shield gate 124G disposed in the shield trench ST between the first pillar 120 and the second pillar 120, a first main gate 124S1 close to the outer side F1 of the first pillar 120, a second main gate 124S2 close to the outer side F1 of the second pillar 120, the bit line 110 disposed in the lower portions of the first and second pillars 120 and electrically connected to the first and second pillars 120, and the capacitors 130 disposed in the upper portions of the first and second pillars 120 and electrically connected to the first and second pillars 120, respectively.

[0224] As described above, by forming the main gate 124S and the shield gate 124G as a single structure, the space between the neighboring vertical channel transistors TR may be reduced. Accordingly, since the space between the neighboring cells may be reduced, the cell area may be decreased, thereby increasing the degree of integration or securing the process margin.

[0225] FIGS. 16A to 16I illustrate a method for fabricating a semiconductor device in accordance with an embodiment of the present disclosure. FIGS. 16A to 16I are cross-sectional views simultaneously illustrating the fabrication method according to the A-A' and B-B' lines shown in FIG. 15A.

[0226] Referring to FIG. 16A, a buffer layer 12 may be formed over a substrate 11. The substrate 11 may be a material appropriate for semiconductor processing. The substrate 11 may include a semiconductor substrate. The substrate 11 may be formed of a silicon-containing material. The substrate 11 may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate 11 may include another semiconductor material such as germanium. The substrate 11 may include a III/V-group semiconductor substrate, for example, a compound semiconductor substrate, such as gallium arsenide (GaAs). The substrate 11 may include a Silicon-On-Insulator (SOI) substrate. The buffer layer 12 may include a dielectric material. The buffer layer 12 may include silicon oxide, silicon nitride, or a combination thereof.

[0227] A conductive layer 13A may be formed over the buffer layer 12. The conductive layer 13A may include a metal-based material. The conductive layer 13A may include a metal, a metal nitride, a metal silicide, or a combination thereof. The conductive layer 13A may be formed by an Atomic Layer Deposition (ALD) process, a Physical Vapor Deposition (PVD) process, or a Chemical Vapor Deposition (CVD) process. The conductive layer 13A may be formed to have a thickness of approximately 100 to 400 Å. For example, the conductive layer 13A may be formed by depositing a tungsten layer to a thickness of approximately 200 Å through a Physical Vapor Deposition (PVD) process. The buffer layer 12 may be formed of silicon oxide to reduce the parasitic capacitance between the substrate 11 and the conductive layer 13A. For example, the buffer layer 12 may include Tetraethyl Orthosilicate (TEOS).

[0228] A barrier material layer 14A may be formed over the conductive layer 13A. The barrier material layer 14A may include a metal-based material. The barrier material layer 14A may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier material layer 14A may include titanium nitride, molybdenum, or ruthenium. The barrier material layer 14A may be formed by an Atomic Layer Deposition (ALD) process, a Physical Vapor Deposition (PVD) process, or a Chemical Vapor Deposition (CVD) process. The barrier material layer 14A may be formed to have a thickness of approximately 10 to 50 Å. For example, the barrier material layer 14A may be formed by depositing titanium nitride to a thickness of approximately 20 Å through a physical vapor deposition (PVD) process.

[0229] An oxide semiconductor stack may be formed over the barrier material layer 14A. For example, the oxide semiconductor stack may include a first interface material layer 15A, a channel material layer 16A, and a second interface material layer 17A.

[0230] The first interface material layer 15A may be formed over the barrier material layer 14A. The first interface material layer 15A may include a conductive material. The first interface material layer 15A may include an oxide semiconductor material. The first interface material layer 15A may contain indium. The first interface material layer 15A may include an indium-rich oxide semiconductor material. For example, the first interface material layer 15A may include indium-rich IGZO. The first interface material layer 15A may be formed to have a thickness of approximately 10 to 50 Å.

[0231] The channel material layer 16A may be formed over the first interface material layer 15A. The channel material layer 16A may include a conductive material. The channel material layer 16A may include an oxide semiconductor material. The channel material layer 16A may contain indium. The channel material layer 16A may include indium gallium zinc oxide (IGZO). The channel material layer 16A may be formed to have a thickness of approximately 200 to 1000 Å.

[0232] The second interface material layer 17A may be formed over the channel material layer 16A. The second interface material layer 17A may include a conductive material. The second interface material layer 17A may include an oxide semiconductor material. The second interface material layer 17A may contain indium. The second interface material layer 17A may include an indium-rich oxide semiconductor material. For example, the second

interface material layer 17A may include indium-rich IGZO. The second interface material layer 17A may be formed to have a thickness of approximately 10 to 50 Å.

[0233] As described above, the first interface material layer 15A, the channel material layer 16A, and the second interface material layer 17A may be vertically stacked over the barrier material layer 14A. The first interface material layer 15A, the channel material layer 16A, and the second interface material layer 17A may all include an oxide semiconductor material. The first interface material layer 15A, the channel material layer 16A, and the second interface material layer 17A may all include indium gallium zinc oxide (IGZO). The first interface material layer 15A and the second interface material layer 17A may have a higher indium concentration than that of the channel material layer 16A. The channel material layer 16A may be IGZO, and the first interface material layer 15A and the second interface material layer 17A may be indium-rich IGZO. Since the first interface material layer 15A and the second interface material layer 17A contain a high concentration of indium, the resistance may be reduced compared to that of the channel material layer 16A, and the seamless interconnection of the channel material layer 16A may be possible.

[0234] Subsequently, a sacrificial layer 18A may be formed over the second interface material layer 17A. The sacrificial layer 18A may include a stack of different materials. The sacrificial layer 18A may include silicon nitride.

[0235] Referring to FIG. 16B, sacrificial lines 18 may be formed over the second interface material layer 17A. The sacrificial lines 18 may be formed by etching the sacrificial layer 18A. The sacrificial lines 18 may function to protect the first interface material layer 15A, the channel material layer 16A, and the second interface material layer 17A from a subsequent process. For example, the sacrificial lines 18 may be used as an etching barrier during the etching process of the first interface material layer 15A, the channel material layer 16A, and the second interface material layer 17A.

[0236] Each of the sacrificial lines 18 may include a stack of different materials. The sacrificial lines 18 may include silicon nitride. The etching process of the sacrificial layer 18A for forming the sacrificial lines 18 may include a double patterning process.

[0237] Subsequently, the second interface material layer 17A, the channel material layer 16A, and the first interface material layer 15A may be etched by using the sacrificial lines 18 as an etching barrier. Further, the barrier material layer 14A and the conductive layer 13A may be etched by using the sacrificial lines 18 as an etching barrier.

[0238] A plurality of line structures 19L and first trenches 19T may be formed over the buffer layer 12 by a series of the etching processes as described above. Each of the line structures 19L may include a stack of a bit line 13, a bit line barrier layer 14, a first oxide semiconductor interface layer 15B, a channel material layer 16B, a second oxide semiconductor interface layer 17B, and a sacrificial line 18. The bit line barrier layer 14 may be formed by etching the barrier material layer 14A, and the bit line 13 may be formed by etching the conductive layer 13A. The first oxide semiconductor interface layer 15B, the channel material layer 16B, and the second oxide semiconductor interface layer 17B may be formed by etching the first interface material layer 15A, the channel material layer 16A, and the second interface material layer 17A, respectively. The first trenches 19T may be disposed between the line structures 19L. The stack

of the first oxide semiconductor interface layer **15B**, the channel material layer **16B**, and the second oxide semiconductor interface layer **17B** may be referred to as an ‘oxide semiconductor line’.

[0239] Referring to FIG. **16C**, dielectric lines **20** may be formed between the line structures **19L**. The dielectric lines **20** may include a dielectric material. For example, the dielectric lines **20** may include silicon oxide, silicon nitride, silicon carbon oxide (SiCO), a spin-on-dielectric layer, or a combination thereof. The dielectric lines **20** may be formed by depositing a dielectric material to fill the first trenches **19T** between the line structures **19L** and then performing a planarization process of the dielectric material. The dielectric lines **20** may fill the first trenches **19T** respectively. The dielectric lines **20** may be formed by sequentially forming silicon nitride, silicon carbon oxide, and a spin-on-dielectric layer and then performing a planarization process.

[0240] Referring to FIG. **16D**, a portion of the line structures **19L** may be selectively etched to form sacrificial pillars **18P** and oxide semiconductor pillars **21P**. The bit line barrier layer **14** and the bit line **13** may be disposed in the lower portions of the oxide semiconductor pillars **21P**.

[0241] Each of the oxide semiconductor pillars **21P** may include a stack of a first oxide semiconductor interface layer **15**, a channel layer **16**, and a second oxide semiconductor interface layer **17**. The first oxide semiconductor interface layer **15**, the channel layer **16**, and the second oxide semiconductor interface layer **17** may be formed by etching the first oxide semiconductor interface layer **15B**, the channel material layer **16B**, and the second oxide semiconductor interface layer **17B**, respectively. The sacrificial pillars **18P** may be formed over the second oxide semiconductor interface layer **17**.

[0242] Second trenches **22** and **22A** may be formed between the oxide semiconductor pillars **21P**. The bottom surfaces of the second trenches **22** and **22A** may expose the upper surface of the bit line barrier layer **14**. The first trenches **19T** illustrated in FIG. **16B** and the second trenches **22** and **22A** illustrated in FIG. **16C** may intersect with each other. The first trenches **19T** may be deeper than the second trenches **22** and **22A**. In the B-B' direction, the dielectric lines **20** may be cut by the second trenches **22** and **22A**. Hereinafter, the dielectric lines **20** may be simply referred to as ‘dielectric pillars **20**’. In the A-A' direction, the dielectric pillars **20** may be disposed between the oxide semiconductor pillars **21P**.

[0243] Each of the oxide semiconductor pillars **21P** may include first to fourth sidewalls SW1 to SW4. The first sidewall SW1 and the second sidewall SW2 of each oxide semiconductor pillar **21P** may be exposed by the second trenches **22**. The third sidewall SW3 and the fourth sidewall SW4 of each oxide semiconductor pillar **21P** may not be exposed by the dielectric pillars **20**. The sacrificial pillar **18P** may also include the exposed sidewalls and the non-exposed sidewalls, just like the oxide semiconductor pillar **21P**.

[0244] The second trenches **22** and **22A** may include the first sub-trench **22** and the second sub-trench **22A**. The first sub-trench **22** and the second sub-trench **22A** may define the space between the neighboring oxide semiconductor pillars **21P**. The width of the first sub-trench **22** may be larger than the width of each second sub-trench **22A**. The first sub-trench **22** may correspond to the first and second spaces S1 and S2 shown in FIGS. **15A** and **15C** and the shield trench ST shown in FIG. **15D**. The second sub-trenches **22A** may

correspond to the third space S3 shown in FIG. **15A** and may be referred to as ‘word line trenches’.

[0245] As described above, the space between the neighboring oxide semiconductor pillars **21P** may be asymmetric due to the difference between the width of the first sub-trench **22** and the width of the second sub-trench **22A**.

[0246] Referring to FIG. **16E**, a gate dielectric layer **23** may be formed over the exposed first and second sidewalls SW1 and SW2 of the oxide semiconductor pillars **21P**. The gate dielectric layer **23** may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The high-k material may include a material having a greater dielectric constant than the dielectric constant of silicon oxide. For example, the high-k material may include a material having a greater dielectric constant than approximately 3.9. For another example, the high-k material may include a material having a dielectric constant greater than approximately 10. For yet another example, the high-k material may include a material having a dielectric constant of approximately 10 to 30. The high-k material may include at least one metallic element. The high-k material may include a hafnium-containing material. The hafnium-containing material may include hafnium oxide, hafnium silicon oxide, hafnium silicon oxynitride, or a combination thereof. According to another embodiment of the present disclosure, the high-k material may include lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, zirconium silicon oxynitride, aluminum oxide, or a combination thereof. Other known high-k materials may be optionally used as the high-k material. The gate dielectric layer **23** may include a metal oxide. The gate dielectric layer **23** may be formed conformally on the first and second sidewalls SW1 and SW2 of the oxide semiconductor pillar **21P**. The gate dielectric layer **23** may also conformally cover the exposed sidewalls and upper surface of the sacrificial pillar **18P**. The gate dielectric layer **23** and the first oxide semiconductor interface layer **15** may have the same thickness. According to another embodiment of the present disclosure, the gate dielectric layer **23** may be thinner than the first oxide semiconductor interface layer **15**.

[0247] A word line conductive layer **24A** may be formed over the gate dielectric layer **23**. The word line conductive layer **24A** may be formed conformally over the gate dielectric layer **23**. The word line conductive layer **24A** may include a metal-based material. The word line conductive layer **24A** may include a metal, a metal nitride, or a combination thereof. The word line conductive layer **24A** may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0248] The word line conductive layer **24A** may fully gap-fill the second sub-trenches **22A** over the gate dielectric layer **23**. The word line conductive layer **24A** may be conformally formed over the gate dielectric layer **23**. The word line conductive layer **24A** may partially fill the first sub-trench **22**.

[0249] Spacers **25** may be formed over the word line conductive layer **24A**. The spacers **25** may include an oxide. The spacers **25** may be formed by depositing silicon oxide over the word line conductive layer **24A** and then performing an etch-back process. The upper surfaces of the spacers **25** may be disposed at a lower level than the upper surfaces of the sacrificial pillars **18P**.

[0250] Referring to FIG. 16F, tapered vertical word lines 24S may be formed. To form the tapered vertical word lines 24S, the word line conductive layer 24A may be selectively etched by using the spacer 25 as an etching barrier. The spacers 25 may function to protect the tapered vertical word lines 24S during the etching process of the word line conductive layer 24A. The etching process of the word line conductive layer 24A may include an etch-back process and a wet etching process that are performed sequentially. Non-tapered vertical word lines may be formed by an etch-back process, and the tapered vertical word lines 24S may be formed by the subsequent wet etching process. The bottom portion of the tapered vertical word lines 24S may become thin by the wet etching process. The tapered vertical word lines 24S may correspond to the main gate 124S of FIGS. 15A, 15B, 15C, and 15D, and the bottom portions of the tapered vertical word lines 24S may correspond to the lower level portion 124L.

[0251] The tapered vertical word lines 24S may have a single structure. The tapered vertical word lines 24S may be disposed on one side of the oxide semiconductor pillars 21P.

[0252] While the tapered vertical word lines 24S are formed, a common gate, i.e., a shield gate 24G, may be formed. The shield gate 24G may be formed in the second sub-trench 22A.

[0253] The shield gate 24G may correspond to the shield gate 124G of FIGS. 15A, 15B, 15C, and 15D.

[0254] Referring to FIG. 16G, the spacers 25 of FIG. 16F may be removed. Each of the tapered vertical word lines 24S may be a single structure and may be formed on one sidewall among the first and second sidewalls of the oxide semiconductor pillars 21P. Referring to FIGS. 15A, 15B, 15C, and 15D, the tapered vertical word lines 24S may extend in the second direction D2, and the bit lines 13 may extend in the first direction D1. The oxide semiconductor pillars 21P and the dielectric pillars 20 may be alternately disposed in the second direction D2. The tapered vertical word lines 24S may extend along the profile of the exposed sidewalls of the oxide semiconductor pillars 21P and the dielectric pillars 20.

[0255] As described above, the space between the neighboring oxide semiconductor pillars 21P may be asymmetric due to the difference between the width of the first sub-trench 22 and the width of each second sub-trench 22A. Accordingly, when the word line conductive layer 24A is deposited, the second sub-trenches 22A may be completely buried and the first sub-trench 22 may be deposited in the form of a spacer. As a result, the shield gate 24G may have a single buried structure, and only the main gate 124S may be separated.

[0256] Since the shield gate 24G is shared, the space of the main gate 24S may be widened, thereby securing additional process margin.

[0257] Referring to FIG. 16H, an inter-layer dielectric layer 26 may be formed over the tapered vertical word line 24S. The inter-layer dielectric layer 26 may be planarized to expose the upper surfaces of the sacrificial pillars 18P of FIG. 16G. The inter-layer dielectric layer 26 may include silicon oxide, such as a spin-on-dielectric (SOD) layer. The sacrificial pillars 18P may serve as a stop layer during the planarization process of the inter-layer dielectric layer 26.

[0258] Subsequently, the sacrificial pillars 18P may be selectively removed. As a result, hole-shaped recesses 18R may be formed. The hole-shaped recesses 18R may selectively expose the surfaces of the second oxide semiconduc-

tor interface layers 17. The sacrificial pillars 18P may be removed by performing a wet etching process.

[0259] Referring to FIG. 16I, first contact plugs 27 filling the hole-shaped recesses 18R may be formed. The first contact plugs 27 may directly contact the second oxide semiconductor interface layers 17. The first contact plugs 27 may include a metal, a metal nitride, or a combination thereof. For example, the first contact plugs 27 may be formed by depositing titanium nitride to fill the hole-shaped recesses 18R of FIG. 16H, and then planarizing the titanium nitride to expose the surface of the inter-layer dielectric layer 26.

[0260] Subsequently, referring to FIG. 2J, second contact plugs 28 may be formed over the first contact plugs 27.

[0261] Subsequently, referring to FIG. 2K, a spacer material layer 29 may be formed between the second contact plugs 28. The spacer material layer 29 may include silicon nitride. The spacer material layer 29 may be formed between the neighboring second contact plugs 28. A capacitor 30 may be formed over each of the second contact plugs 28.

[0262] This technology may be able to reduce the space between the neighboring cells by forming a main gate and a shield gate each having a single structure, thereby reducing the cell area and increasing the degree of integration or securing the process margin.

[0263] According to the embodiment of the present disclosure, since a bit line and a transistor are sequentially formed by using a depositable channel material, procedural difficulty may be reduced.

[0264] According to the embodiment of the present disclosure, the degree of integration may be increased by stacking a memory cell array without wafer bonding, and the degree of integration may be improved through a peripheral-under-cell (PUC) structure where a memory cell array is formed over a peripheral circuit portion.

[0265] According to the embodiment of the present disclosure, a metal oxide, particularly IGZO, may be used as a channel material to suppress gate-induced drain leakage and junction leakage to improve retention characteristics.

[0266] The effects desired to be obtained in the embodiments of the present disclosure are not limited to the effects mentioned above, and other effects not mentioned above may also be clearly understood by those of ordinary skill in the art to which the present disclosure pertains from the description above.

[0267] While the embodiments of the present disclosure have been described with respect to specific embodiments, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the present disclosure as defined in the following claims. Furthermore, the embodiments may be combined to form additional embodiments.

What is claimed is:

1. A semiconductor device comprising:

- a first pillar and a second pillar spaced apart from each other by a shield trench, each pillar including an inner side that defines inner sidewalls of the shield trench and an outer side that faces the inner side;
- a shield gate formed in the shield trench;
- a first main gate formed on the outer side of the first pillar;
- a second main gate formed on the outer side of the second pillar;
- a bit line formed on lower portions of the first and second pillars; and

- a capacitor formed on an upper portion of each of the first and second pillars.
2. The semiconductor device of claim 1, wherein each of the first and second main gates has a tapered shape.
3. The semiconductor device of claim 2, wherein each of the first and second main gates includes:
- a first sidewall of a straight profile; and
  - a second sidewall of a tapered profile.
4. The semiconductor device of claim 3, wherein the first sidewalls of the first and second main gates are formed on the outer sides of the first and second pillars, respectively.
5. The semiconductor device of claim 1, wherein each of the first and second main gates includes:
- an upper level portion; and
  - a lower level portion which is thinner than the upper level portion.
6. The semiconductor device of claim 1, further comprising:
- a gate dielectric layer formed between the outer sides of the first and second pillars and the first and second main gates; and
  - a shield gate dielectric layer formed between the shield gate and the inner sides of the first and second pillars.
7. The semiconductor device of claim 1, wherein each of the first and second pillars includes:
- a first oxide semiconductor interface layer formed over the bit line;
  - a second oxide semiconductor interface layer formed below the capacitor; and
  - an oxide semiconductor channel layer between the first oxide semiconductor interface layer and the second oxide semiconductor interface layer.
8. The semiconductor device of claim 7, wherein each of the oxide semiconductor channel layer, the first oxide semiconductor interface layer, and the second oxide semiconductor interface layer includes an oxide semiconductor material, and
- each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes an oxide semiconductor material having a lower resistance than the oxide semiconductor channel layer.
9. The semiconductor device of claim 7, wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes
- a metal-rich oxide semiconductor material having a higher metal content than the oxide semiconductor channel layer.
10. The semiconductor device of claim 7, wherein the oxide semiconductor channel layer includes indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO) or zinc tin oxide (ZTO),
- wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes indium-rich IGZO, and
  - wherein the indium-rich IGZO includes IGZO whose indium content is higher than gallium and zinc contents.
11. A semiconductor device comprising:
- a trench;
  - a first row array of first oxide semiconductor pillars and
  - a second row array of second oxide semiconductor pillars spaced apart from each other by the trench, each
- pillar including an inner side that defines inner side-walls of the trench and an outer side that faces the inner side;
  - a common gate extending along the first row array and the second row array, and formed in the trench while being formed on the inner sides of the first and second oxide semiconductor pillars;
  - a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and
  - a second gate extending along the outer sides of the second oxide semiconductor pillars of the second array.
12. The semiconductor device of claim 11, wherein the first and second oxide semiconductor pillars of the first and second row arrays form a column array in which the first oxide semiconductor pillars and the second oxide semiconductor pillars are alternately disposed in a direction intersecting with the first and second row arrays.
13. The semiconductor device of claim 12, further comprising:
- a bit line formed on lower portions of the first and second oxide semiconductor pillars of the column array; and
  - capacitors formed in upper portions of the first and second oxide semiconductor pillars of the row arrays and the column array.
14. The semiconductor device of claim 11, wherein each of the first and second gates has a tapered shape.
15. The semiconductor device of claim 11, wherein each of the first and second gates includes:
- a first sidewall of a straight profile; and
  - a second sidewall of a tapered profile.
16. The semiconductor device of claim 15, wherein the first sidewalls of the first and second gates are formed on the outer sides of the first and second oxide semiconductor pillars, respectively.
17. The semiconductor device of claim 11, further comprising:
- a gate dielectric layer formed between the outer sides of the first and second oxide semiconductor pillars and the first and second gates; and
  - a common gate dielectric layer formed between the common gate and the inner sides of the first and second oxide semiconductor pillars.
18. The semiconductor device of claim 11, wherein each of the first and second oxide semiconductor pillars includes:
- a first oxide semiconductor interface layer;
  - a second oxide semiconductor interface layer; and
  - an oxide semiconductor channel layer between the first oxide semiconductor interface layer and the second oxide semiconductor interface layer.
19. The semiconductor device of claim 18, wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes
- a metal-rich oxide semiconductor material having a higher metal content than the oxide semiconductor channel layer.
20. The semiconductor device of claim 18, wherein the oxide semiconductor channel layer includes indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO), or zinc tin oxide (ZTO),
- wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes indium-rich IGZO, and

wherein the indium-rich IGZO includes IGZO whose indium content is higher than gallium and zinc contents.

**21.** A method for fabricating a semiconductor device, the method comprising:

forming a bit line;

forming a first row array of first oxide semiconductor pillars and a second row array of second oxide semiconductor pillars spaced apart from each other by a trench, each pillar including an inner side that defines inner sidewalls of the trench and an outer side that faces the inner side over the bit line;

forming a common gate extending along the first row array and the second row array on the inner sides of the first and second oxide semiconductor pillars;

forming a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and

forming a second gate extending along the outer sides of the second oxide semiconductor pillars of the second row array.

**22.** The method of claim **21**, wherein forming the first row array of the first oxide semiconductor pillars and the second row array of the second oxide semiconductor pillars includes:

forming an oxide semiconductor stack;

forming a plurality of oxide semiconductor lines by etching the oxide semiconductor stack;

forming dielectric lines between the oxide semiconductor lines; and

etching the oxide semiconductor lines to form the first and second oxide semiconductor pillars.

**23.** The method of claim **22**, wherein the oxide semiconductor stack includes a first oxide semiconductor interface layer, an oxide semiconductor channel layer, and a second oxide semiconductor interface layer that are sequentially stacked.

**24.** The method of claim **21**, wherein forming the common gate, the first gate and the second gate includes:

forming a preliminary conductive layer between the oxide semiconductor pillars;

dry-etching the preliminary conductive layer to form a non-tapered vertical conductive layer; and

wet-etching a lower region of the non-tapered vertical conductive layer to form a vertical conductive layer having a tapered shape.

**25.** The method of claim **21**, wherein each of the first and second oxide semiconductor pillars includes an oxide semiconductor material.

**26.** The method of claim **21**, further comprising:

forming contact plugs over the first and second oxide semiconductor pillars of the row array; and

forming capacitors over the contact plugs.

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